TOPIC – AD DNS

Assignment 1: Windows Server Administration Basics

As part of this assignment, you need to complete below tasks:

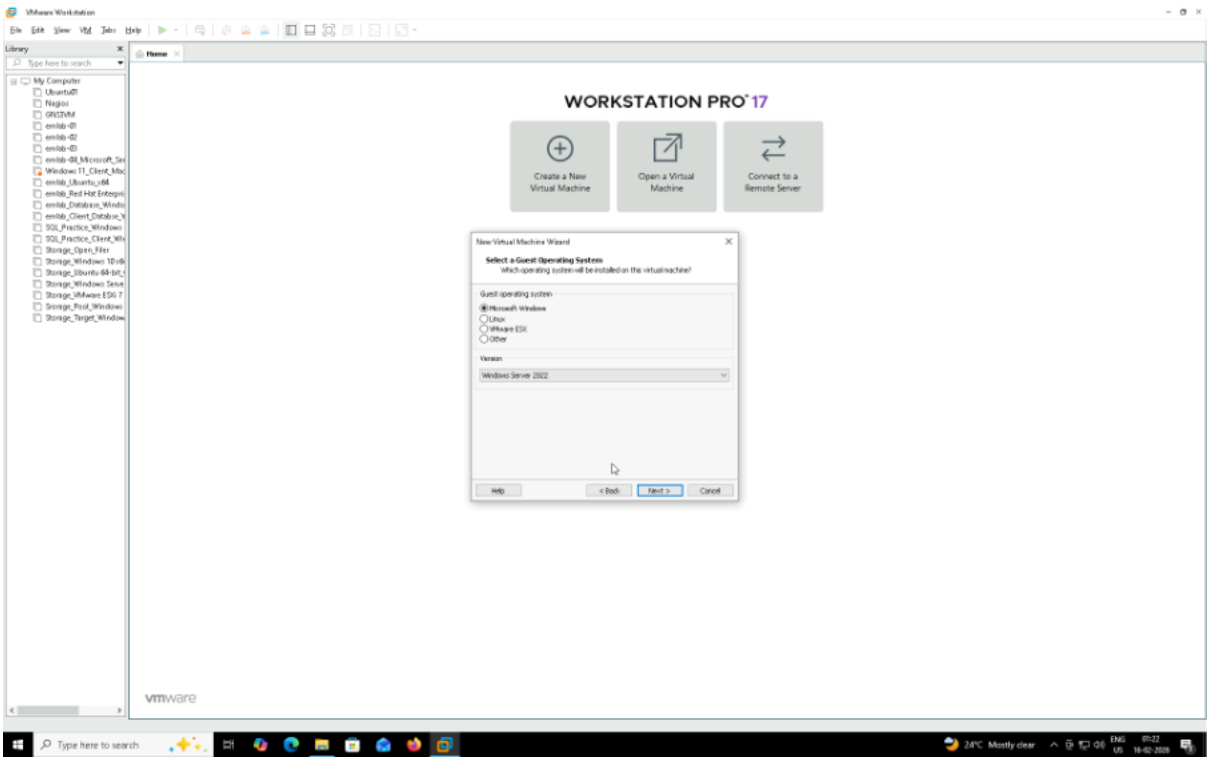
- Install and configure DNS server role on Windows Server
- Create and configure forward and reverse lookup zones
- Practice DNS record management (A, AAAA, CNAME, MX records)

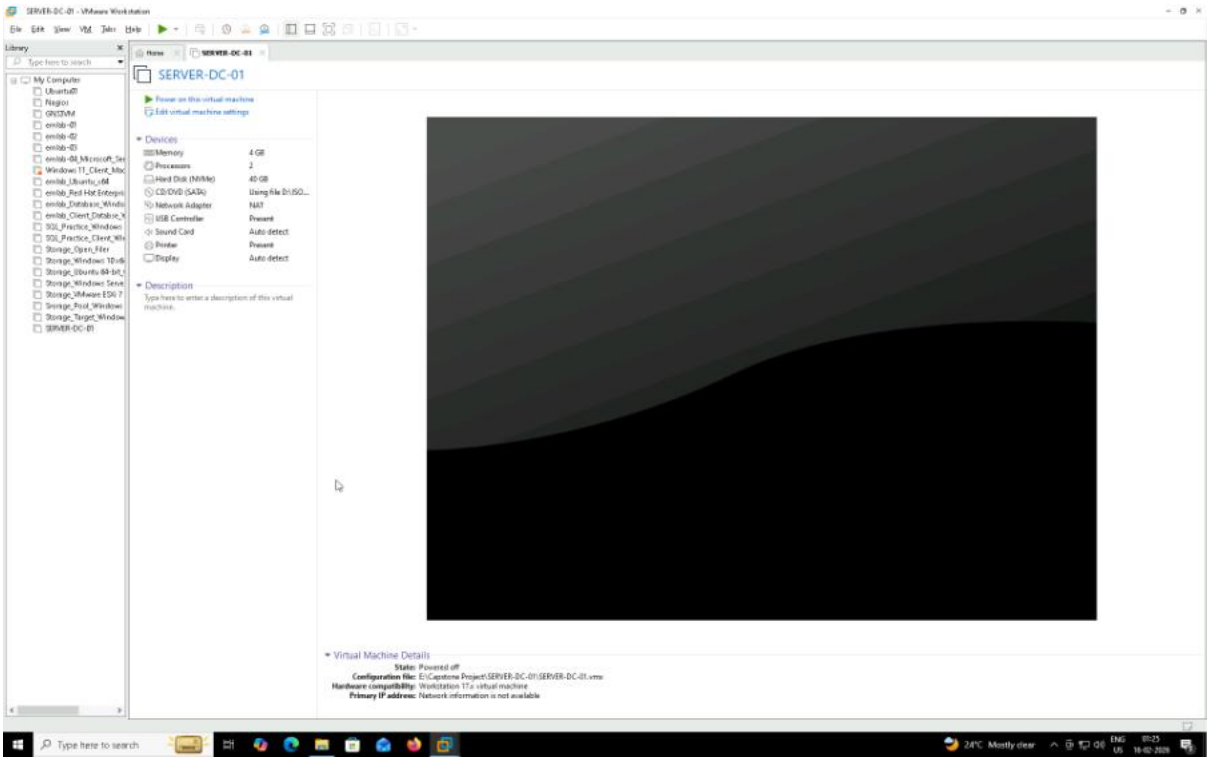
To accomplish this, we will create two VM's: -

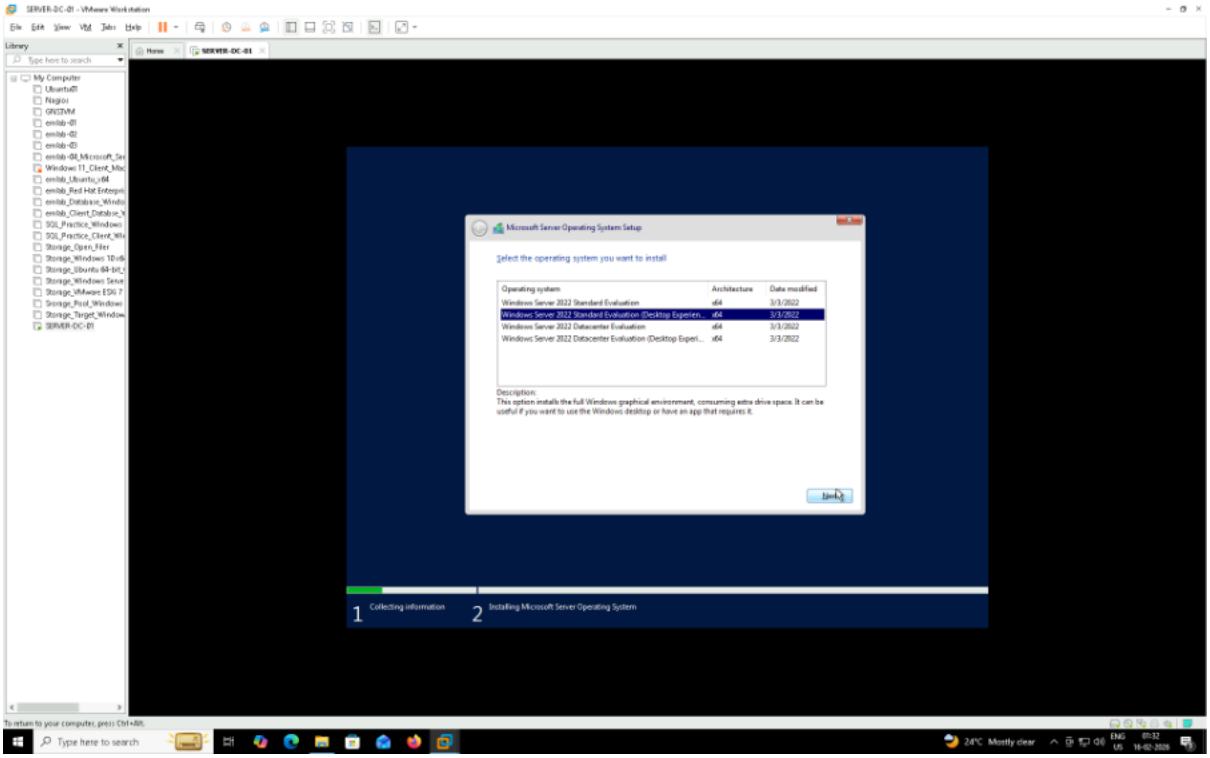
- Server Machine: -
    - Name: - SERVER-DC-01
    - Type: - Microsoft Windows
    - Version: - Windows Server 2022 (64-bit – Standard Edition)
    - RAM: - 4 GB
    - CPU: - 2 Cores
    - Disk: - 40 GB

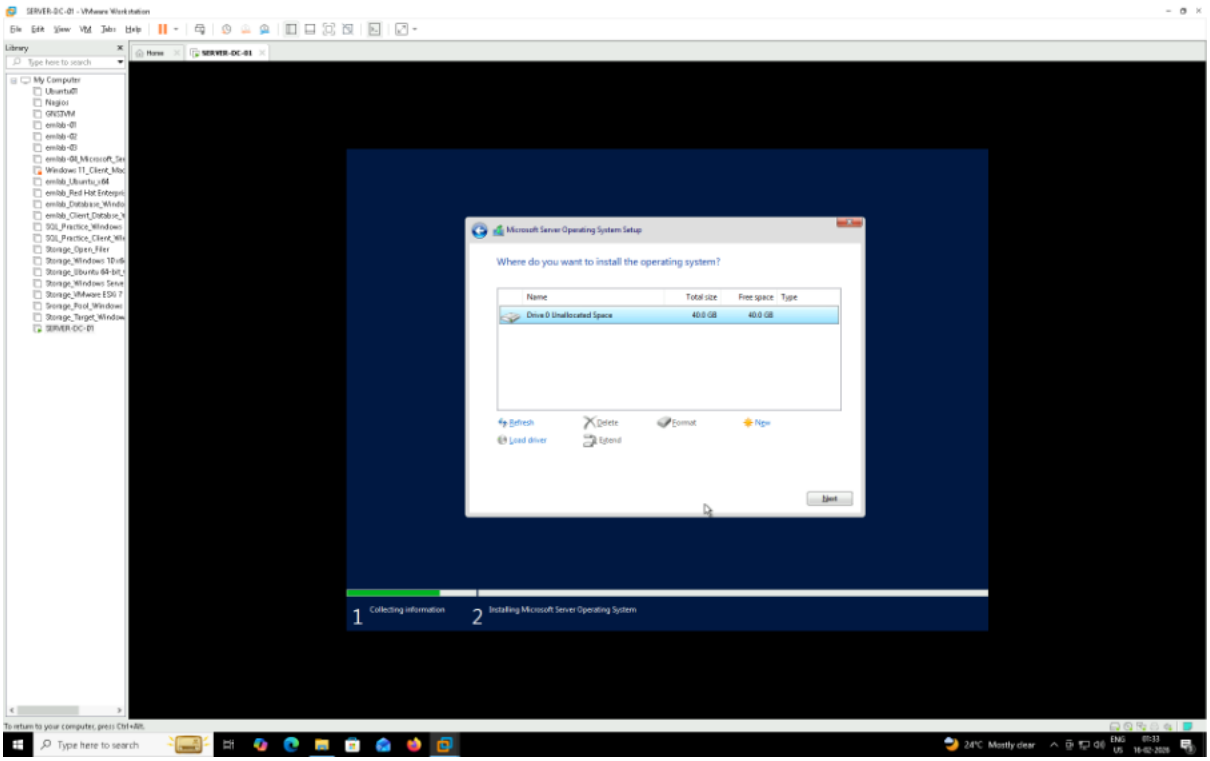
Virtual Machine Preparation: -

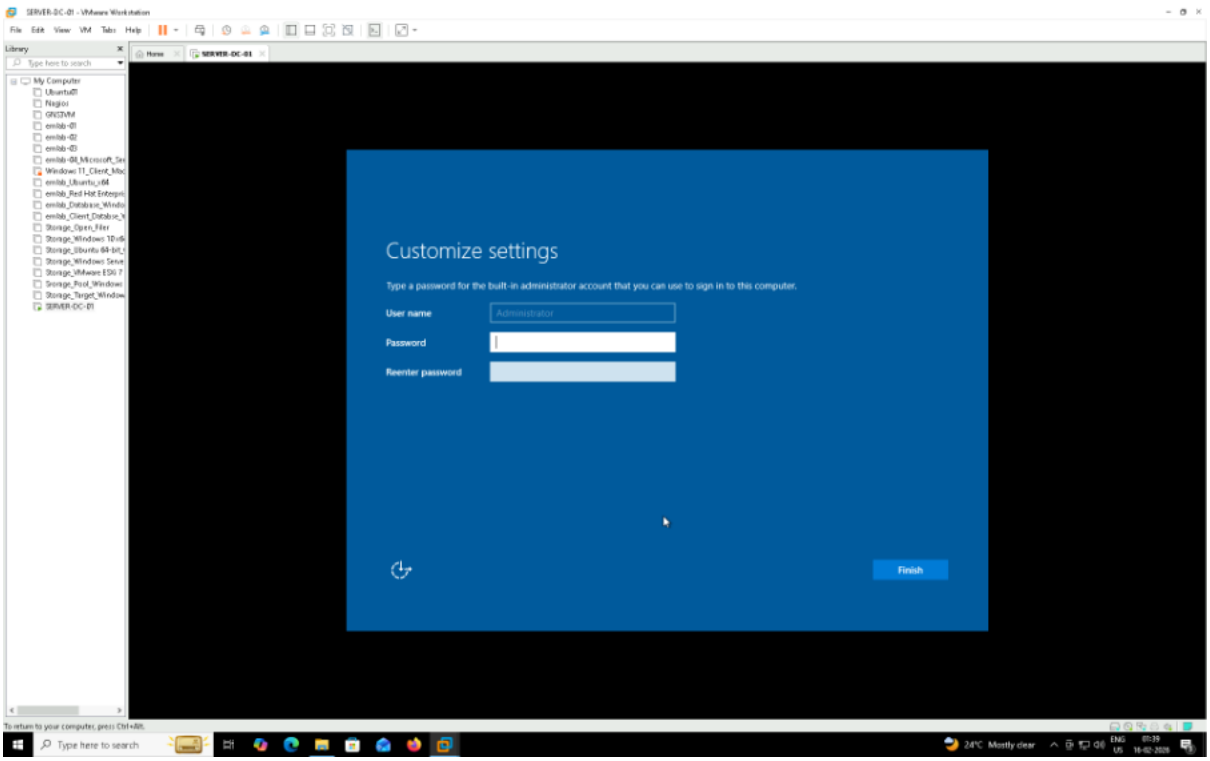
The virtual machine SERVER-DC-01 has been created using VMware Workstation to host ADDS services and DNS. The virtual machine was configured using NAT networking mode to allow internet connectivity for updates and package installation. Since NAT provides automatic IP addressing via VMware DHCP, Static IP configuration would be manually assigned later to ensure reliable Active Directory domain name resolution.

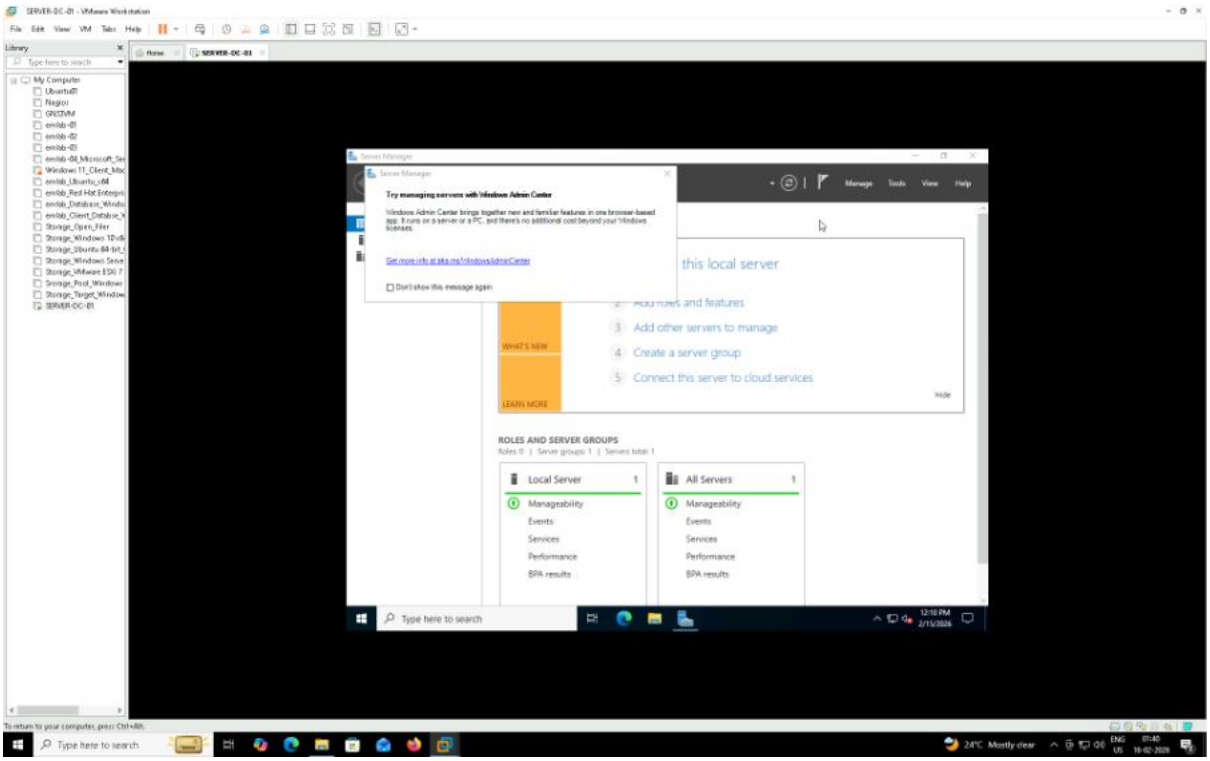

WORKSTATION PRO 17
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server
New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Windows Server 2022
Help
< Back
Next >
Cancel
vmware


SERVER-DC-01
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (NVMe) 40 GB
CD/DVD (SATA) Using file D:\ISO...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: E:\Capstone Project\SERVER-DC-01\SERVER-DC-01.vmx
Hardware compatibility: Workstation 17.x virtual machine
Primary IP address: Network information is not available


Microsoft Server Operating System Setup
Select the operating system you want to install
Operating system
Architecture
Date modified
Windows Server 2022 Standard Evaluation
Windows Server 2022 Standard Evaluation (Desktop Experien...
Windows Server 2022 Datacenter Evaluation
Windows Server 2022 Datacenter Evaluation (Desktop Experi...
Description:
This option installs the full Windows graphical environment, consuming extra drive space. It can be useful if you want to use the Windows desktop or have an app that requires it.
Next
1 Collecting information
2 Installing Microsoft Server Operating System

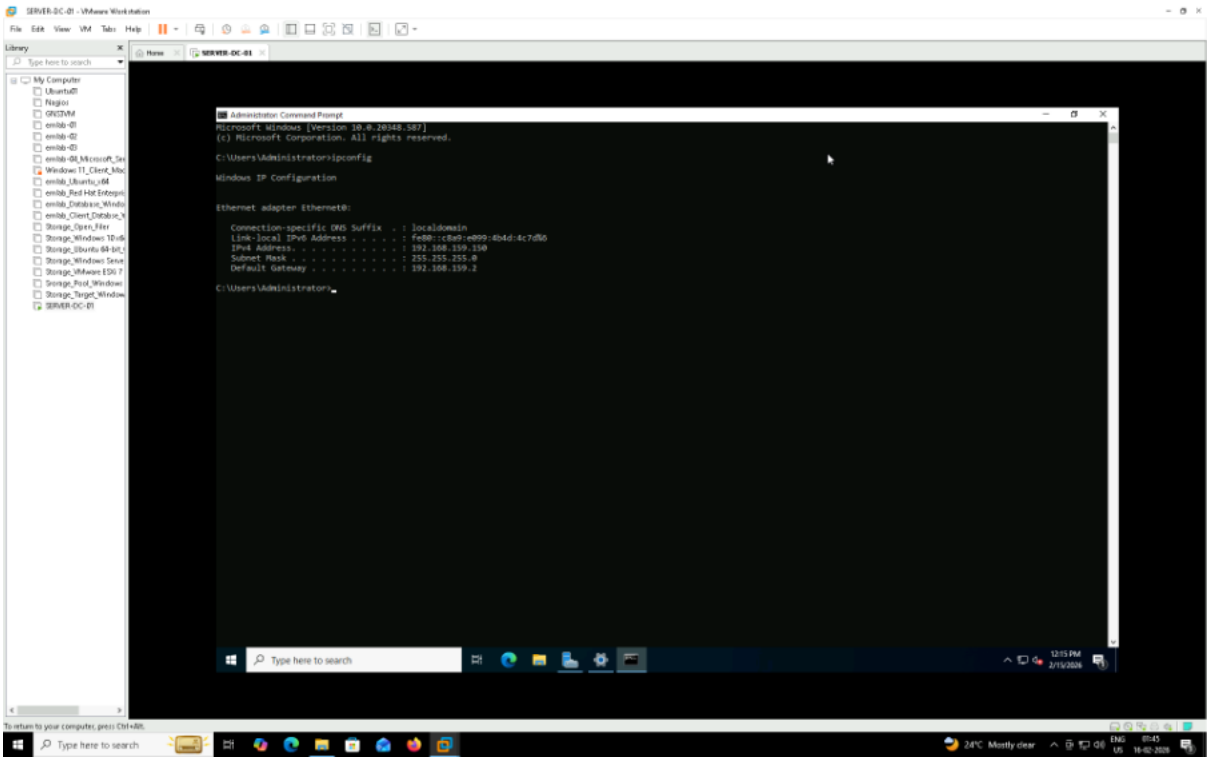

Microsoft Server Operating System Setup
Where do you want to install the operating system?
Name
Total size
Free space
Type
Drive 0 Unallocated Space
40.0 GB
40.0 GB
Refresh
Delete
Format
New
Load driver
Extend
Next
1 Collecting information
2 Installing Microsoft Server Operating System

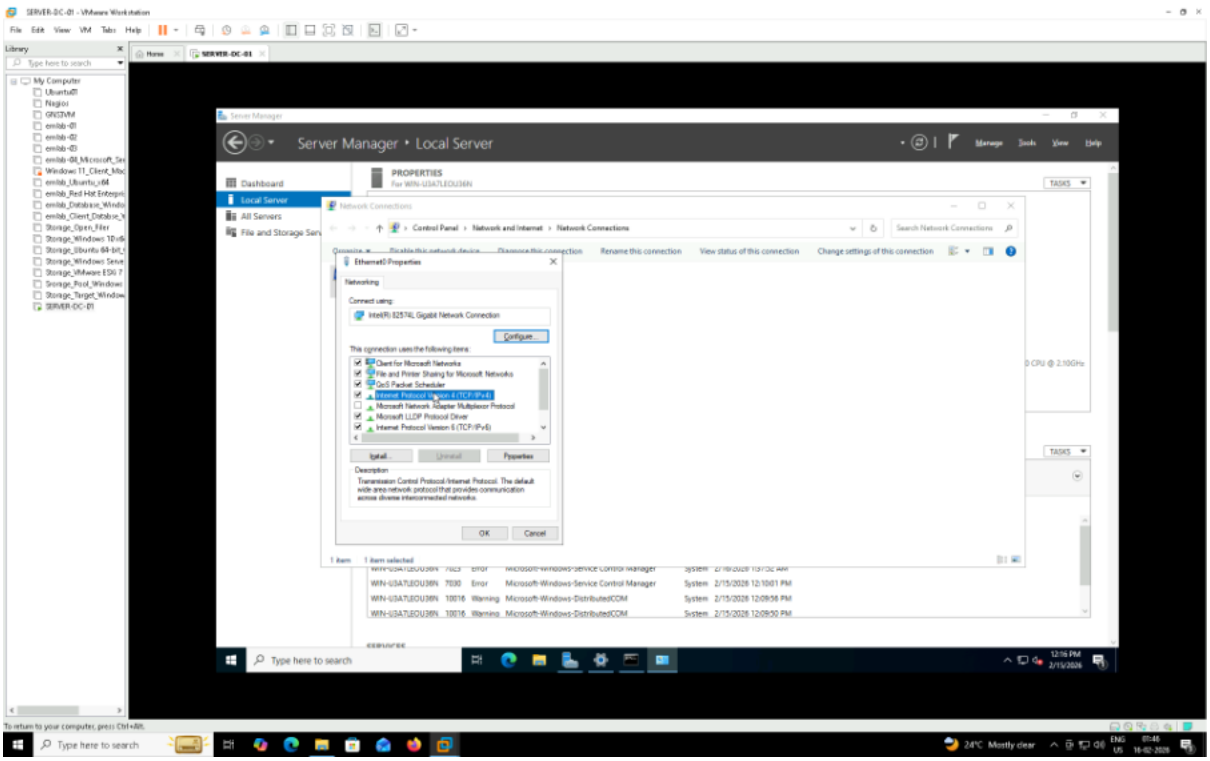

Customize settings
Type a password for the built-in administrator account that you can use to sign in to this computer.
User name
Administrator
Password
Reenter password
Finish

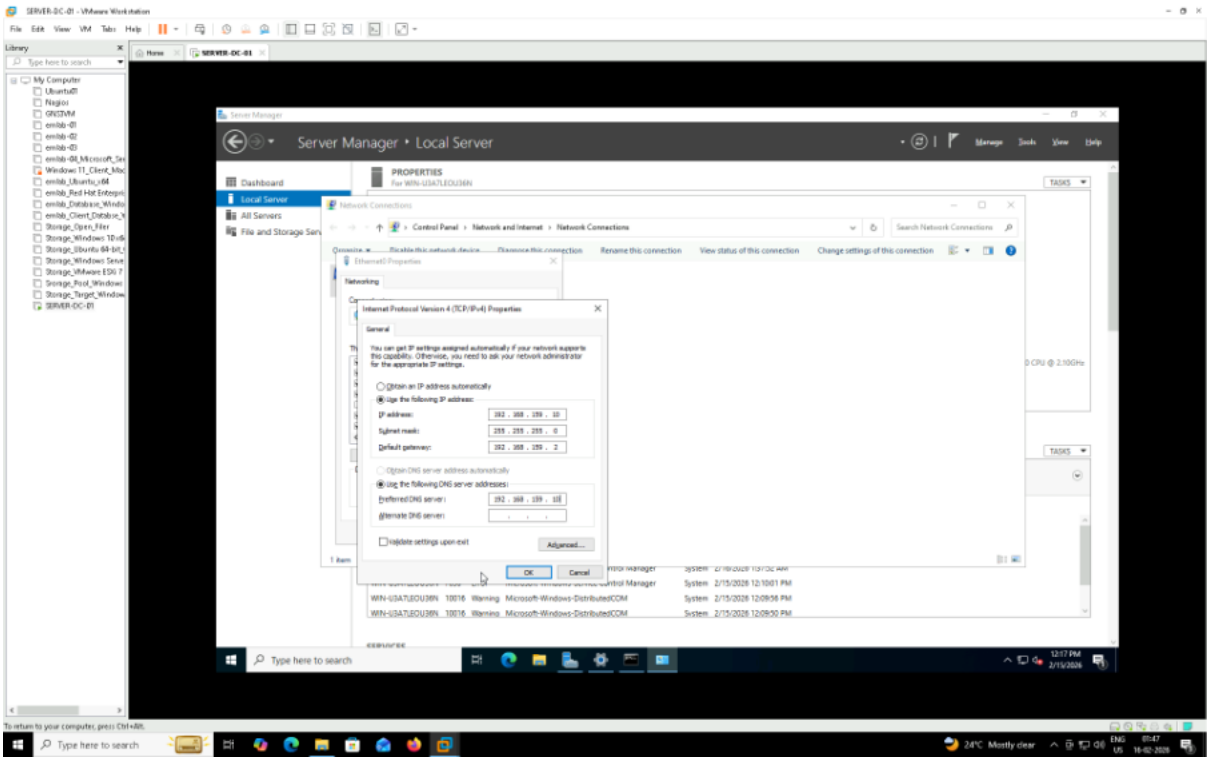

Try managing servers with Windows Admin Center
this local server
Add other servers to manage
Create a server group
Connect this server to cloud services
ROLES AND SERVER GROUPS
Local Server
All Servers
Manageability
Events
Services
Performance
BPA results

Before installing Active Directory, the server must use a static IP address because domain services depend on a fixed DNS location. Since the VM uses NAT, it receives an automatic IP from VMware DHCP Client. So, the current network details were checked

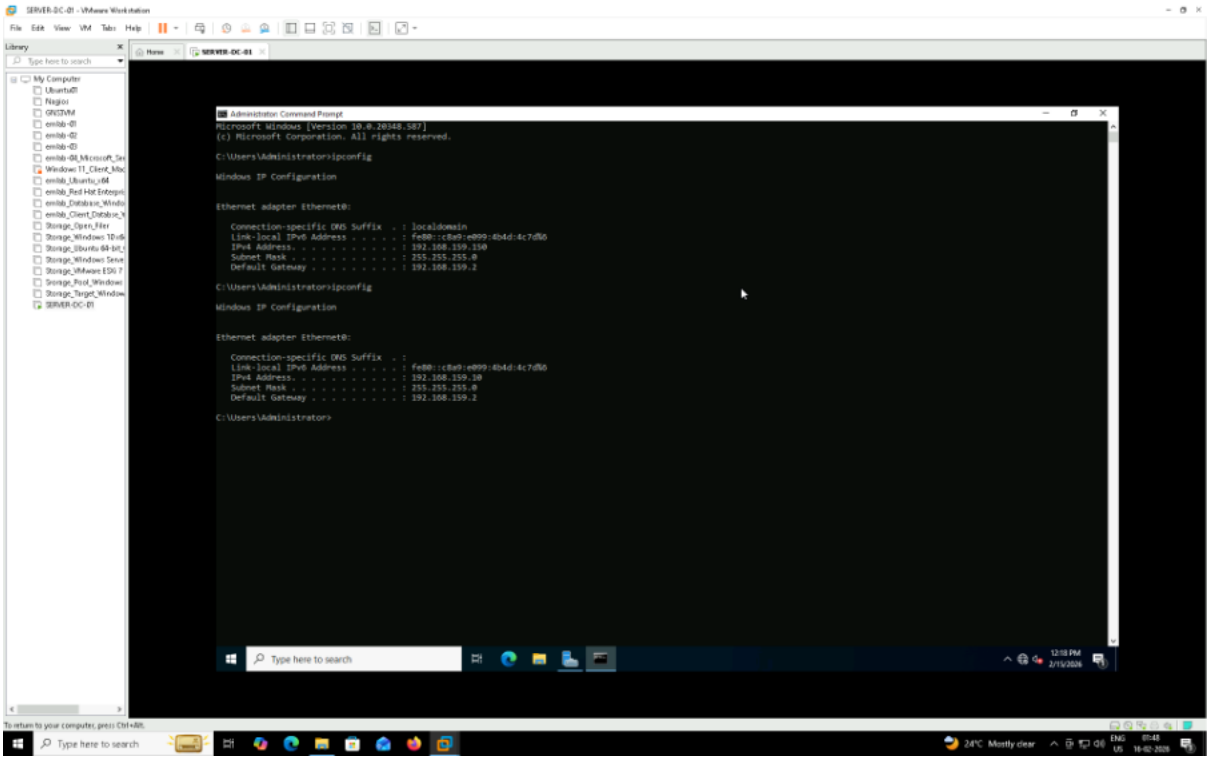
using the ipconfig to identify the subnet and default gateway before assigning a static address



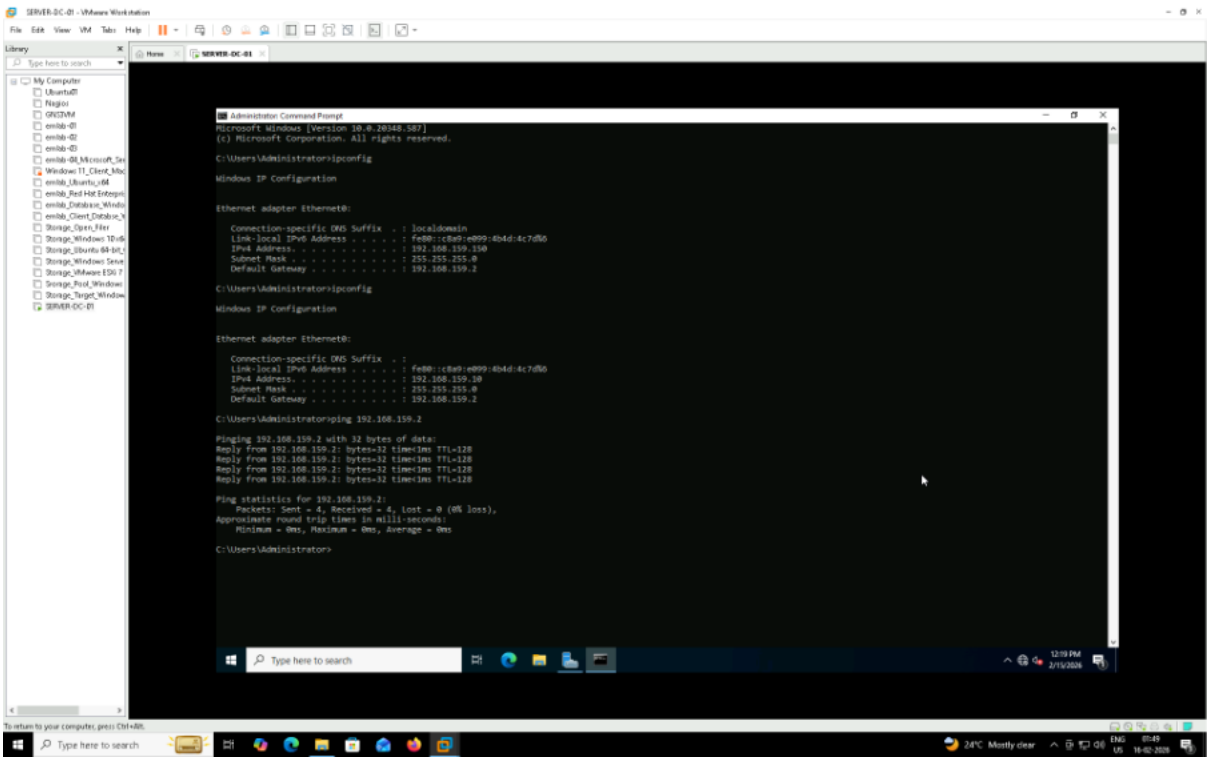

SERVER-DC-01 - VMware Workstation
Server Manager • Local Server
Dashboard
Local Server
All Servers
File and Storage Services
PROPERTIES
Network Connections
Control Panel • Network and Internet • Network Connections
Ethernet0 Properties
Networking
Connect using:
Intel(R) 82574L Gigabit Network Connection
Configure...
This connection uses the following items:
Client for Microsoft Networks
File and Printer Sharing for Microsoft Networks
QoS Packet Scheduler
Internet Protocol Version 4 (TCP/IPv4)
Microsoft Network Adapter Multiplexor Protocol
Microsoft LLDP Protocol Driver
Internet Protocol Version 6 (TCP/IPv6)
Install...
Uninstall
Properties
Description
Transmission Control Protocol/Internet Protocol. The default wide area network protocol that provides communication across diverse interconnected networks.
OK
Cancel
TASKS
Type here to search
24°C Mostly clear


SERVER-DC-01 - VMware Workstation
Server Manager • Local Server
Dashboard
Local Server
All Servers
File and Storage Services
PROPERTIES
Network Connections
Control Panel • Network and Internet • Network Connections
Ethernet0 Properties
Networking
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
Subnet mask:
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
TASKS
Type here to search
24°C Mostly clear

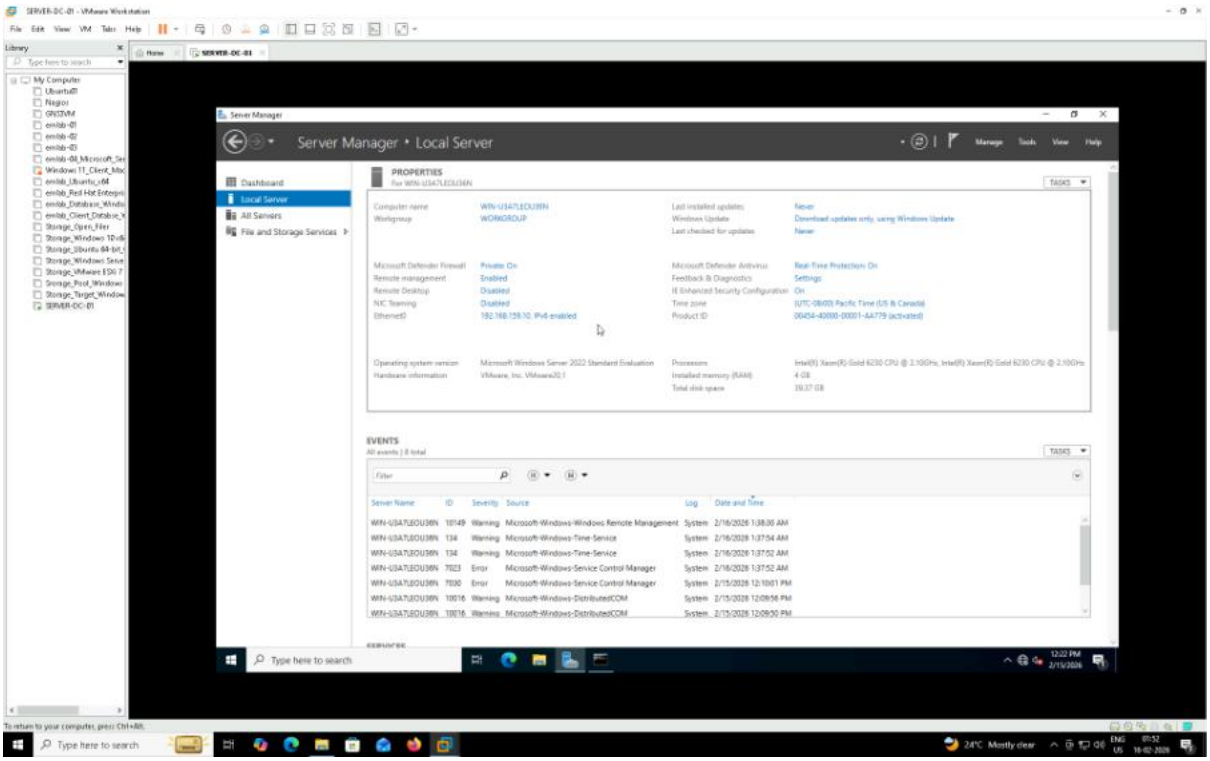
A static IP address was configured to ensure reliable DNS and Active Directory Communication. The server was assigned a fixed address within the NAT subnet and configured to use itself as the DNS Server.



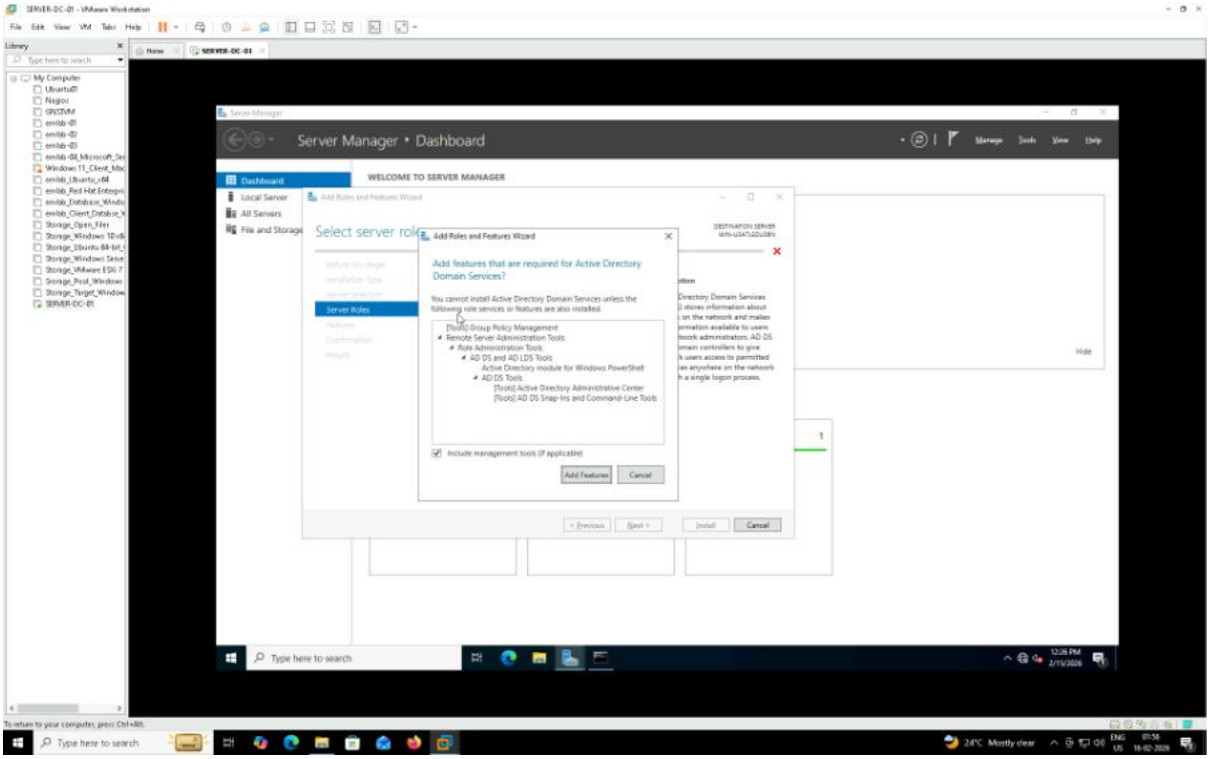
Connectivity with the gateway was verified to confirm network communication.

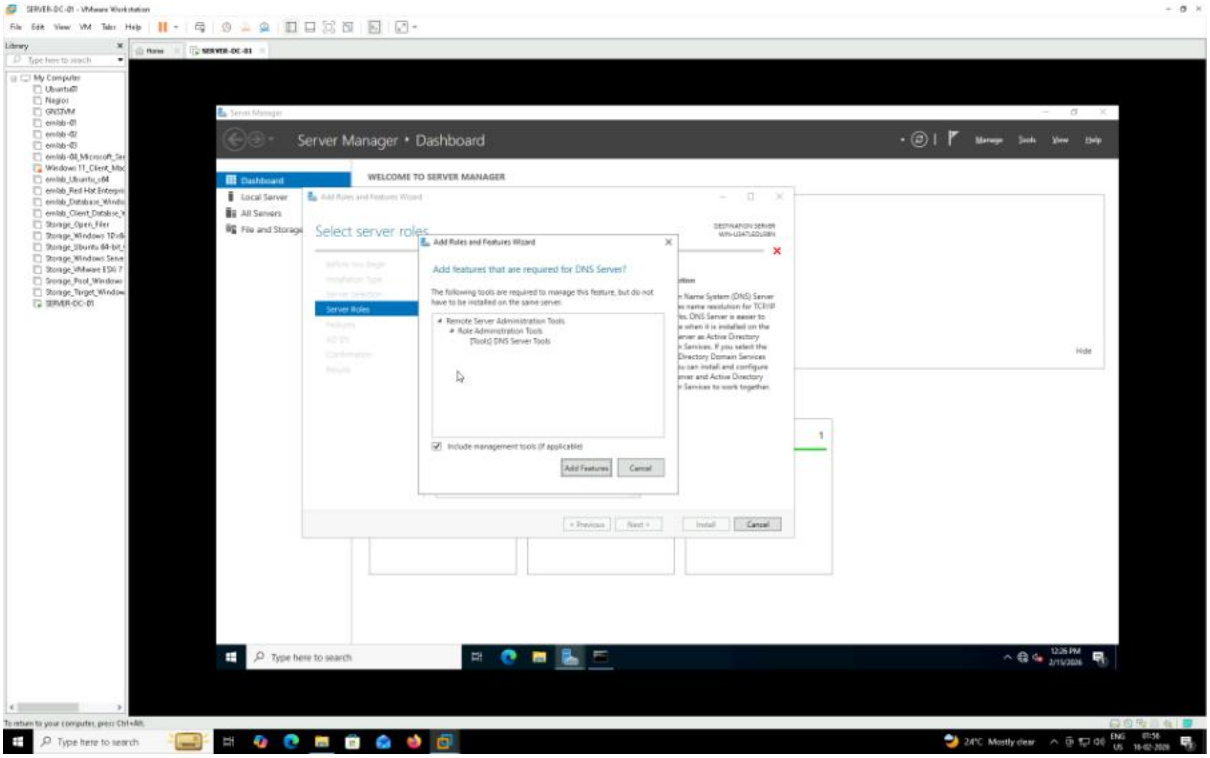


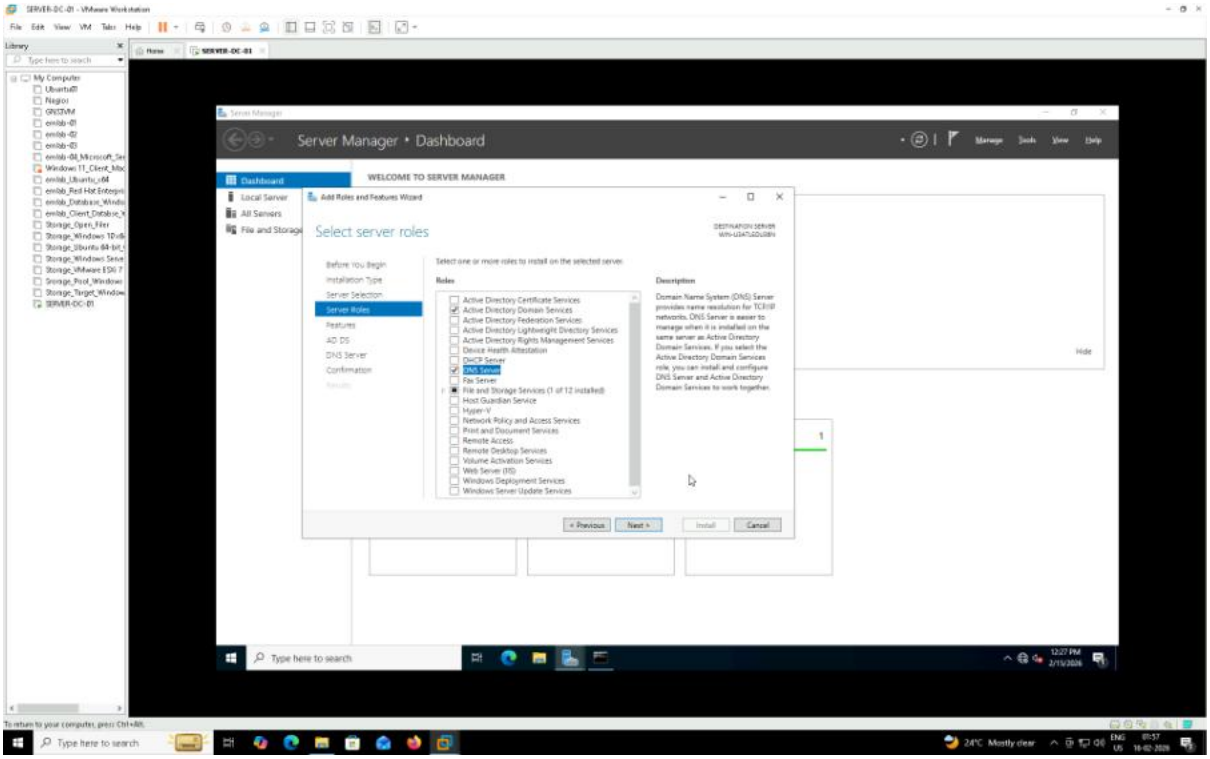
Note that the static IP Address assigned is visible in the Server Manager as well

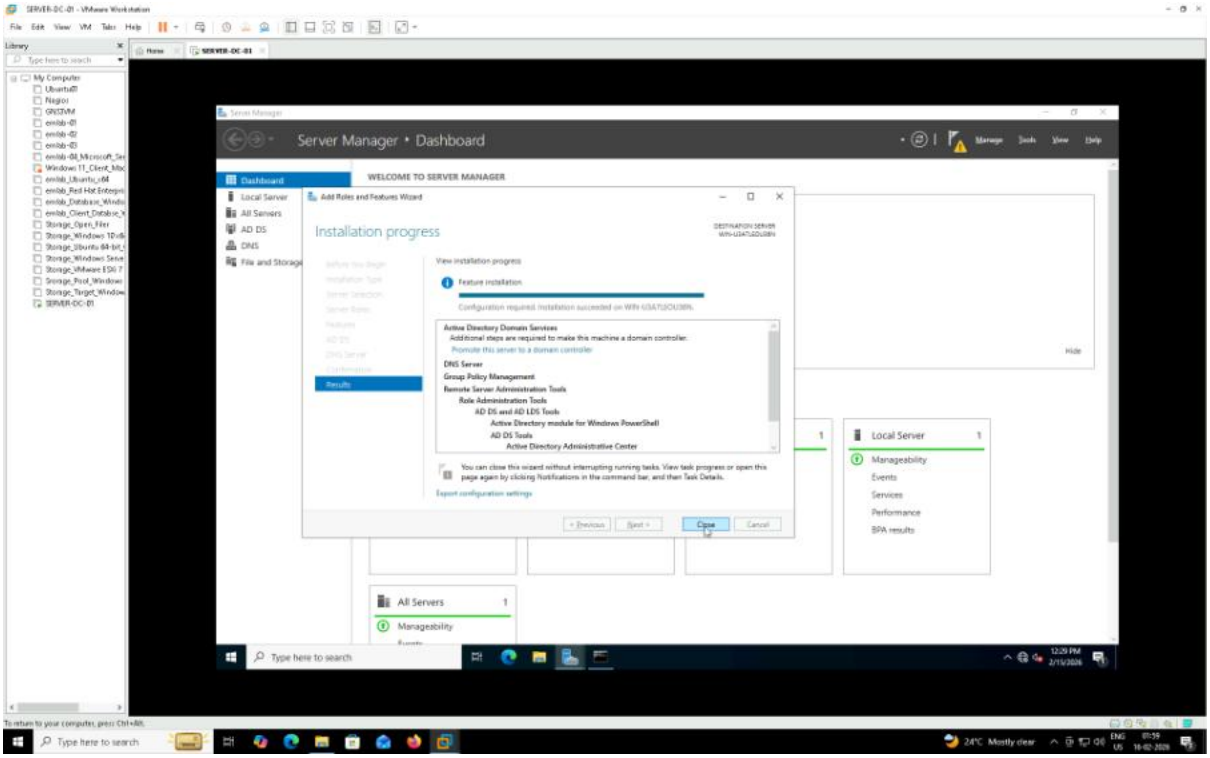


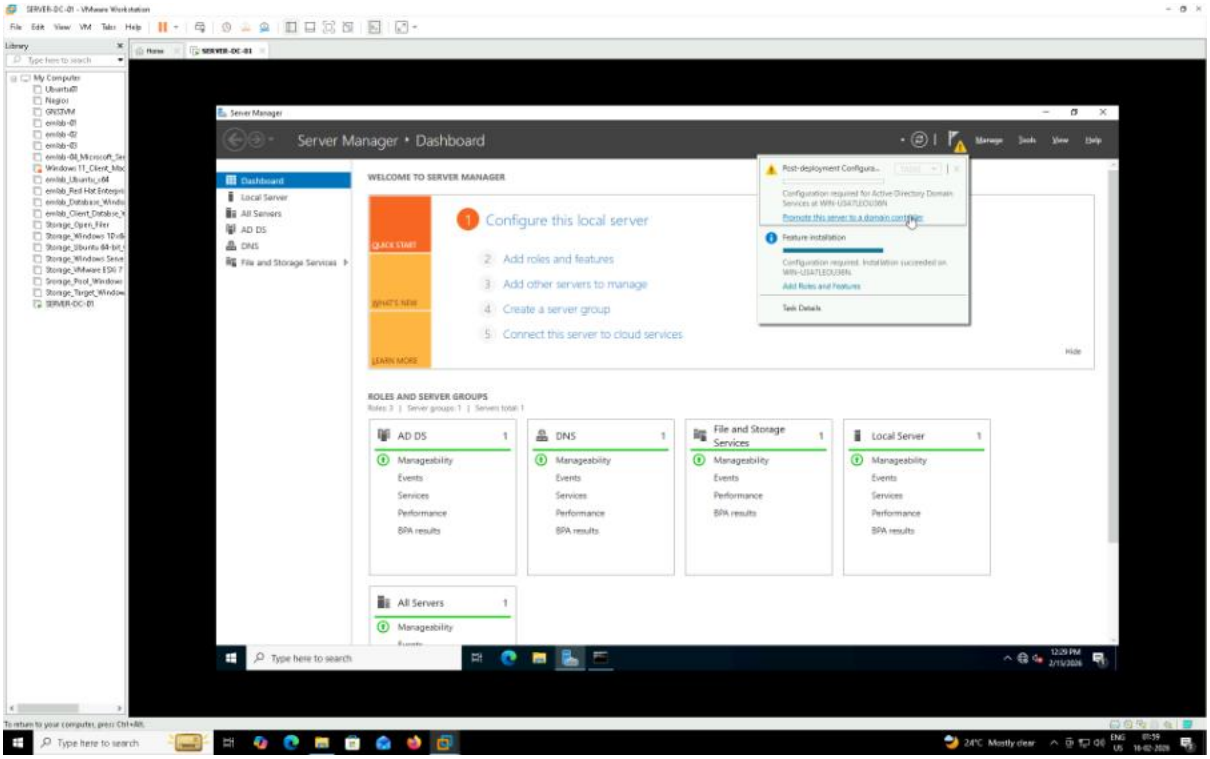
To establish centralized authentication and name resolution, the Active Directory Domain Services and DNS Server roles were installed using the Add Roles and Features Wizard in Server Manager. These roles are essential for domain-based user authentication and hostname resolution within the network.

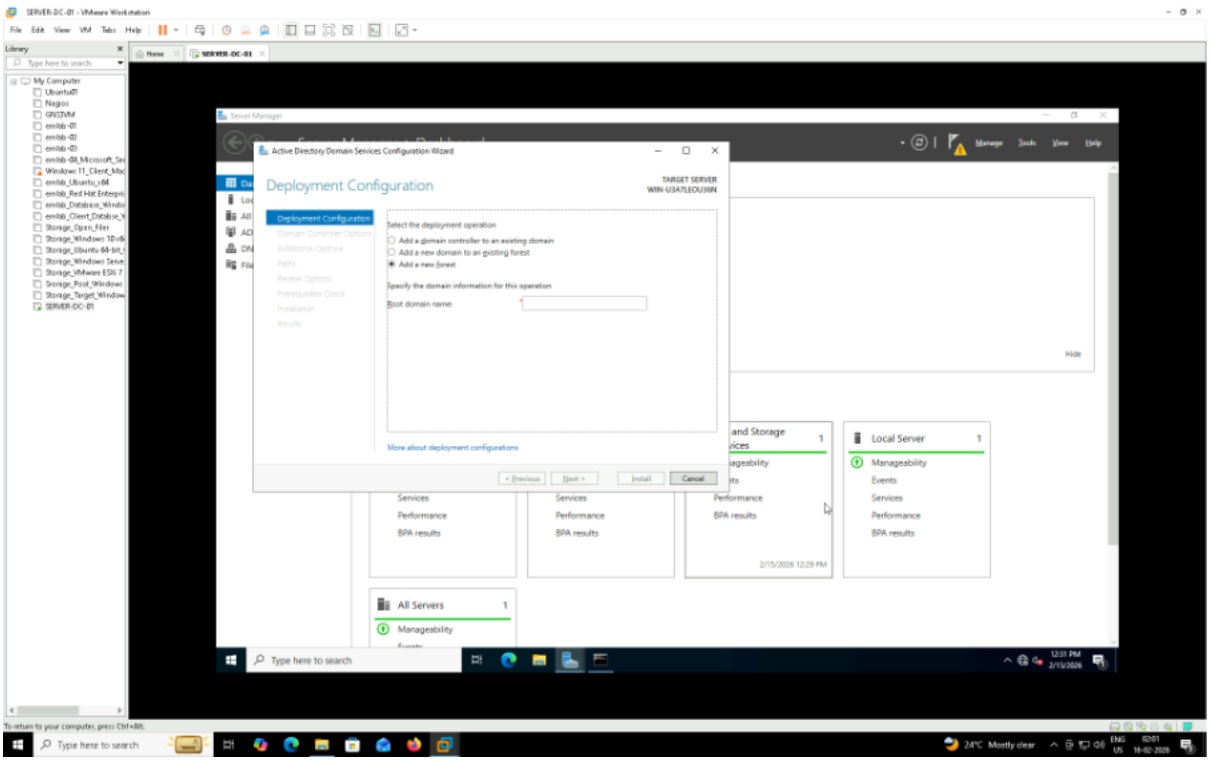



Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Select server roles
Add features that are required for DNS Server?
The following tools are required to manage this feature, but do not have to be installed on the same server.
Remote Server Administration Tools
Role Administration Tools
[Tools] DNS Server Tools
Include management tools (if applicable)
Add Features
Cancel
Server Roles
Previous
Next >
Install
Cancel
Type here to search

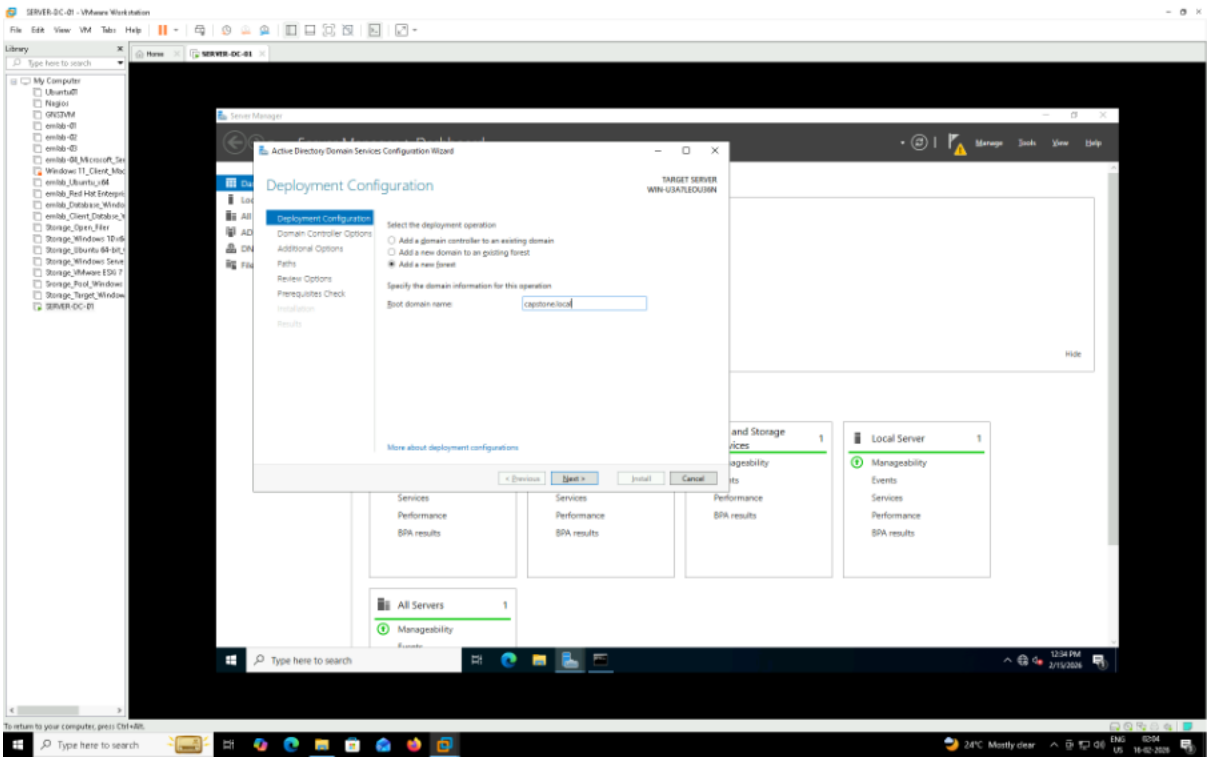

Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Select server roles
Select one or more roles to install on the selected server.
Roles
Active Directory Certificate Services
Active Directory Domain Services
Active Directory Federation Services
Active Directory Lightweight Directory Services
Active Directory Rights Management Services
Device Health Attestation
DHCP Server
DNS Server
File Server
File and Storage Services (1 of 12 installed)
Host Guardian Service
Hyper-V
Network Policy and Access Services
Print and Document Services
Remote Access
Remote Desktop Services
Volume Activation Services
Web Server (IIS)
Windows Deployment Services
Windows Server Update Services
Description
Server Roles
Previous
Next >
Install
Cancel
Type here to search


Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Installation progress
Feature installation
Active Directory Domain Services
DNS Server
Group Policy Management
Remote Server Administration Tools
Role Administration Tools
AD DS and AD LDS Tools
Active Directory module for Windows PowerShell
AD DS Tools
Active Directory Administrative Center
Close
Local Server
Manageability
Events
Services
Performance
BPA results
All Servers
Type here to search

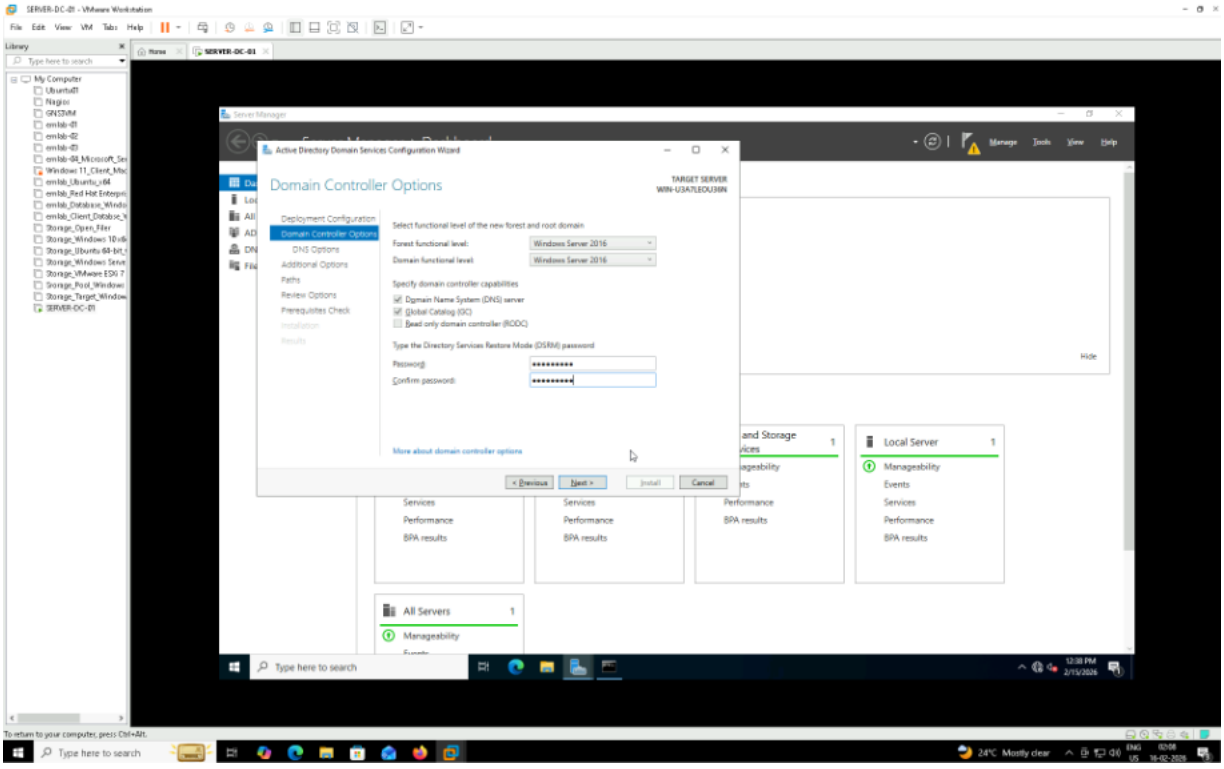

Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Configure this local server
Add roles and features
Add other servers to manage
Create a server group
Connect this server to cloud services
Post-deployment Configura...
Promote this server to a domain controller
Feature installation
Add Roles and Features
Task Details
ROLES AND SERVER GROUPS
AD DS
DNS
File and Storage Services
Local Server
All Servers
Manageability
Events
Services
Performance
BPA results
Type here to search

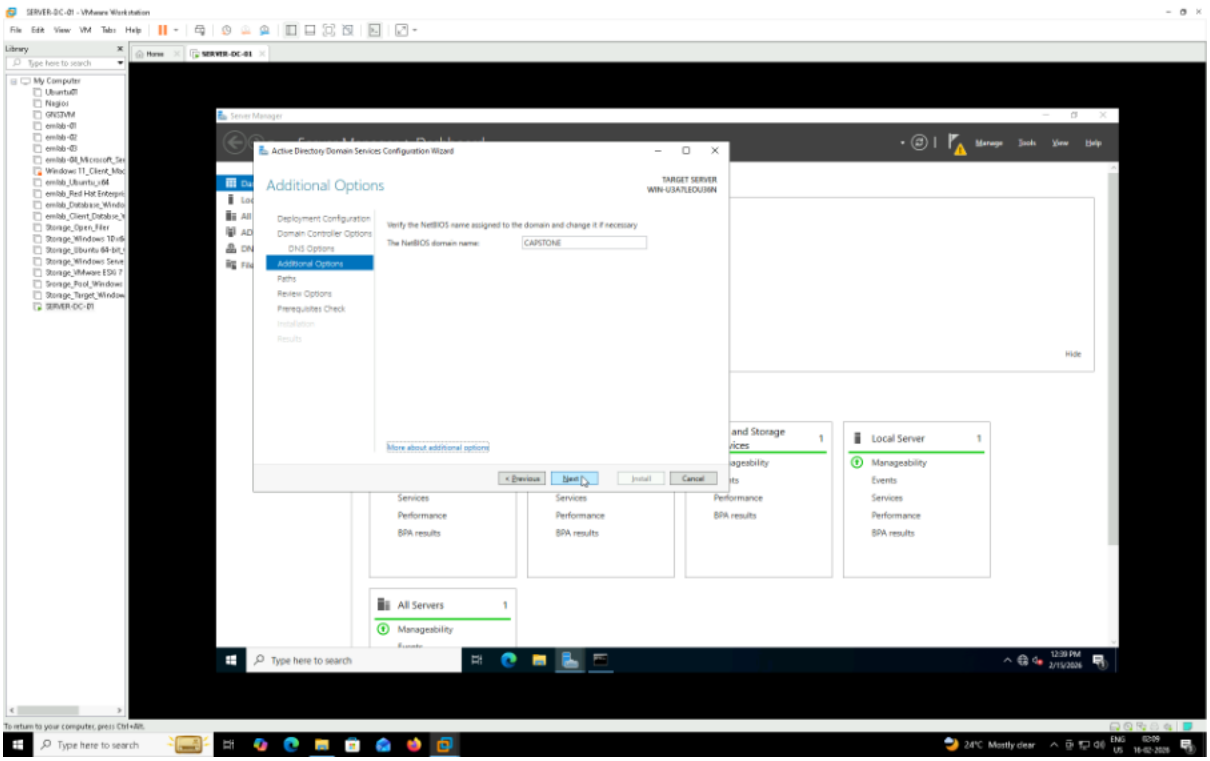


A new Active Directory Forest was created using the domain name **capstone.local** . A private domain suffix was selected to represent an internal organizational network and to prevent conflicts with public DNS namespaces.

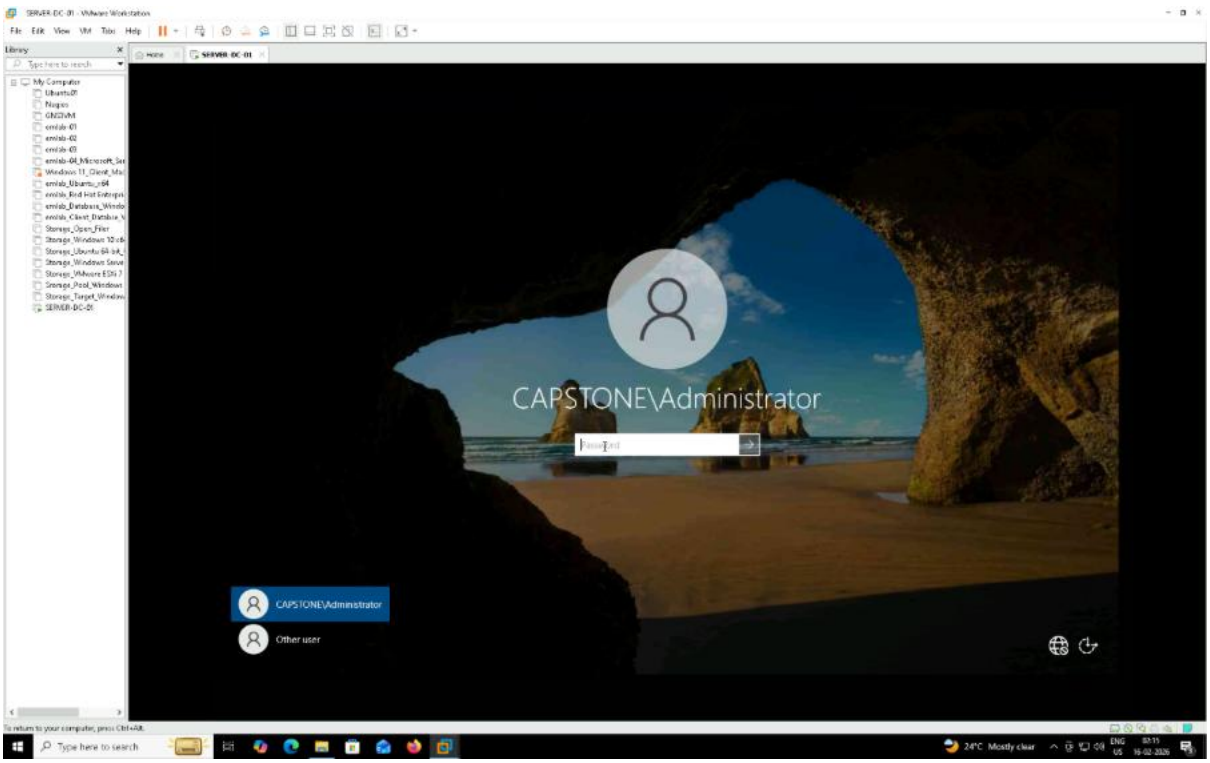


Other options were set to default to maintain compatibility and A Directory Service Restore Mode (DRSM) password was also configured for recovery and maintenance operations.

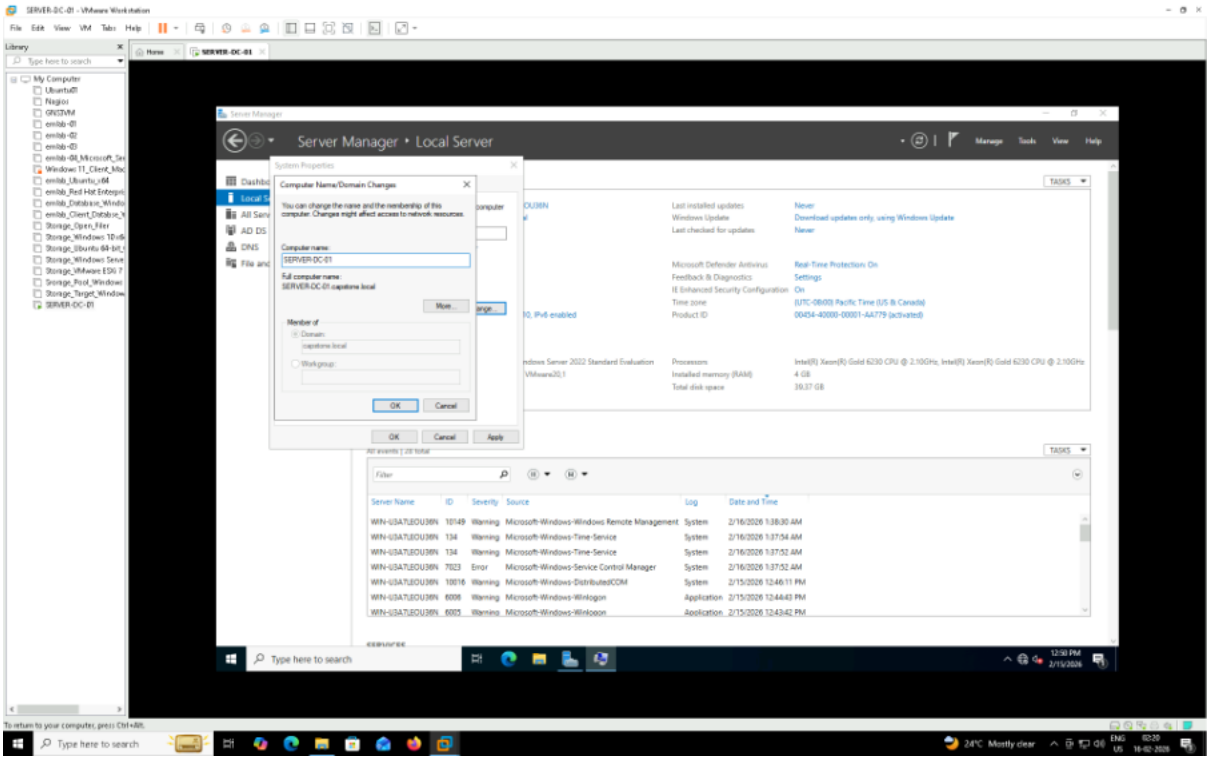




After installation, the server restarted and allowed login using domain credentials. The presence of domain prefix in the login screen confirms that the server was successfully promoted to a Domain Controller for the **capstone.local** domain.

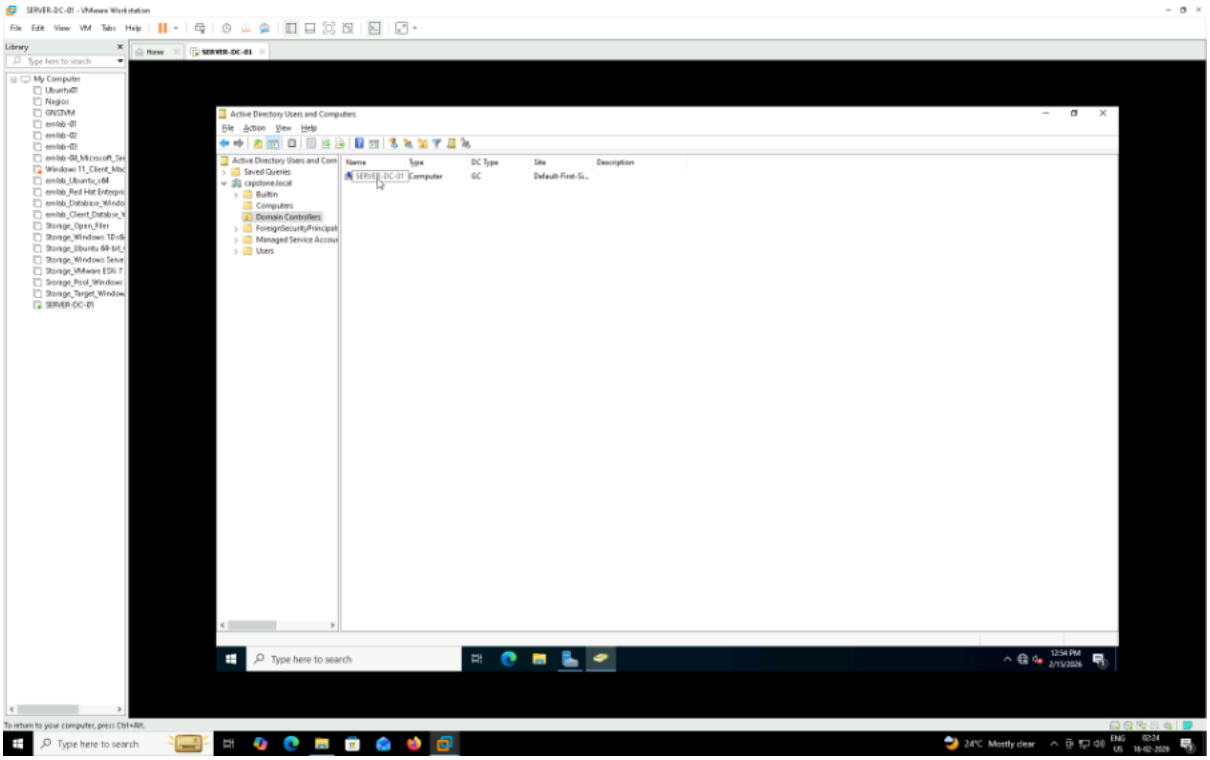


After domain creation, the system generated hostname was replaced with a standard enterprise naming convention SERVER-DC-01 to ensure easier identification and management within the network.

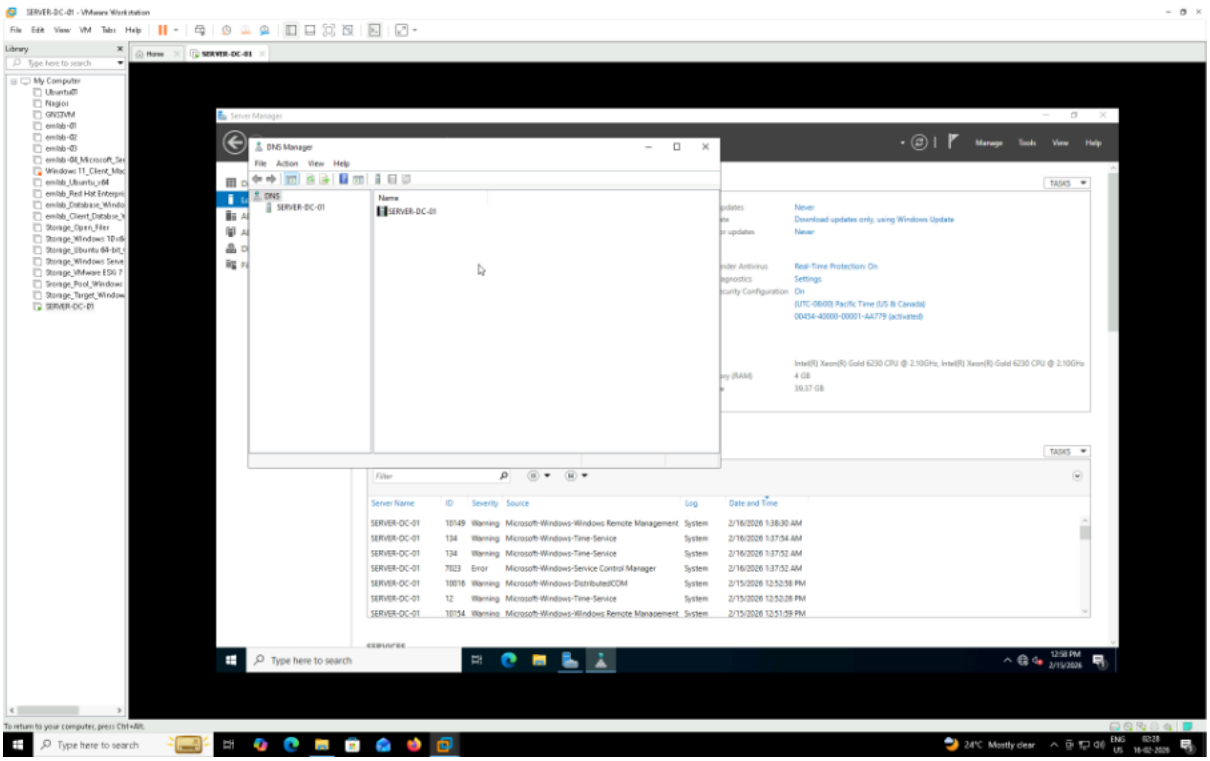


The Active Directory Users and Computers console confirmed successful domain creation and displayed the default organizational structure. The Domain Controller appeared under

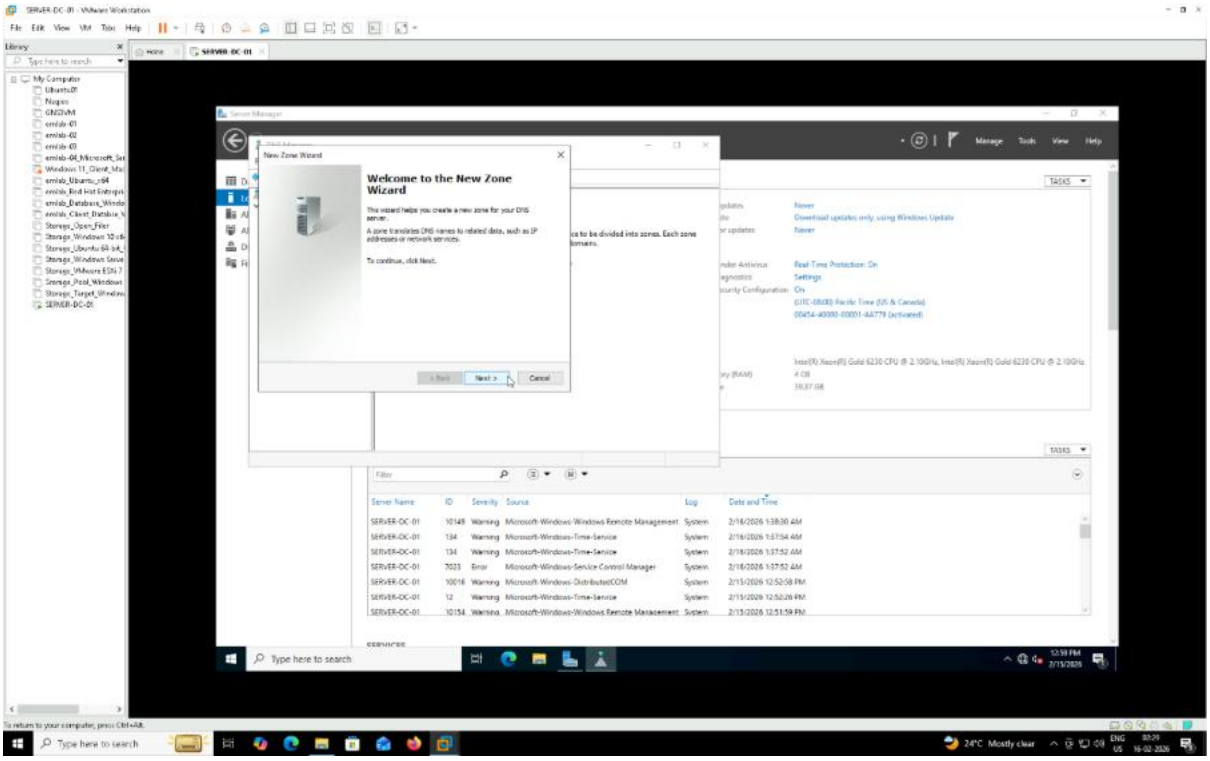
the Domain Controllers container.

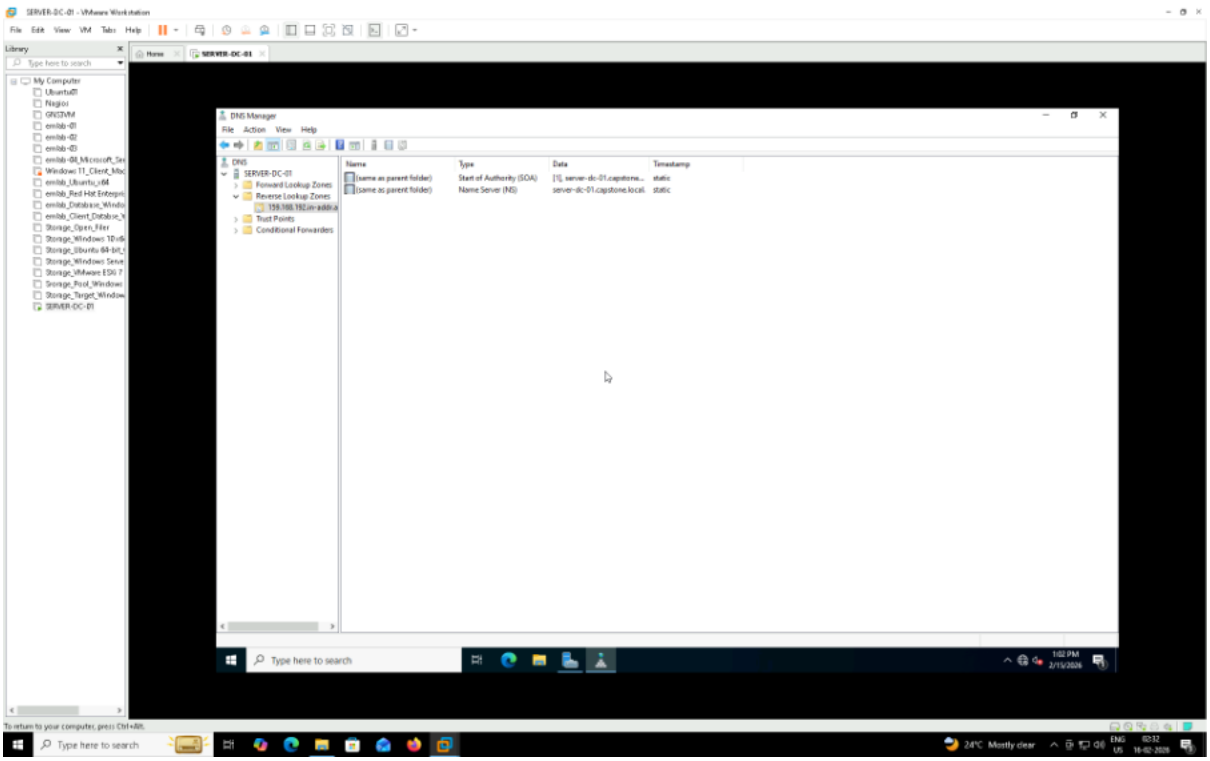


The DNS manager console verified that the forward lookup zone **capstone.local** was automatically created and populated with host records, confirming proper integration between Active Directory and DNS.

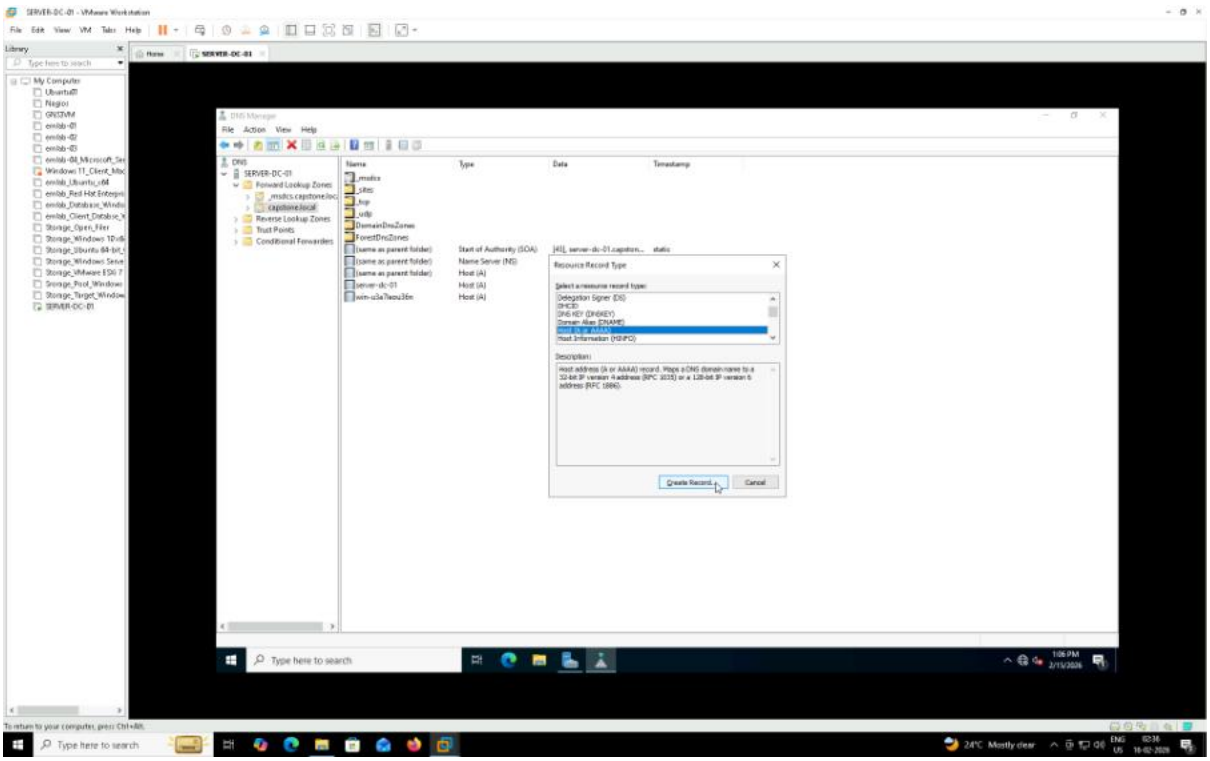


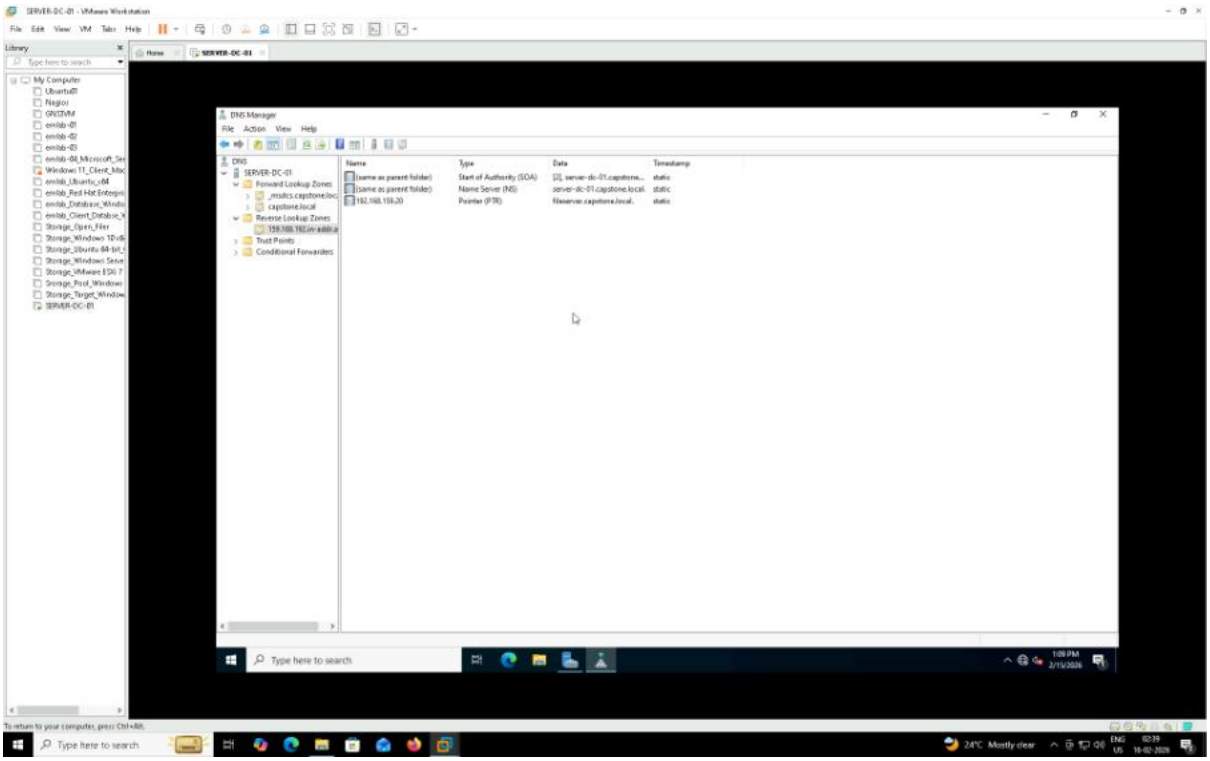
A reverse lookup zone was created to enable IP-to-hostname resolution within the network. The zone was integrated with Active Directory and configured to allow secure dynamic updates to maintain accurate DNS records.



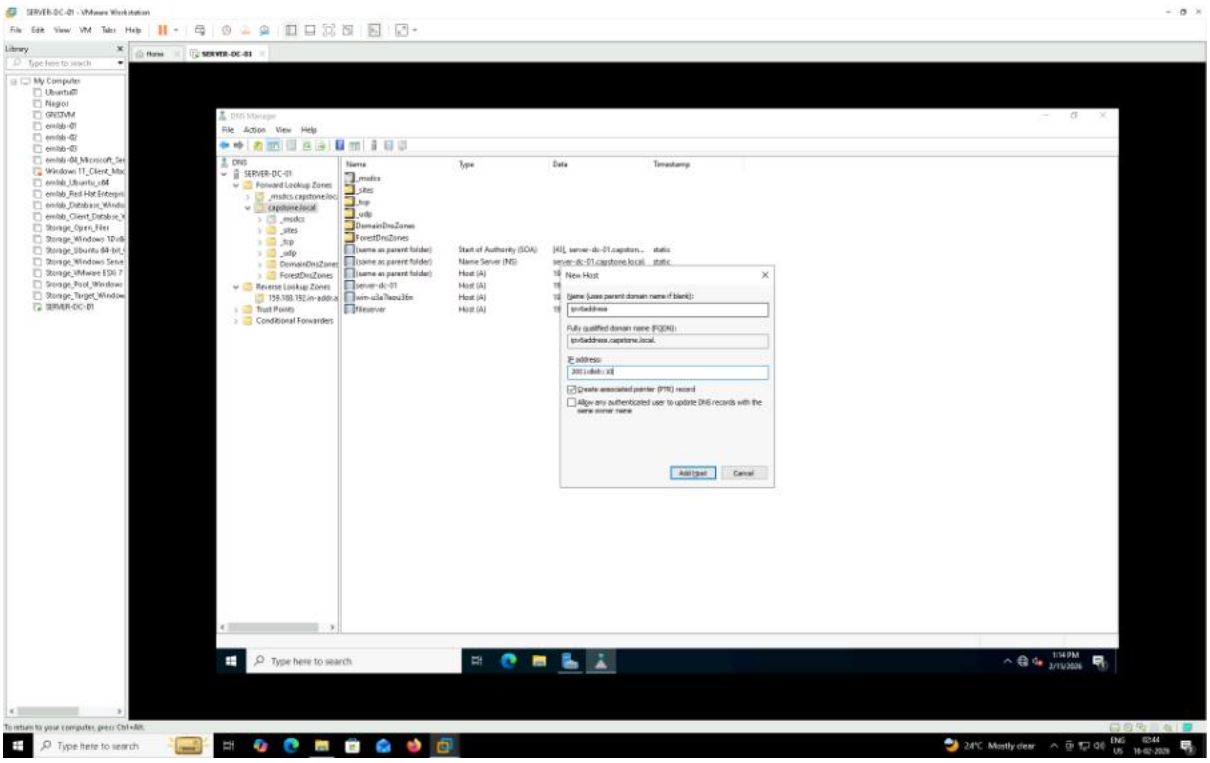


A Record Creation – An A Record was created to map a hostname to an Ipv4 address. The record **fileserver.capstone.local** was configured to resolve to 192.168.159.20. The DNS Server automatically updated the corresponding PTR record in the reverse lookup zone, ensuring bidirectional name resolution between hostname and IP address.



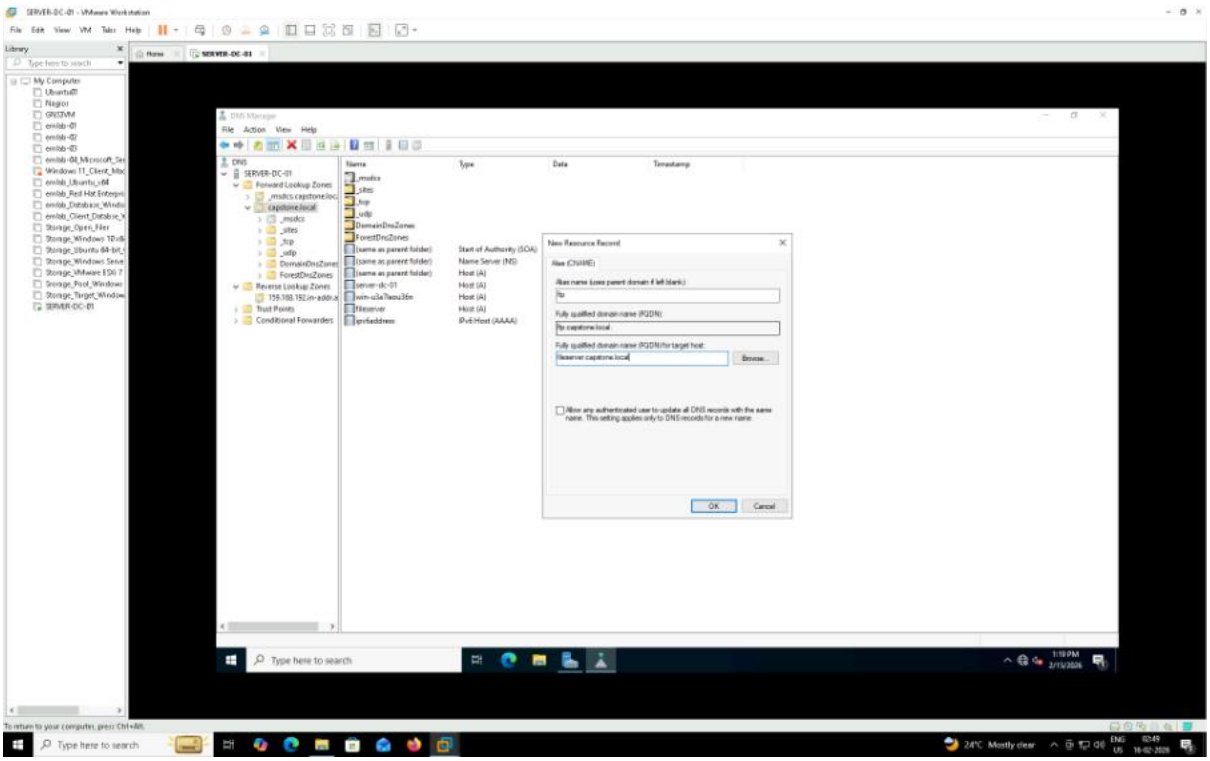


AAAA Record Creation – An AAAA record was configured to provide hostname resolution for IPv6 addressing. The hostname **ipv6server.capstone.local** was mapped to IPv6 address 2001:db8::10.

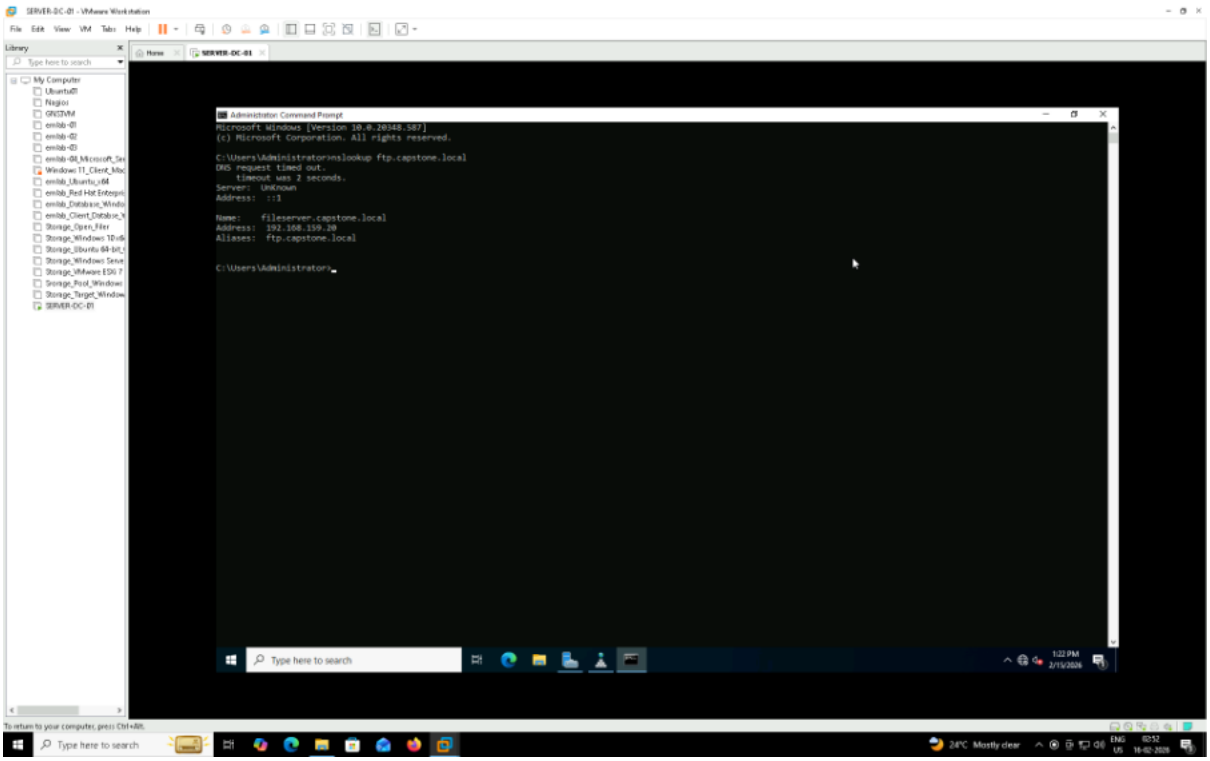


While creating the AAAA record, the DNS server displayed a PTR warning because no Ipv6 reverse lookup zone existed. This does not affect Forward Name Resolution.

A **Canonical name (CNAME)** record was created to map an alias hostname to an existing host record. The alias **ftp.capstone.local** was configured to point to **fileserver.capstone.local**, allowing multiple service names to resolve to the same server.

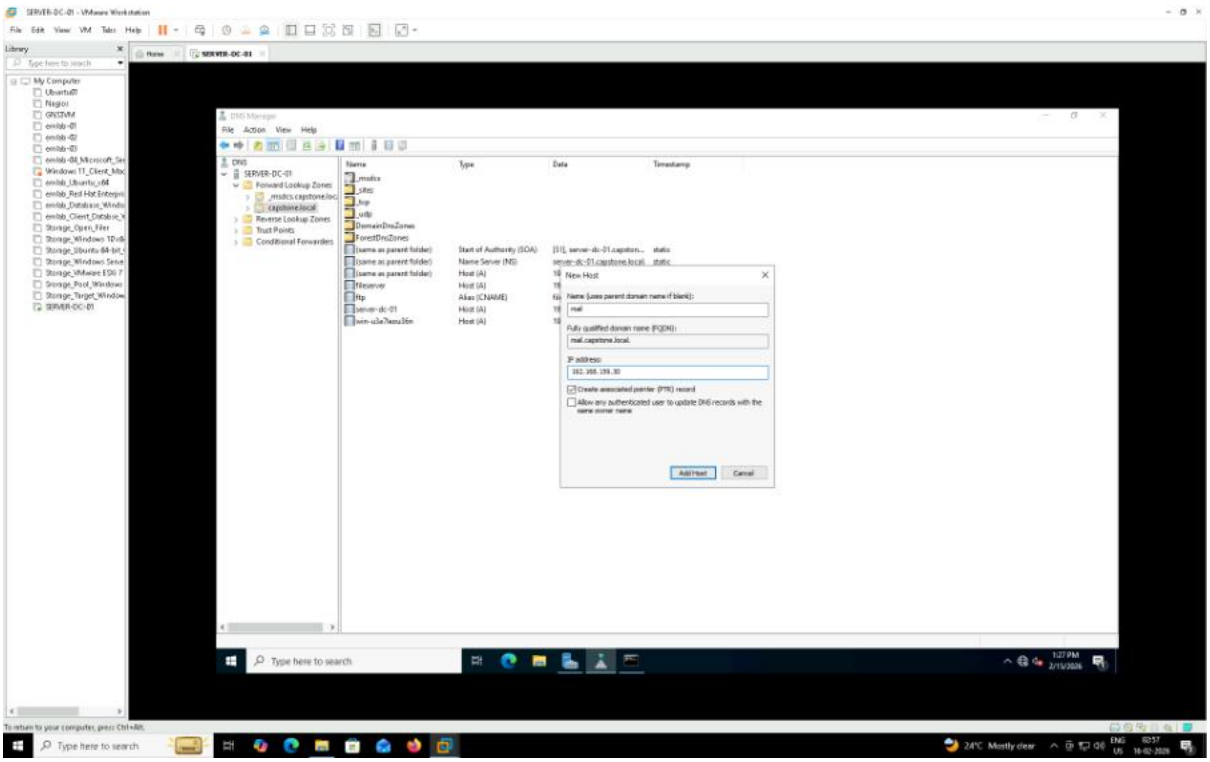


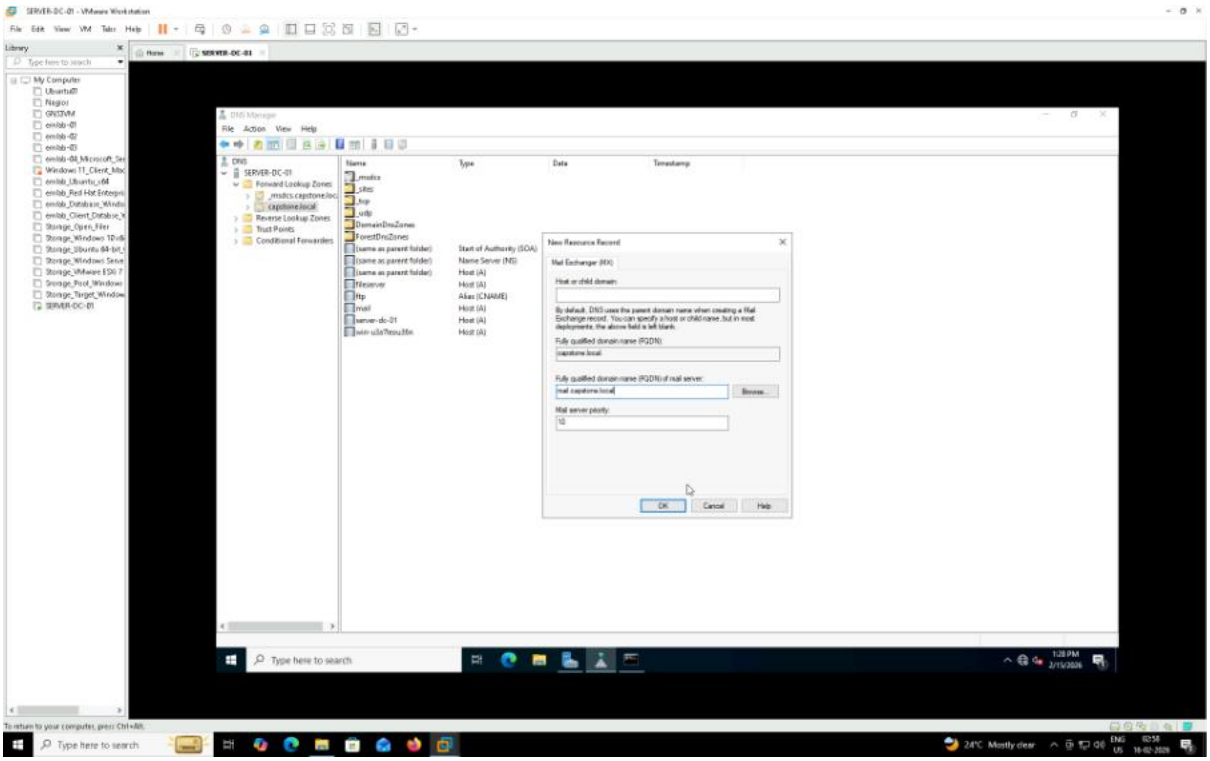
The alias record was verified using the **nslookup** command.

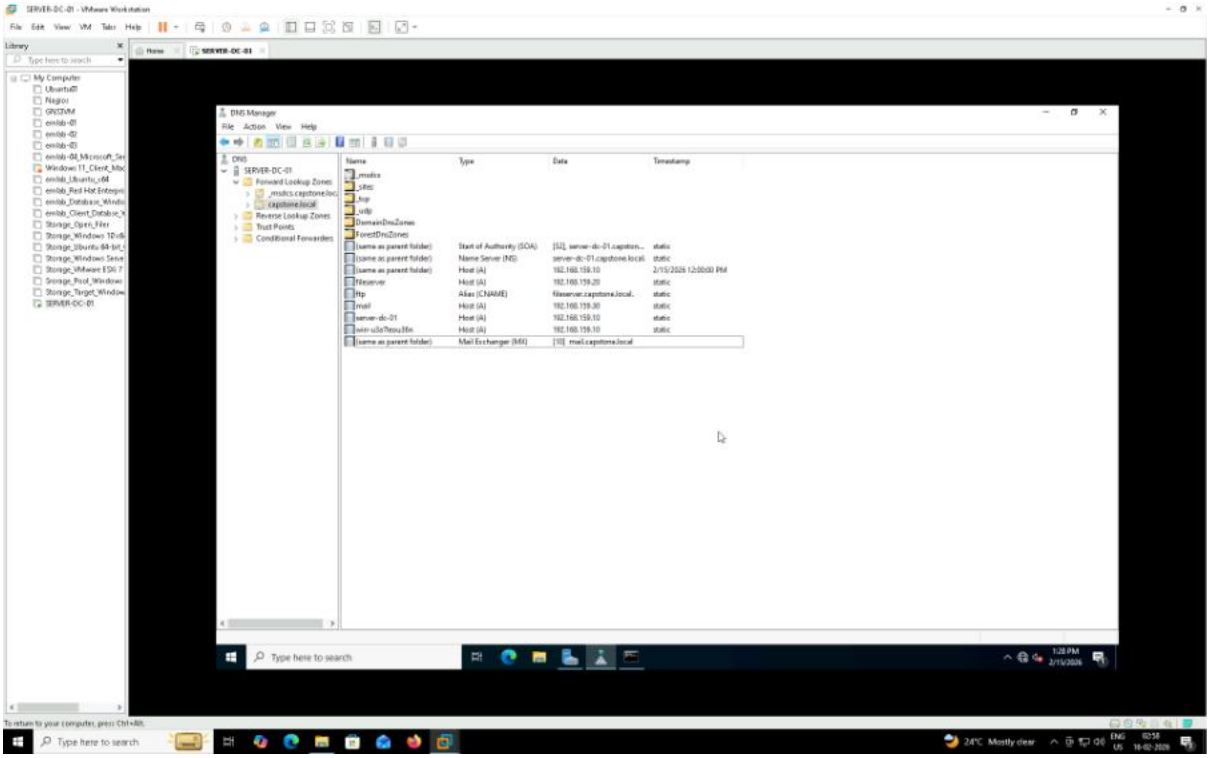


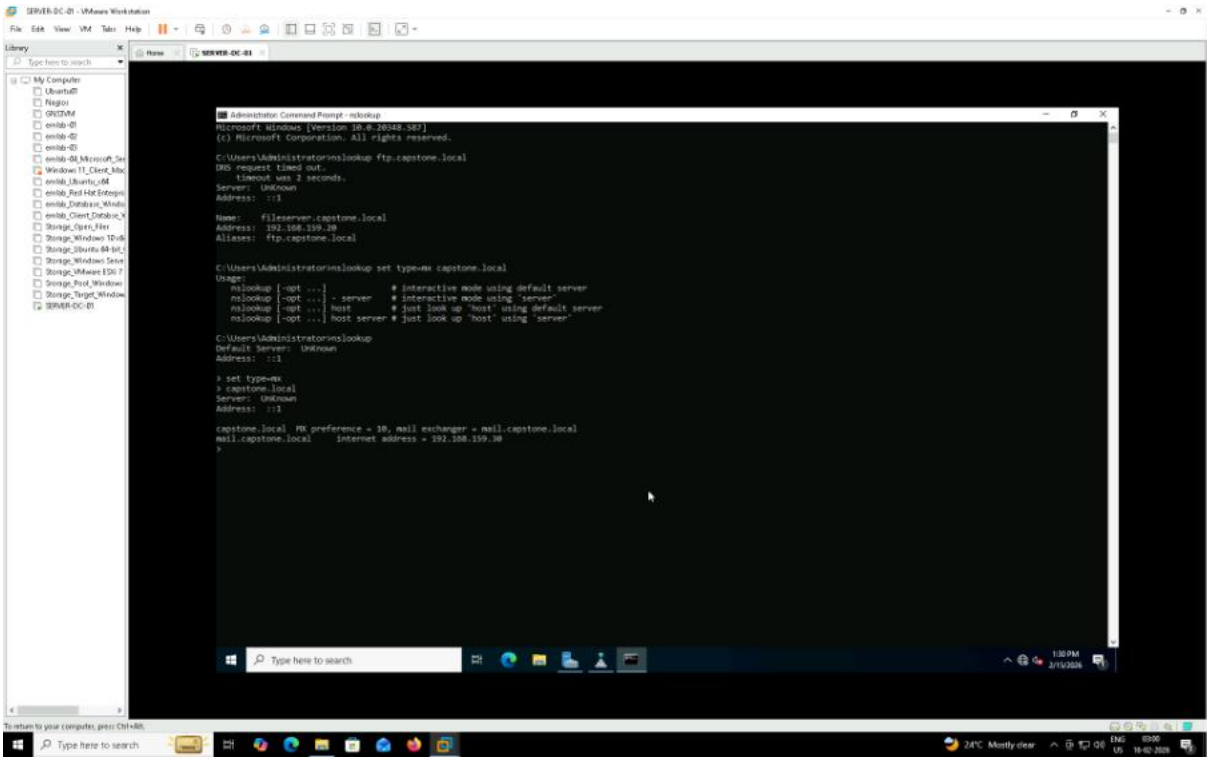
MX Record Creation: - A MC Record was configured to define the mail server responsible for handling email traffic for the domain. The domain **capstone.local** was configured to

use **mail.capstone.local** as its mail exchange server with priority value 10. The configuration was verified using nslookup.






DNS Manager
File Action View Help
SERVER-DC-01
Forward Lookup Zones
capstone.local
Reverse Lookup Zones
Trust Points
Conditional Forwarders
Name Type Data Timestamp
_msdcs
sites
_tcp
_udp
DomainDnsZones
ForestDnsZones
(same as parent folder) Start of Authority (SOA) [52], server-dc-01.capston... static
(same as parent folder) Name Server (NS) server-dc-01.capstone.local. static
(same as parent folder) Host (A) 192.168.159.10 2/15/2026 12:00:00 PM
fileserver Host (A) 192.168.159.20 static
ftp Alias (CNAME) fileserver.capstone.local. static
mail Host (A) 192.168.159.30 static
server-dc-01 Host (A) 192.168.159.10 static
(same as parent folder) Mail Exchanger (MX) [10] mail.capstone.local
Type here to search
1:28 PM
2/15/2026
Type here to search
24°C Mostly clear
ENG
US
02:58
16-02-2026


Administrator: Command Prompt - nslookup
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>nslookup ftp.capstone.local
DNS request timed out.
timeout was 2 seconds.
Server: UnKnown
Address: ::1
Name: fileserver.capstone.local
Address: 192.168.159.20
Aliases: ftp.capstone.local
C:\Users\Administrator>nslookup set type=mx capstone.local
Usage:
nslookup [-opt ...] # interactive mode using default server
nslookup [-opt ...] - server # interactive mode using 'server'
nslookup [-opt ...] host # just look up 'host' using default server
nslookup [-opt ...] host server # just look up 'host' using 'server'
C:\Users\Administrator>nslookup
Default Server: UnKnown
Address: ::1
> set type=mx
> capstone.local
Server: UnKnown
Address: ::1
capstone.local MX preference = 10, mail exchanger = mail.capstone.local
mail.capstone.local internet address = 192.168.159.30
>
Type here to search
1:30 PM
2/15/2026
Type here to search
24°C Mostly clear
ENG
US
03:00
16-02-2026

Assignment 2: Name Resolution Process

As part of this assignment, you need to complete below tasks:

1. Configure DNS client settings on Windows workstations

2. Demonstrate name resolution using nslookup and ipconfig /all

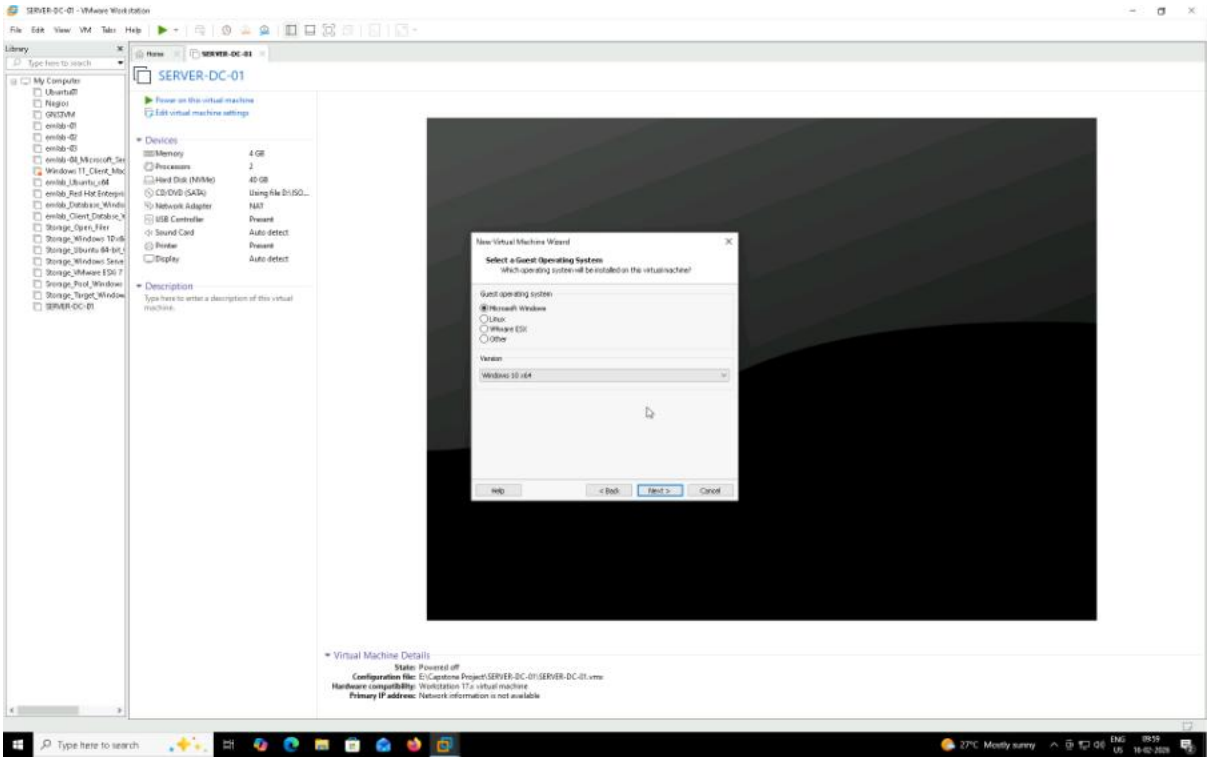
3. Compare DNS resolution methods across different Windows

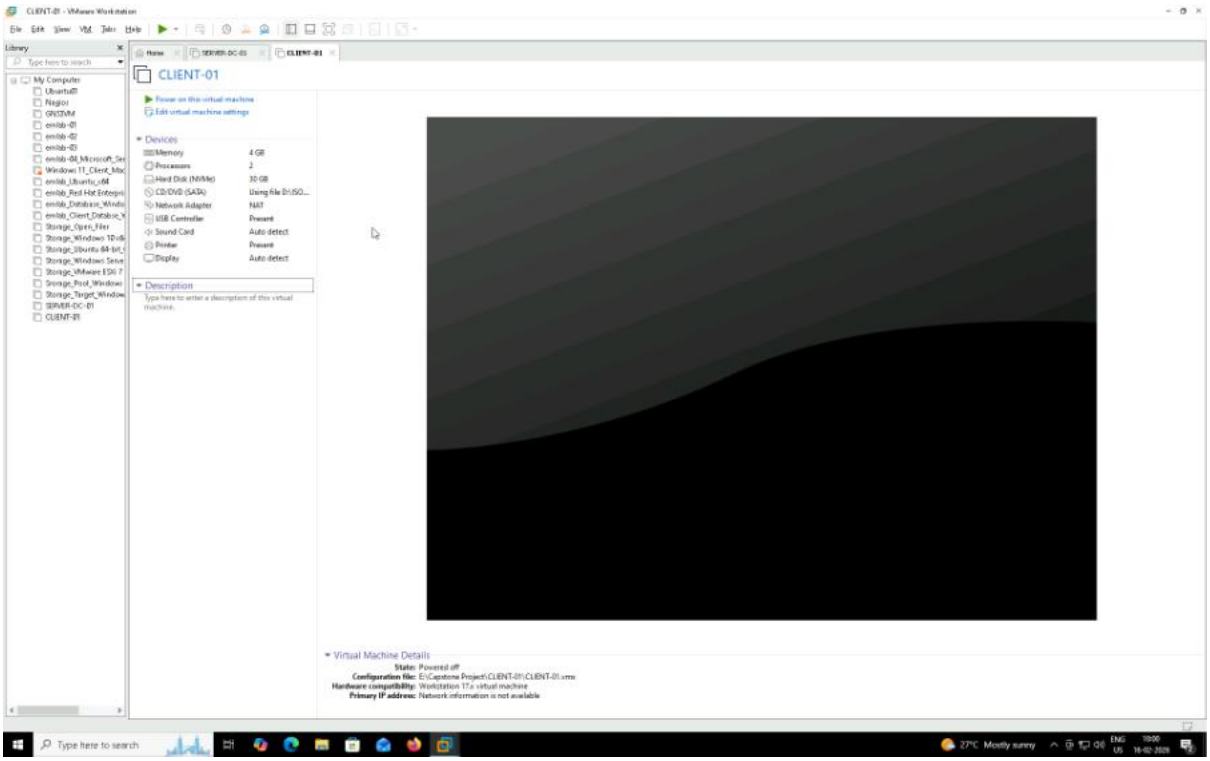
To accomplish this we will need: -

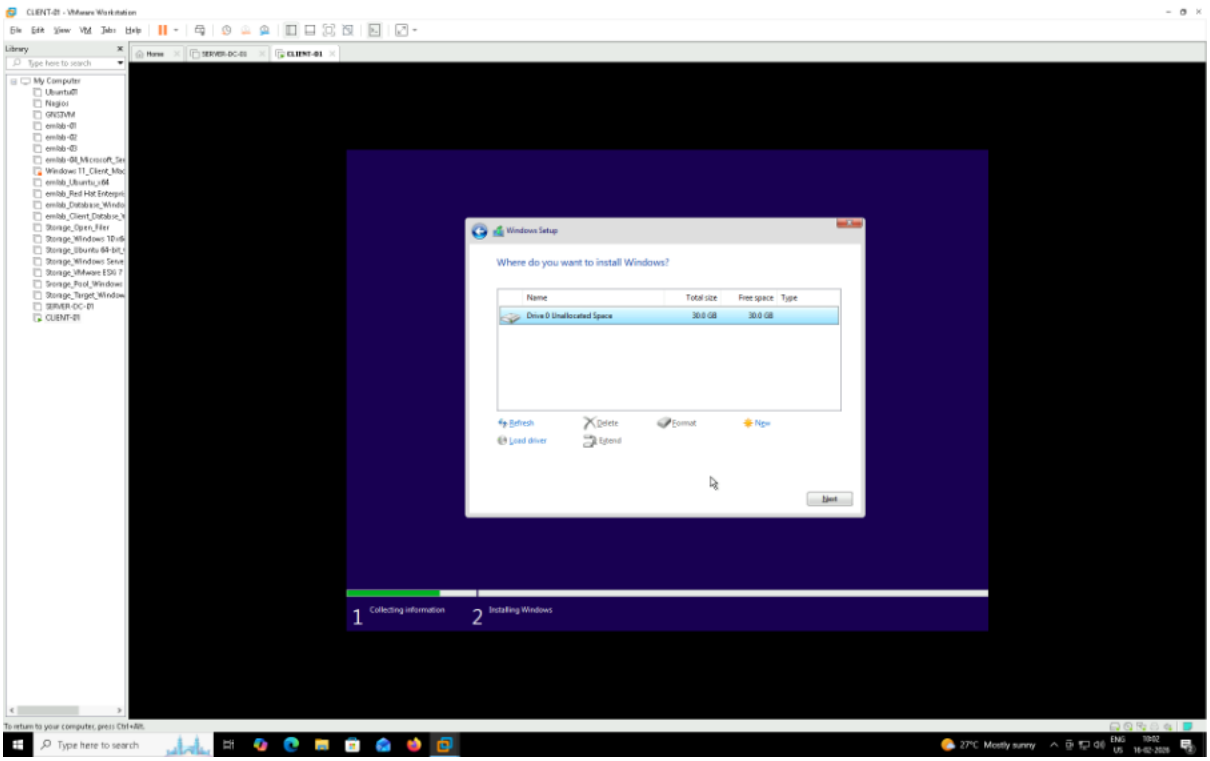
- Server Machine (which we created earlier)
- Client Machine: -
    - Name: - CLIENT-01
    - Type: - Microsoft Windows
    - Version: - Windows 10 Pro (64-bit)
    - RAM: - 4 GB
    - CPU: - 2 Cores
    - Disk: - 30 GB

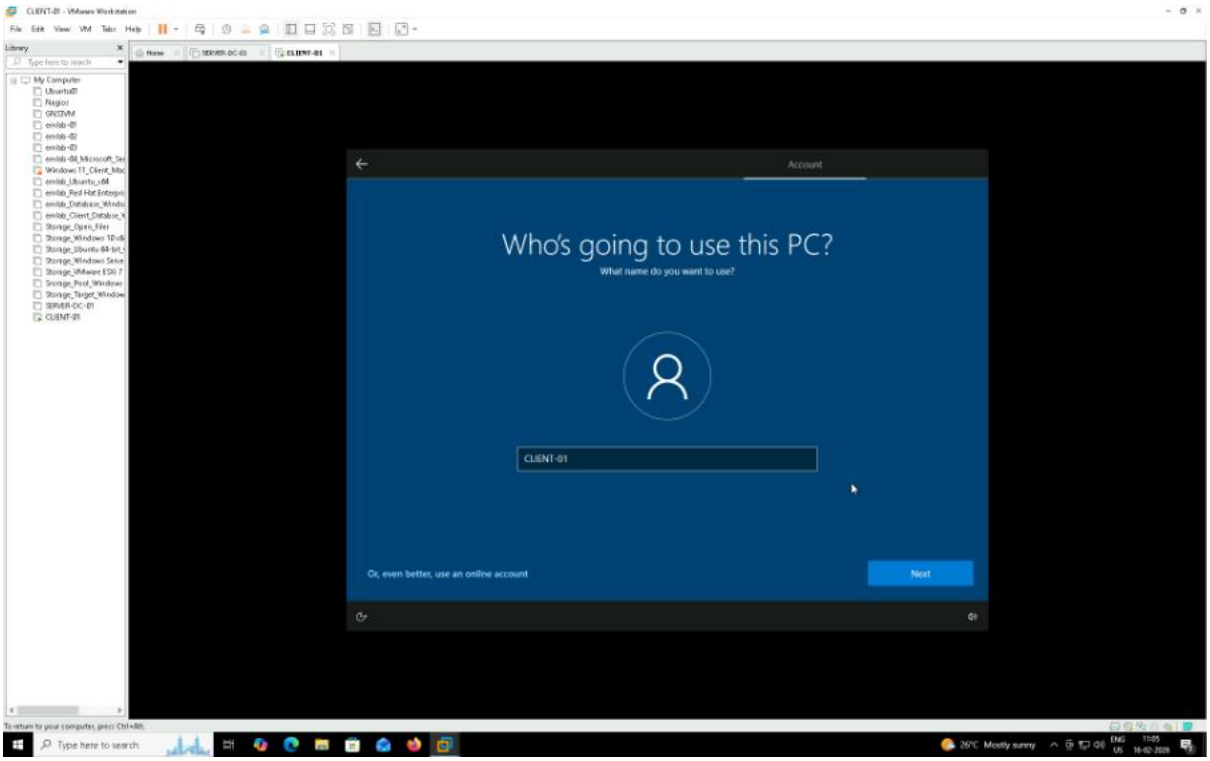
Virtual Machine Properties: -

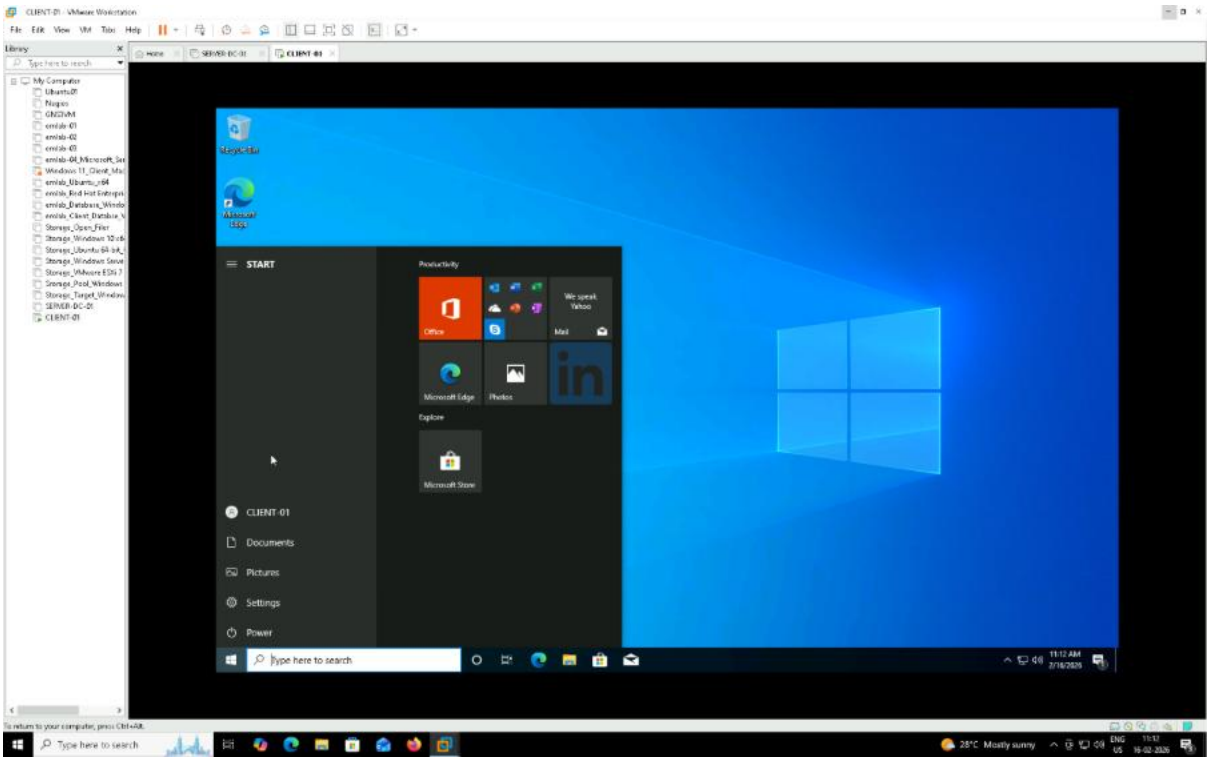
The virtual machine CLIENT- 01 has been created using VMware Workstation to host ADDS services and DNS. The virtual machine was configured using NAT networking mode to allow internet connectivity for updates and package installation. Since NAT provides automatic IP addressing via VMware DHCP, Static IP configuration would be manually assigned later, and DNS would be configured manually to point to Domain Control.


SERVER-DC-01 - VMware Workstation
SERVER-DC-01
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (NVMe) 40 GB
CD/DVD (SATA) Using file D:\ISO...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Windows 10 x64
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: E:\Capstone Project\SERVER-DC-01\SERVER-DC-01.vmx
Hardware compatibility: Workstation 17.x virtual machine
Primary IP address: Network information is not available
Type here to search
27°C Mostly sunny
ENG
US
09:59
16-02-2026

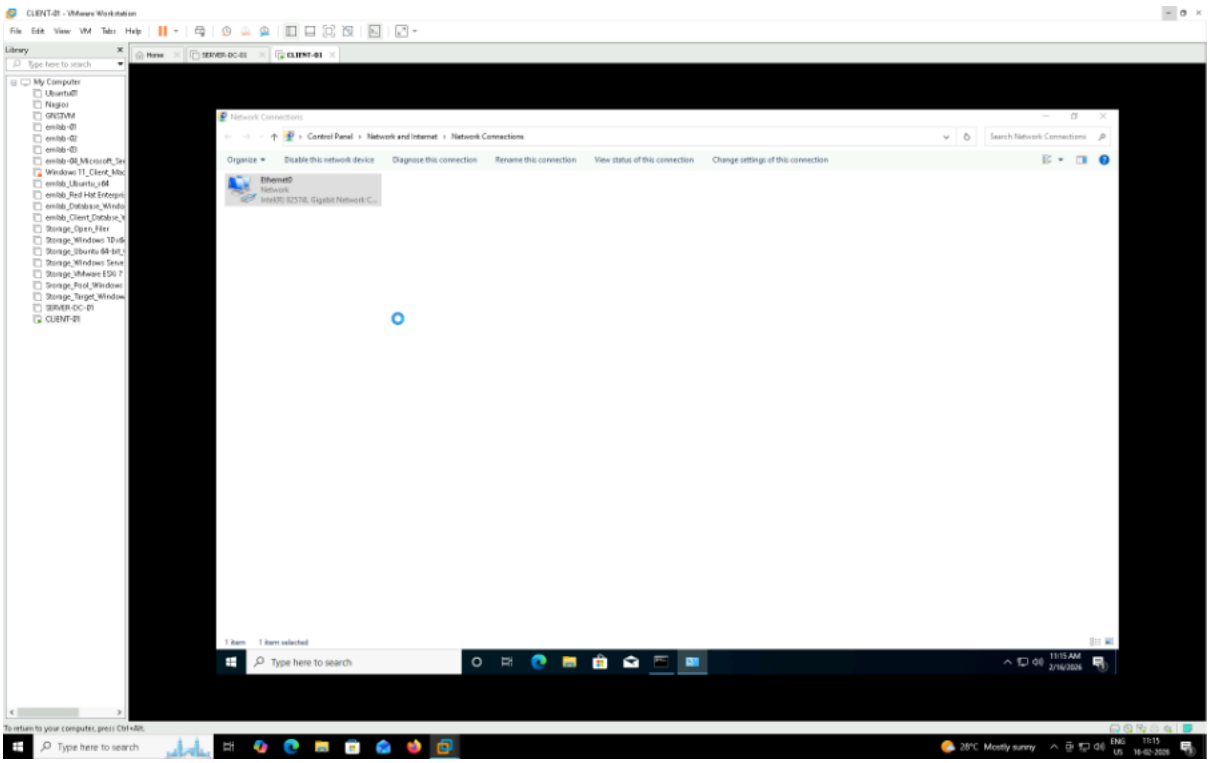

CLIENT-01 - VMware Workstation
CLIENT-01
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (NVMe) 30 GB
CD/DVD (SATA) Using file D:\ISO...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: E:\Capstone Project\CLIENT-01\CLIENT-01.vmx
Hardware compatibility: Workstation 17.x virtual machine
Primary IP address: Network information is not available
Type here to search
27°C Mostly sunny
ENG
US
10:00
16-02-2026

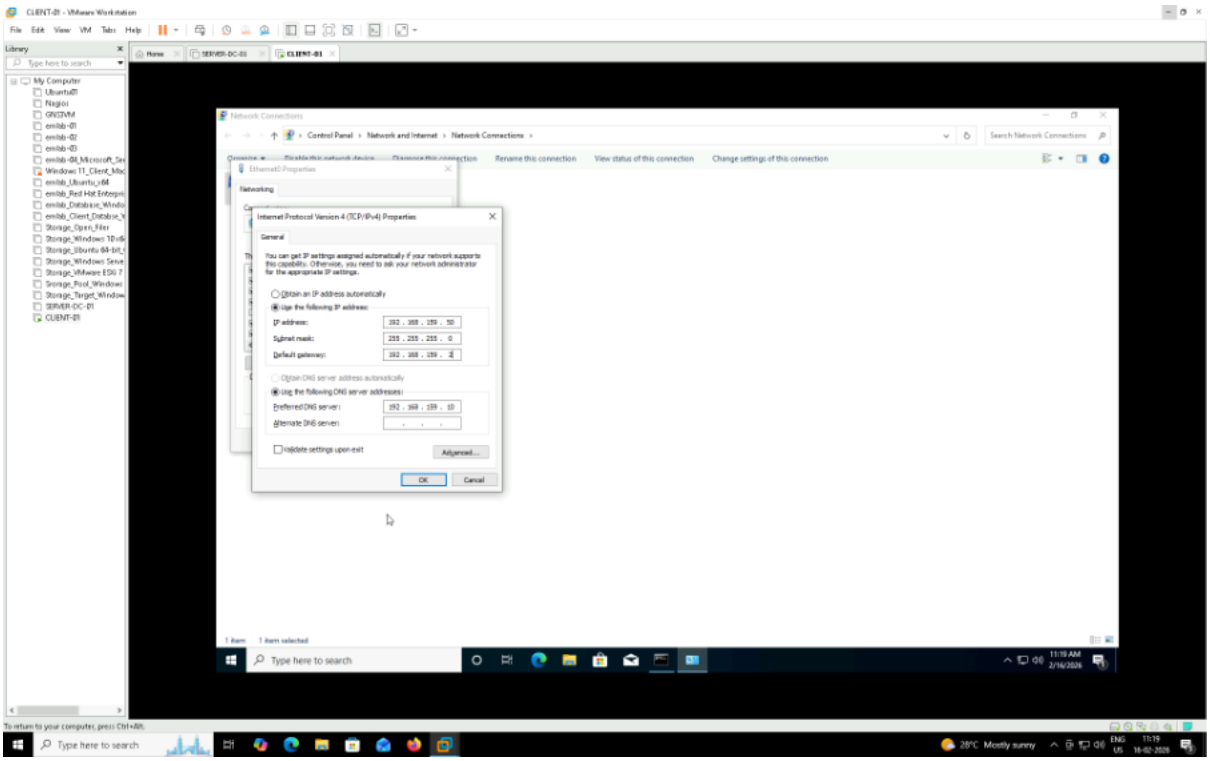

Windows Setup
Where do you want to install Windows?
Name
Total size
Free space
Type
Drive 0 Unallocated Space
30.0 GB
30.0 GB
Refresh
Delete
Format
New
Load driver
Extend
Next
1 Collecting information
2 Installing Windows

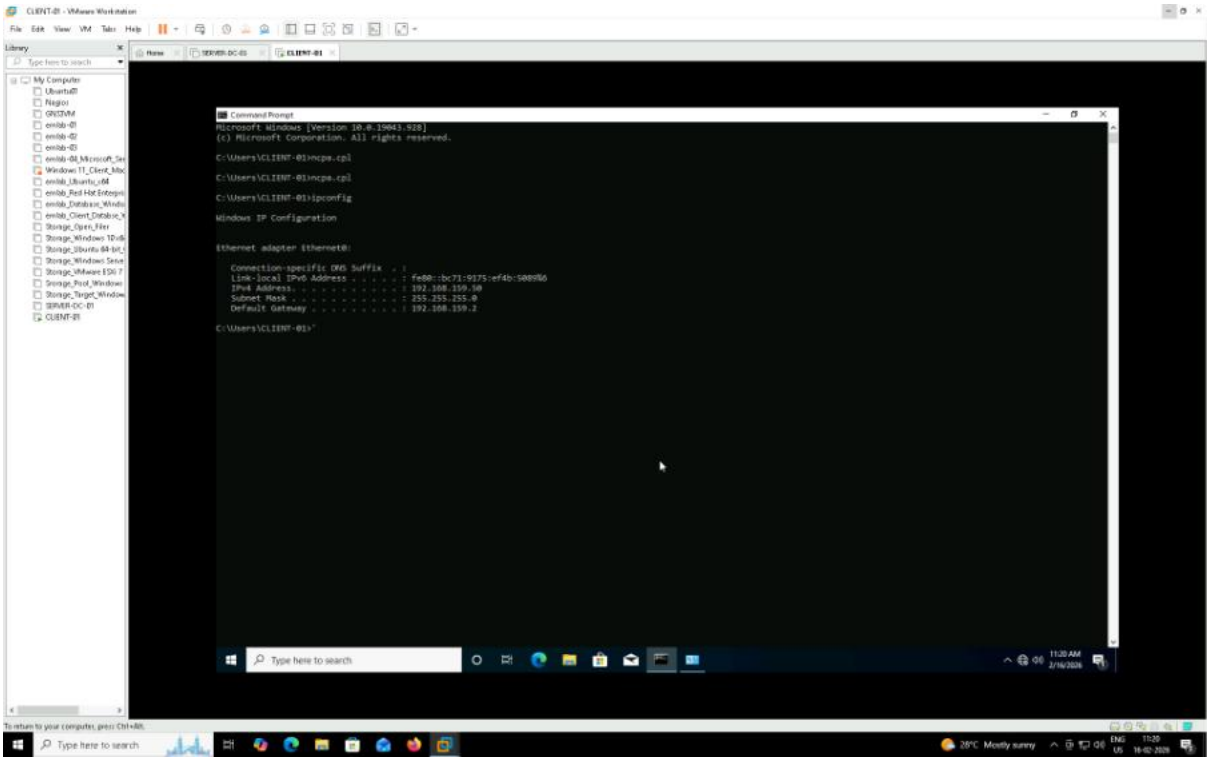

Account
Who's going to use this PC?
What name do you want to use?
CLIENT-01
Or, even better, use an online account
Next

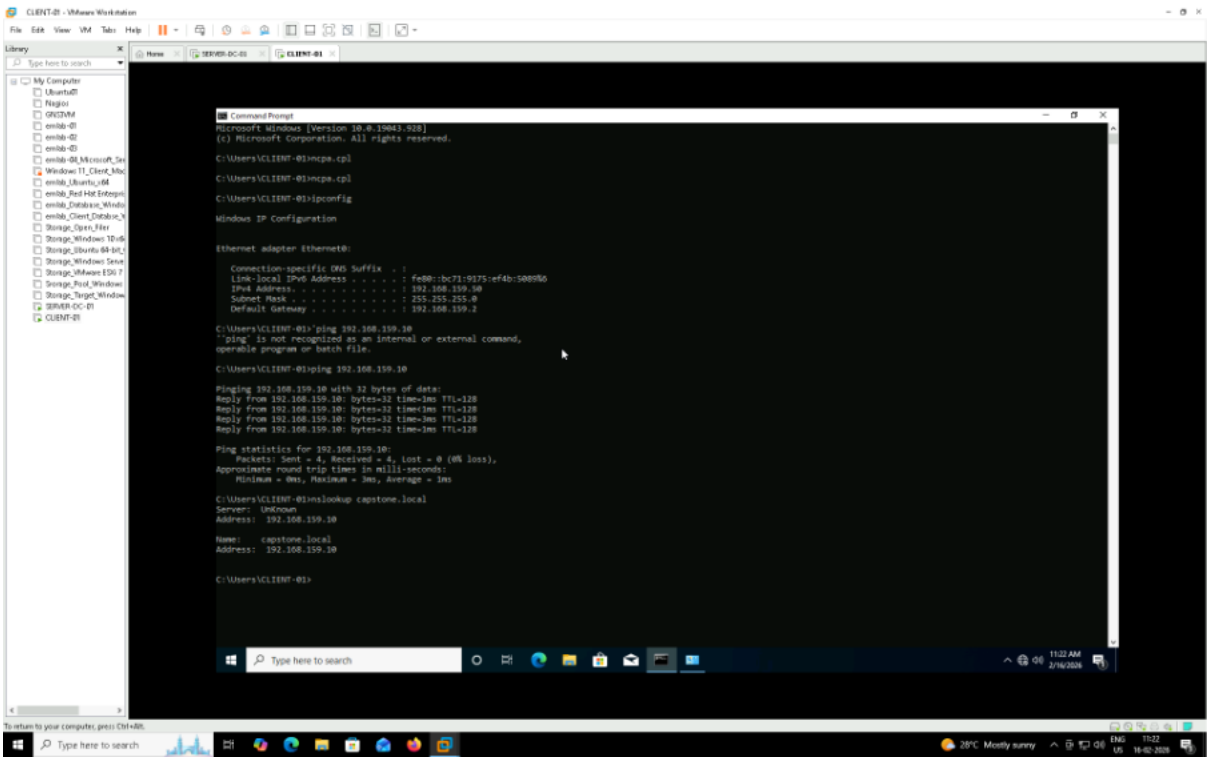


The client machine is configured to use the Domain Controller as its primary DNS Server to enable domain name resolution. The DNS cache was cleared and connectivity to the domain was verified using ping and nslookup commands.

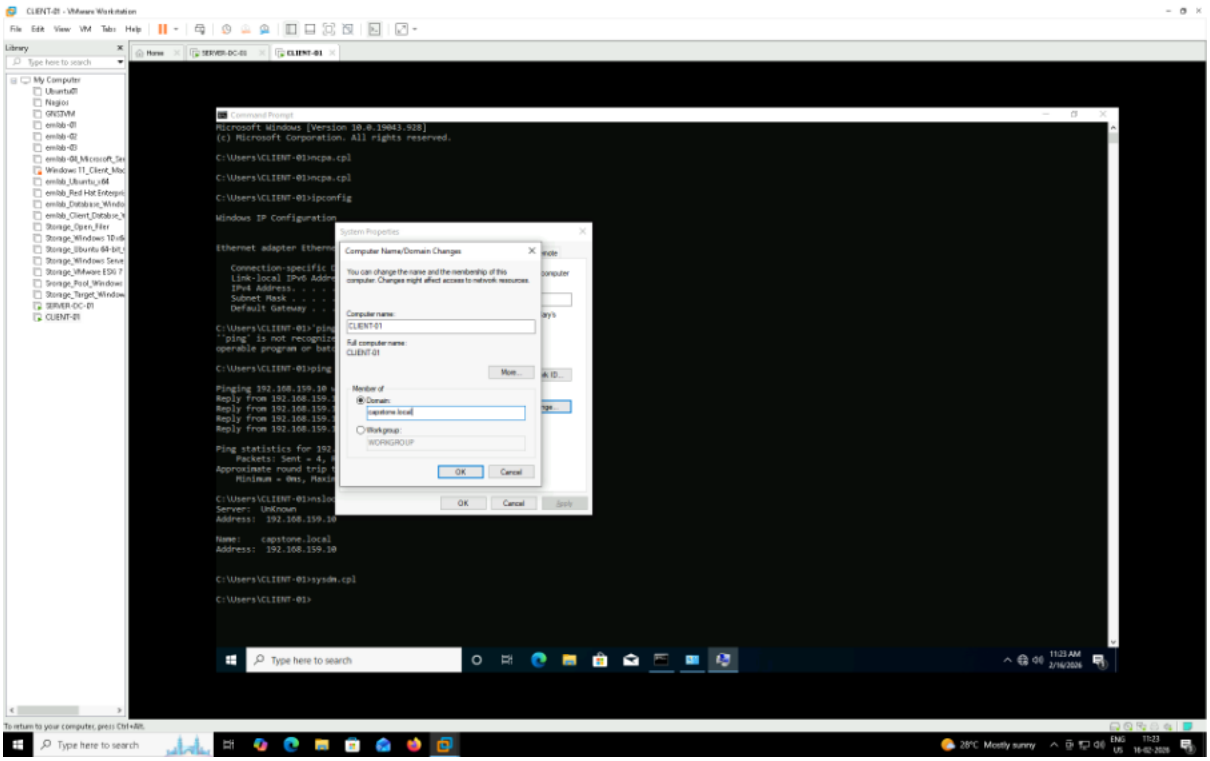


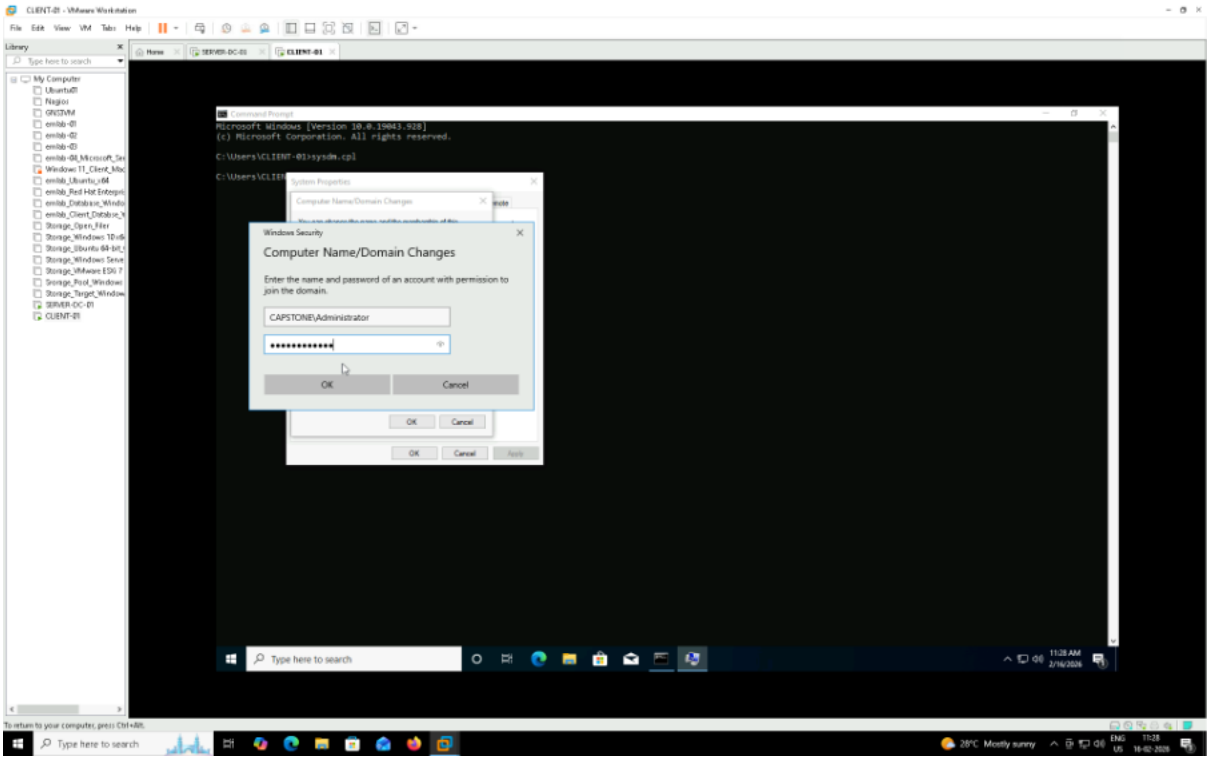

CLIENT-01 - VMware Workstation
Network Connections
Control Panel > Network and Internet > Network Connections
Ethernet0 Properties
Internet Protocol Version 4 (TCP/IPv4) Properties
Type here to search


CLIENT-01 - VMware Workstation
Command Prompt
Microsoft Windows [Version 10.0.19041.928]
(c) Microsoft Corporation. All rights reserved.
C:\Users\CLIENT-01>ncpa.cpl
C:\Users\CLIENT-01>ncpa.cpl
C:\Users\CLIENT-01>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet0:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::bc71:9175:ef4b:589f%6
IPv4 Address. . . . . : 192.168.159.50
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.159.2
C:\Users\CLIENT-01>
Type here to search

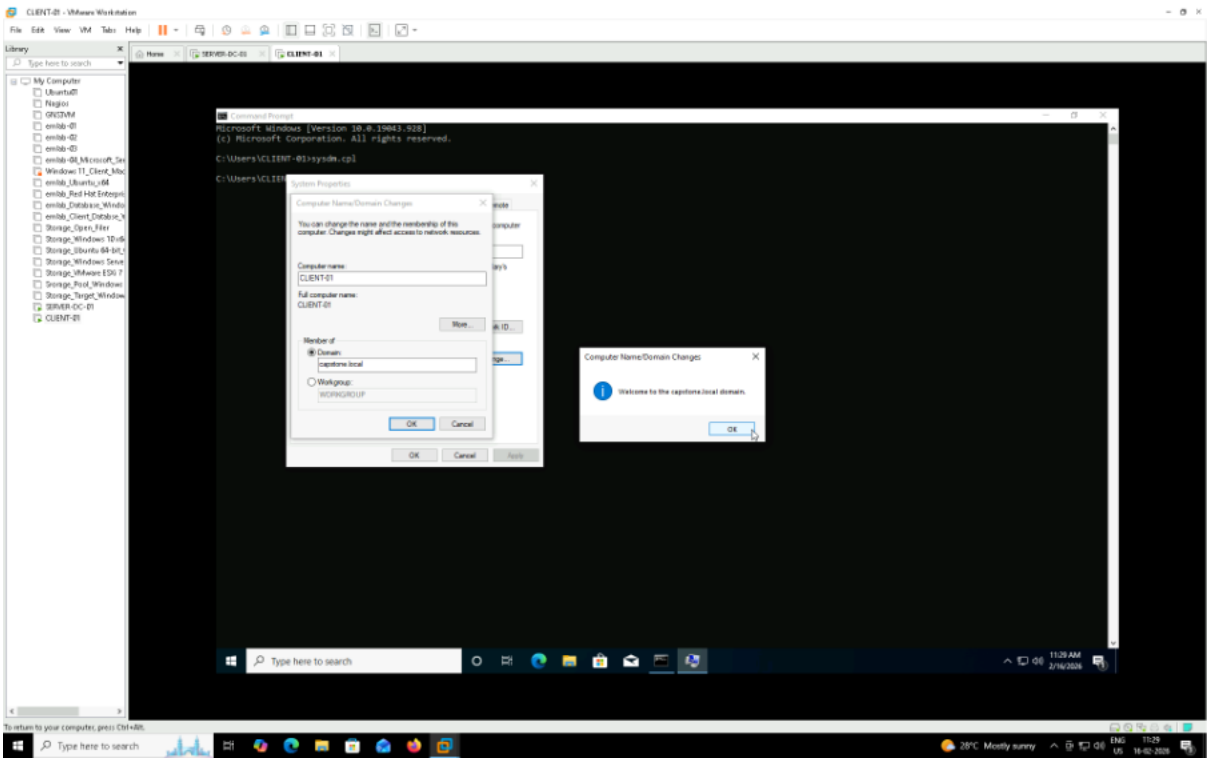


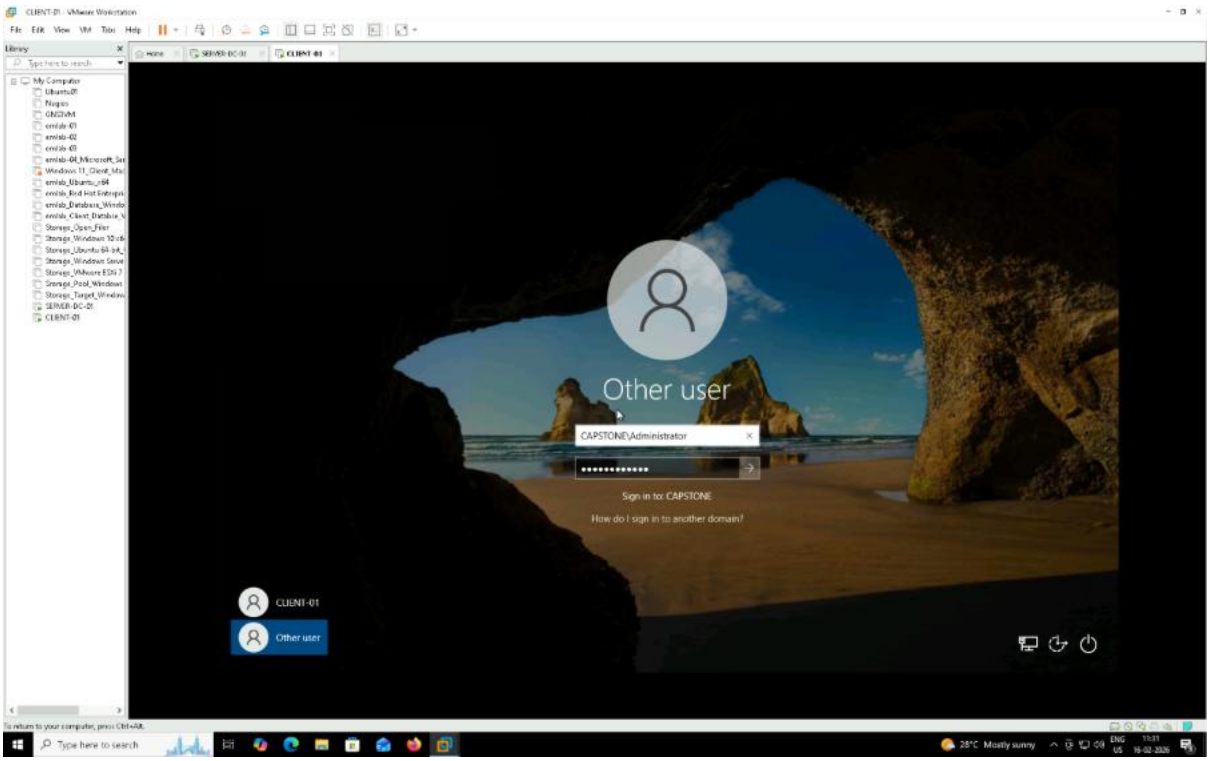
The client machine was joined to the **capstone.local** domain using domain administrator credentials. Successful authentication conformed communication between the client and Domain Controller via DNS and Active Directory Services.



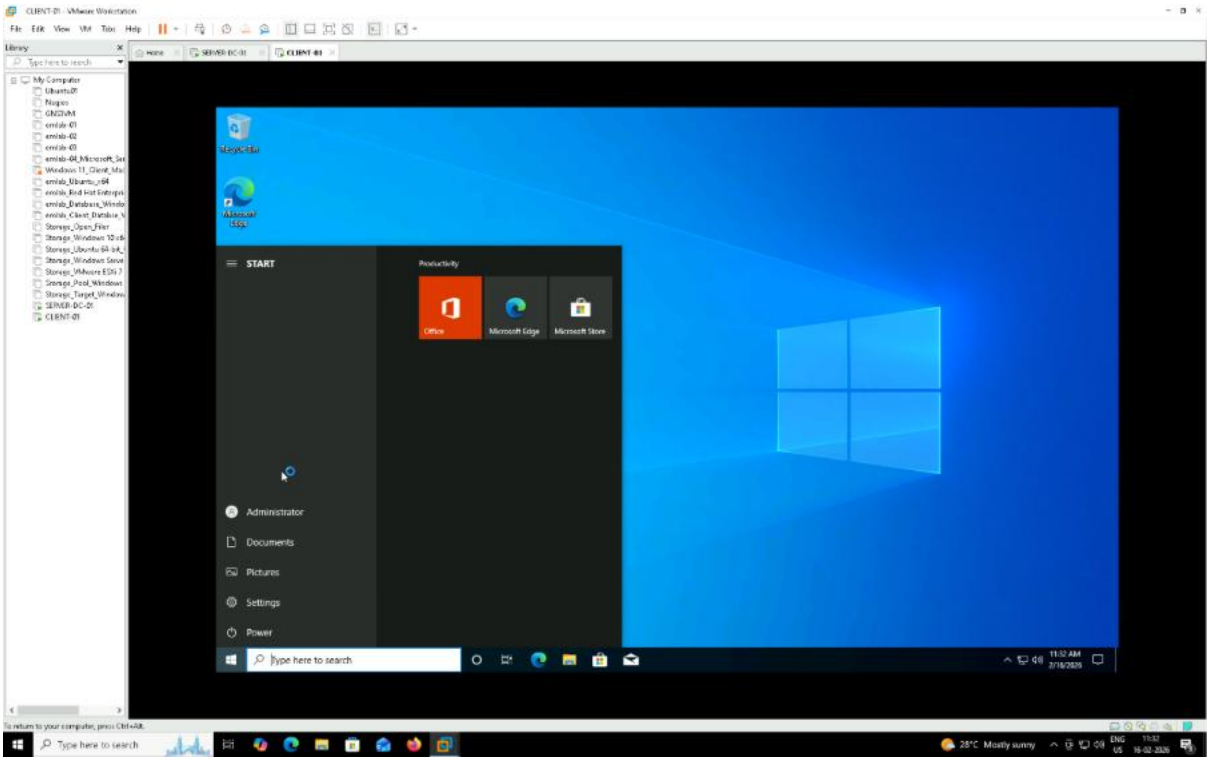


We got a message that our CLINET-01 has been connected successfully to our Domain.

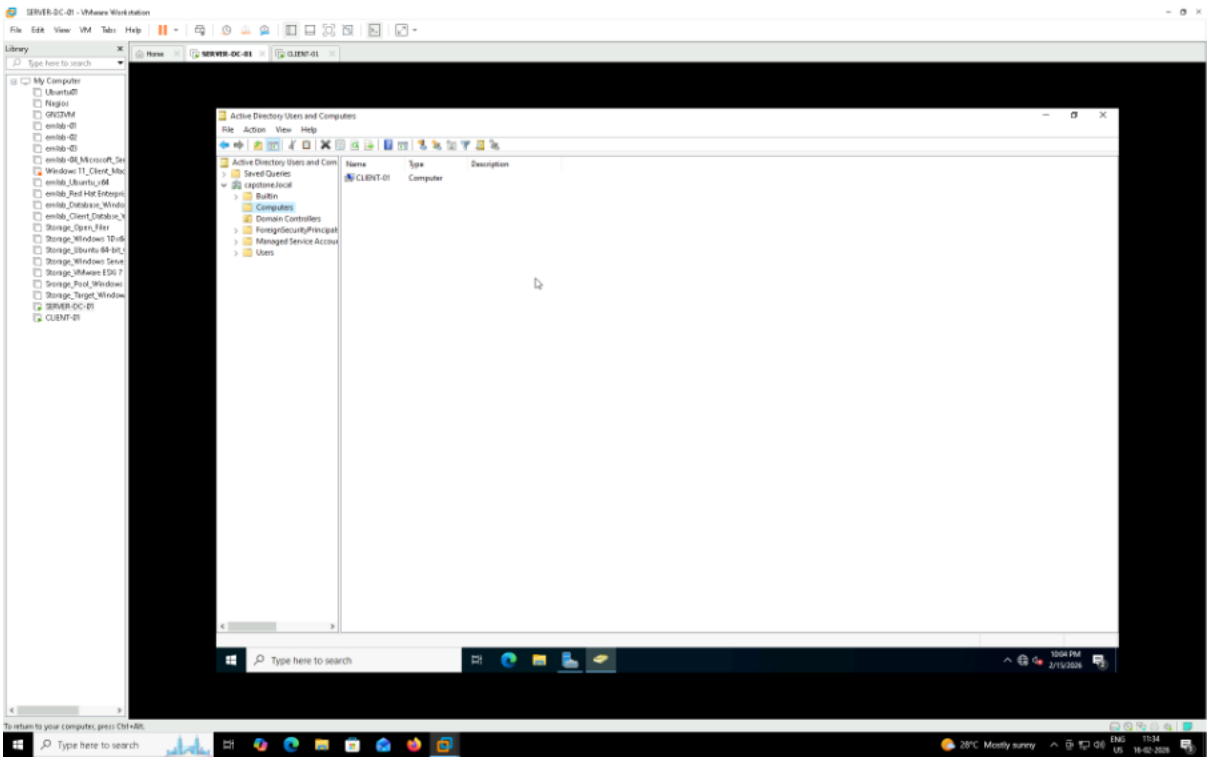




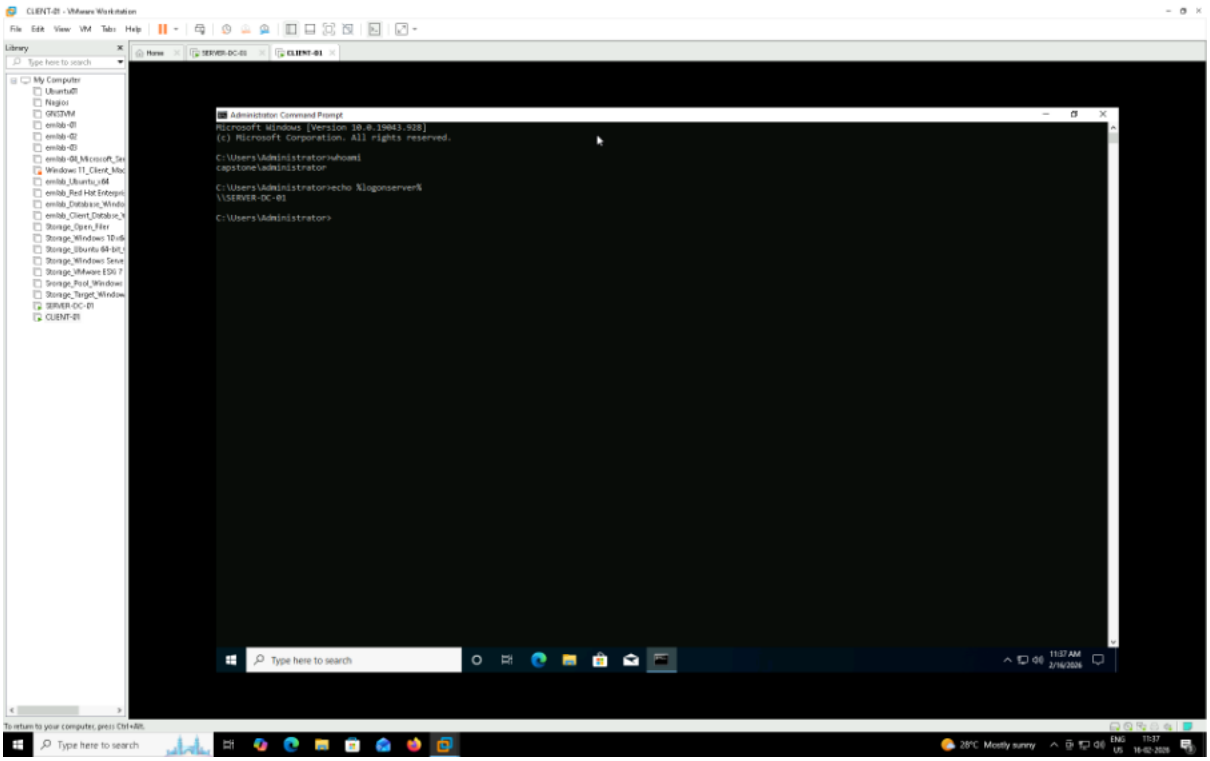
This proves that we have successfully logged into the Administrator account.

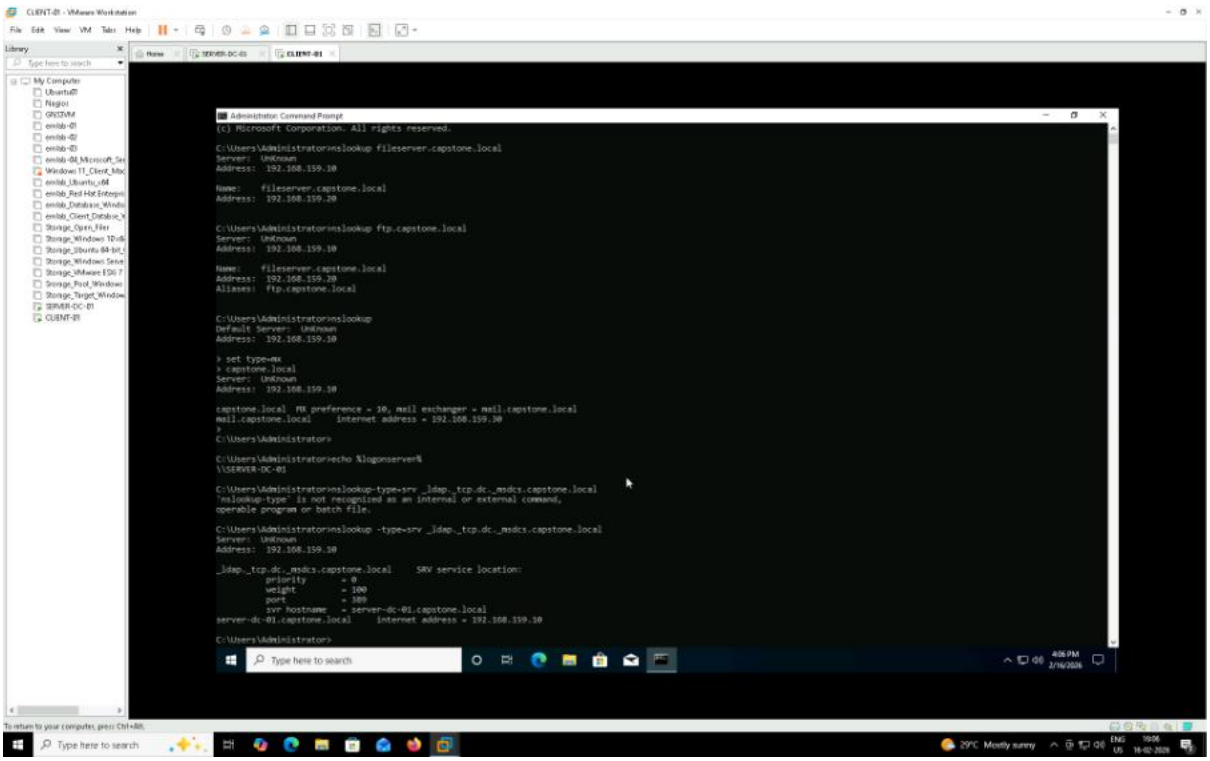


The CLIENT-01 is now visible in the Active Directory Users and Computers, conforming successfully domain join and registration with the Domain Controller.



The client authenticated against the Domain Controller. The whoami and logonserver commands confirmed that authentication requests were processed by SERVER-DC-01.





DNS resolution was verified from the domain client to confirm proper communication with the Domain Controller. Forward lookup, reverse lookup, alias resolution, mail exchange lookup, and domain controller service records were successfully resolved, demonstrating functional Active Directory Integrated DNS.

Assignment 3: Active Directory Structure

As part of this assignment, you need to complete below tasks:

1. Design a logical AD structure for a hypothetical medium-sized

organization

2. Create organizational units (OUs)

3. Implement user and computer account hierarchies

We will complete the Assignment in the following order: -

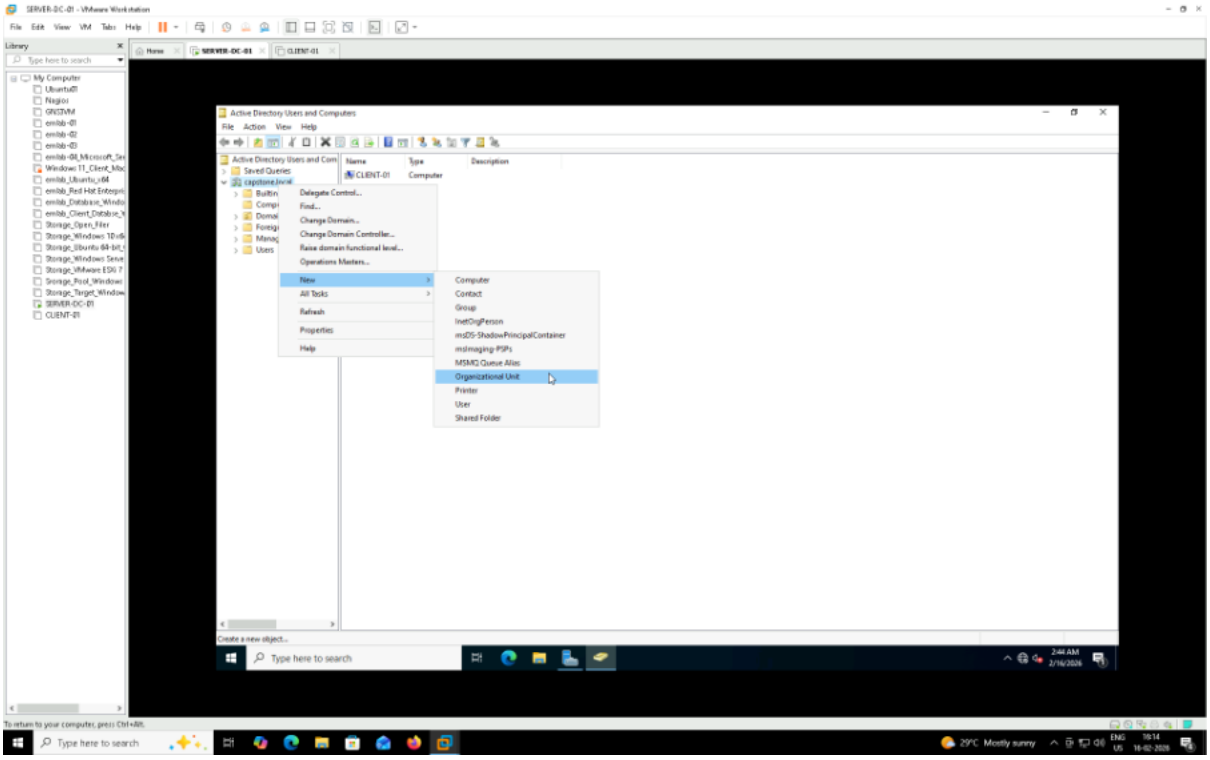
- Create Organizational Units (OU)

- Create Domain Users
- Create Security Groups
- Apply Group Policy (GPO)
- Test policy on client

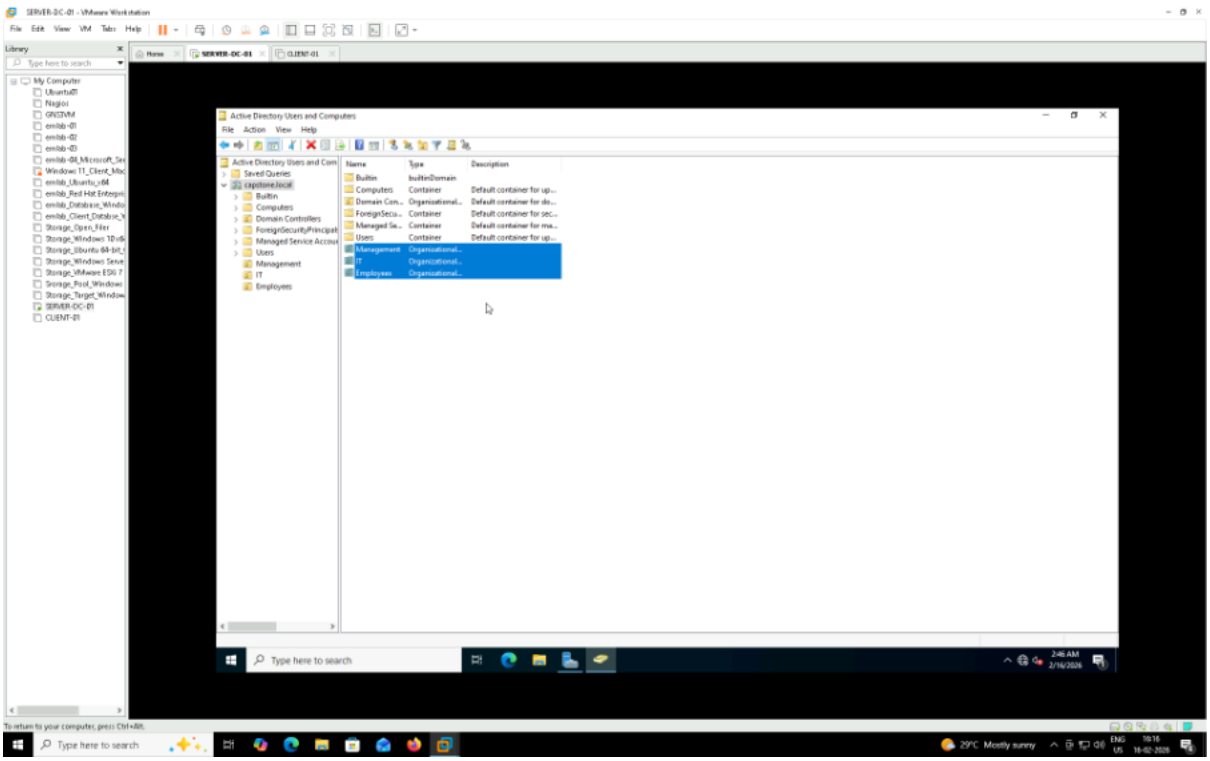
Keeping users directly in user containers is considered a bad practice. Companies organize departments using OUs.

We will create the OUs in the following format: -

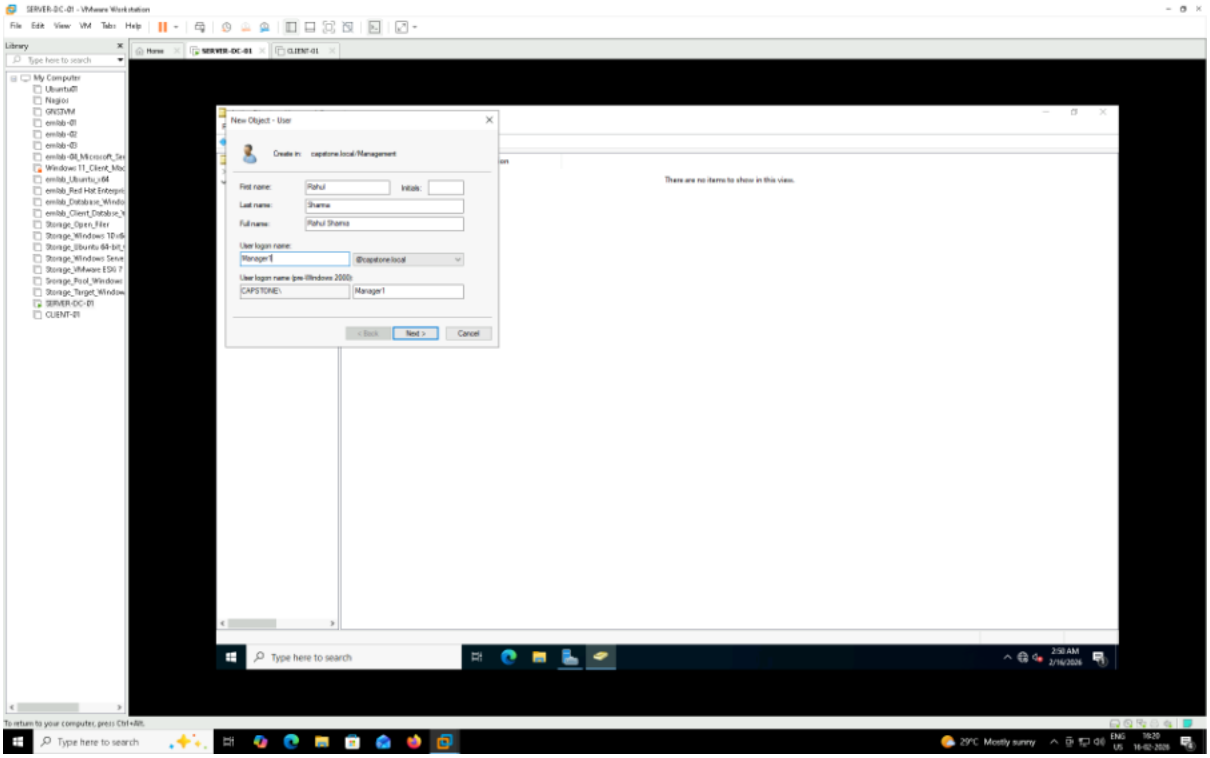
- capstone.local
    - Management
    - IT
    - Employees

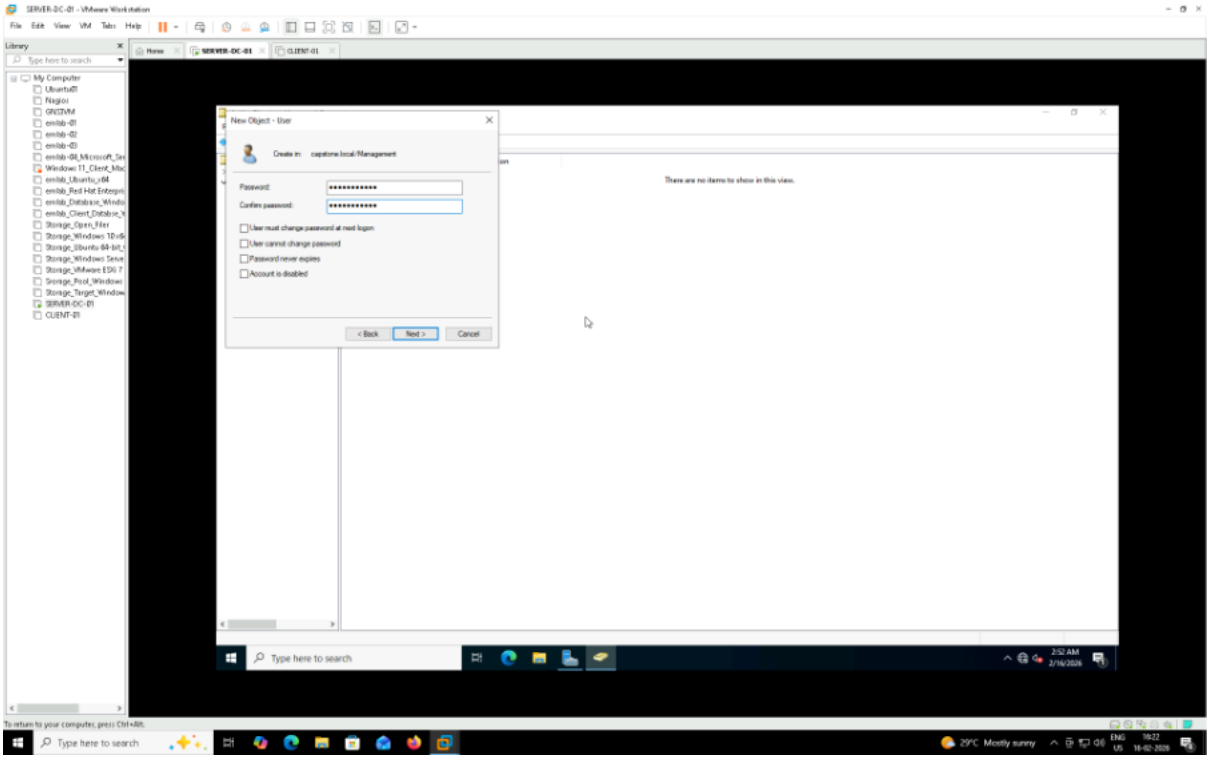


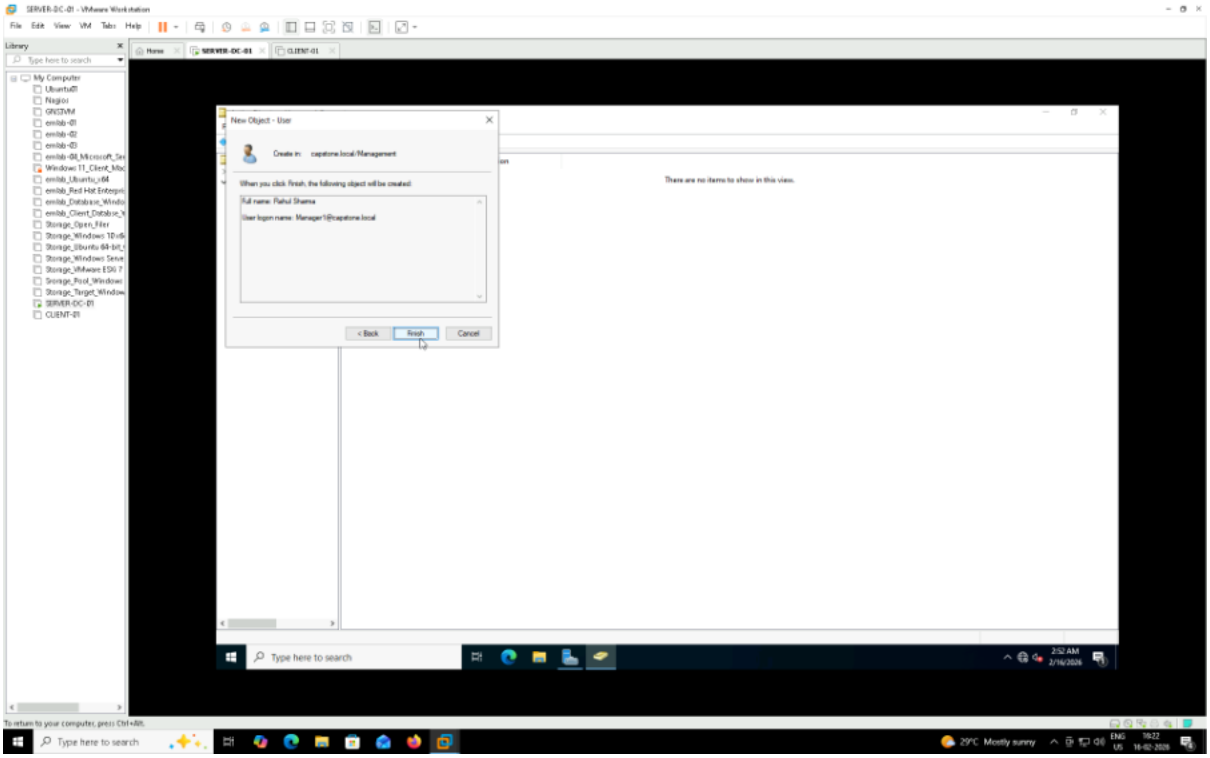
Organizational Units (OUs) were created to logically separate users based on departmental structure. This allows centralized administration and targeted policy application.

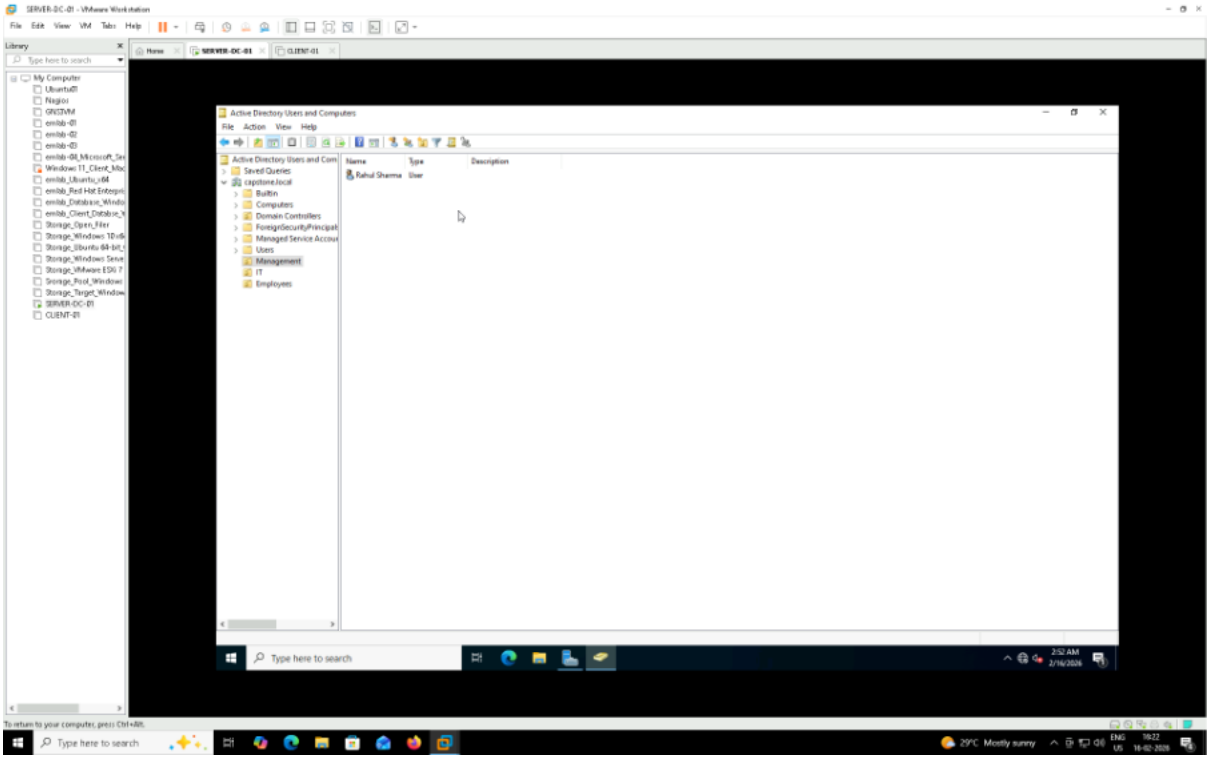


User accounts were created in their respective Organizational Units to represent departmental structure. This enables role-based management and policy enforcement within the domain environment.

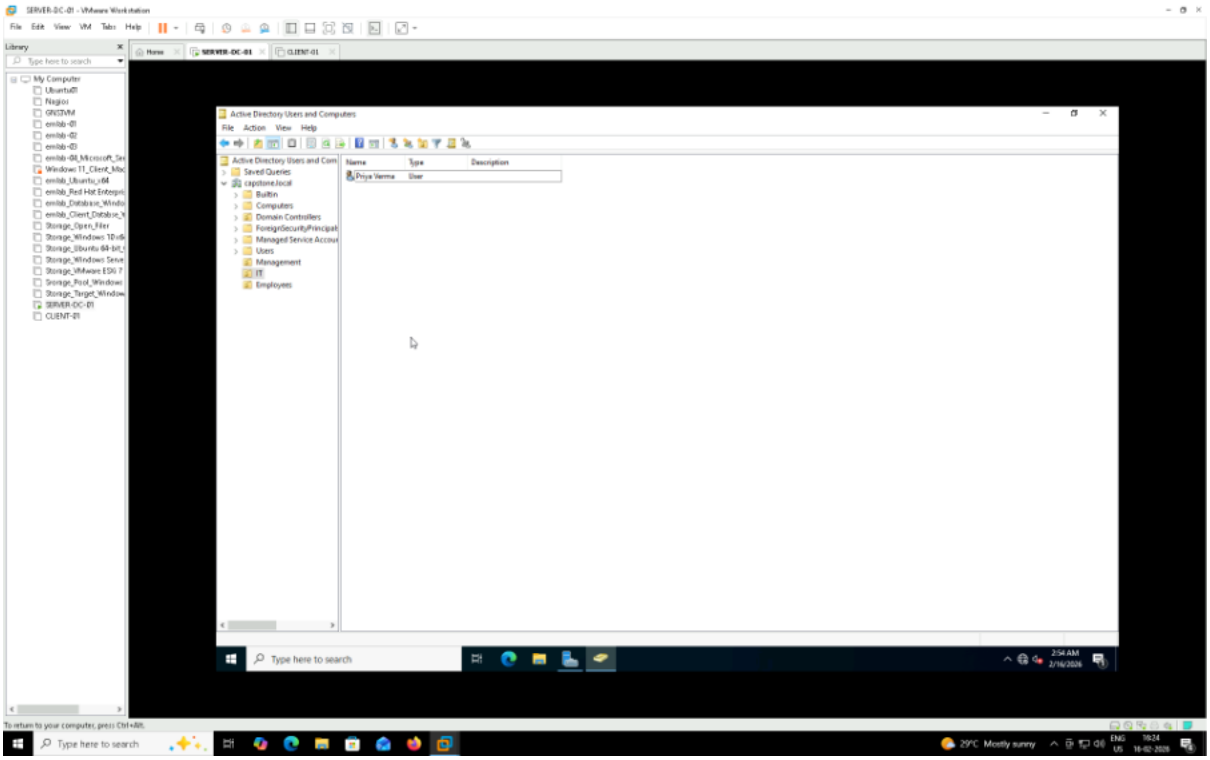

New Object - User
Create in: capstone.local/Management
First name: Rahul
Initials:
Last name: Sharma
Full name: Rahul Sharma
User logon name: Manager1 @capstone.local
User logon name (pre-Windows 2000): CAPSTONE\ Manager1
< Back
Next >
Cancel
There are no items to show in this view.
Type here to search

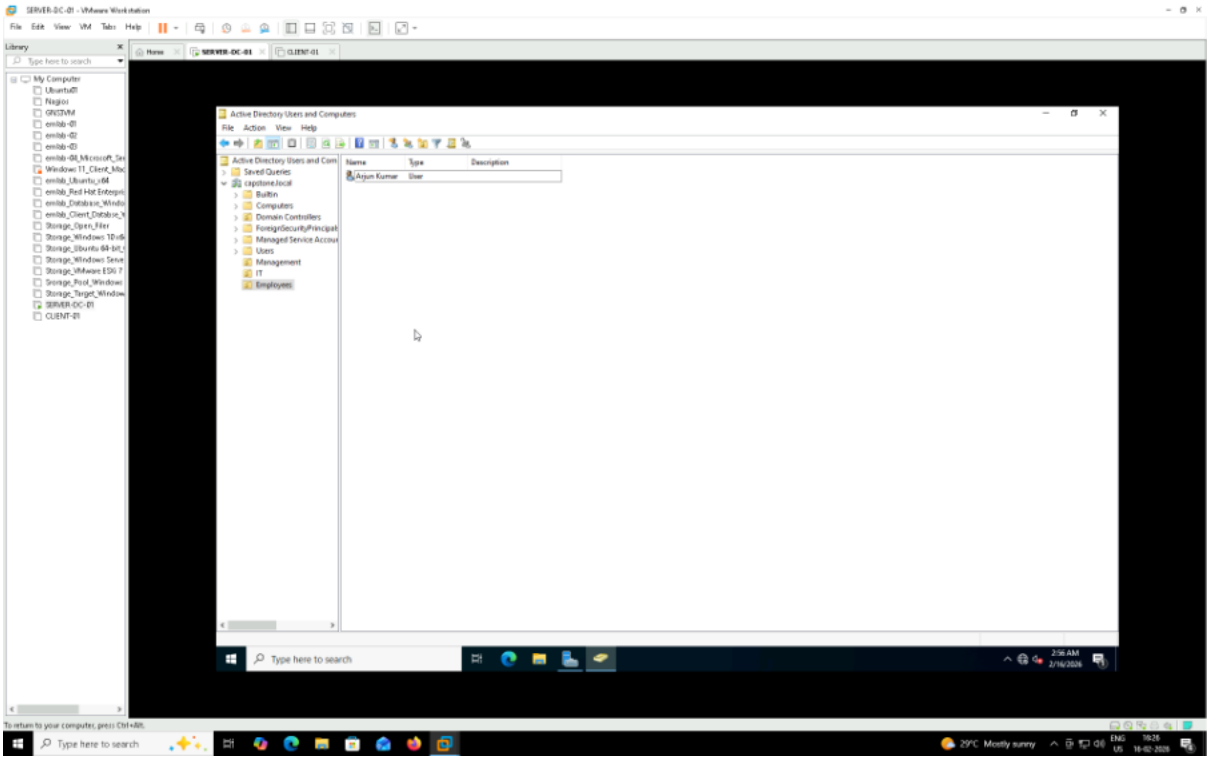

New Object - User
Create in: capstone.local/Management
Password:
Confirm password:
User must change password at next logon
User cannot change password
Password never expires
Account is disabled
< Back
Next >
Cancel
There are no items to show in this view.
Type here to search


New Object - User
Create in: capstone.local/Management
When you click finish, the following object will be created:
Full name: Rahul Sharma
User logon name: Manager1@capstone.local
< Back
Finish
Cancel
There are no items to show in this view.
Type here to search

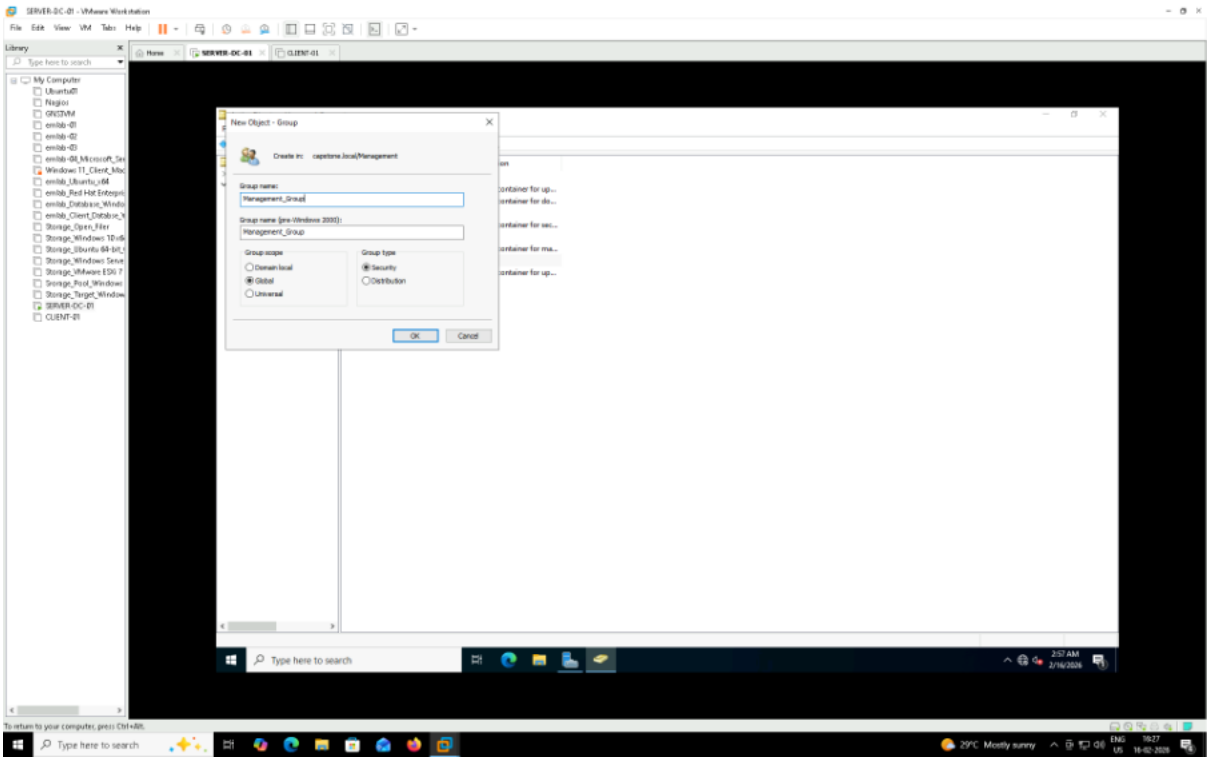

Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
capstone.local
Builtin
Computers
Domain Controllers
ForeignSecurityPrincipal
Managed Service Accou
Users
Management
IT
Employees
Name
Type
Description
Rahul Sharma
User
Type here to search

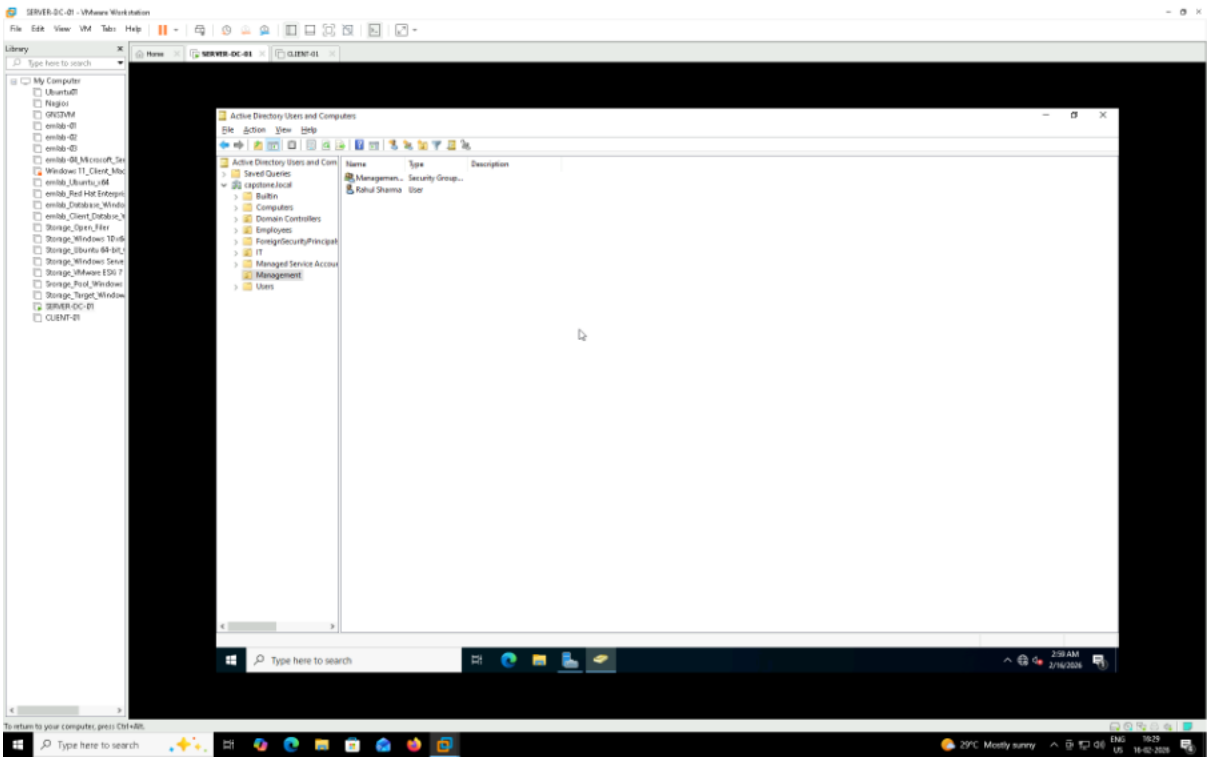
Similarly do the same for the other two Organizational Units.



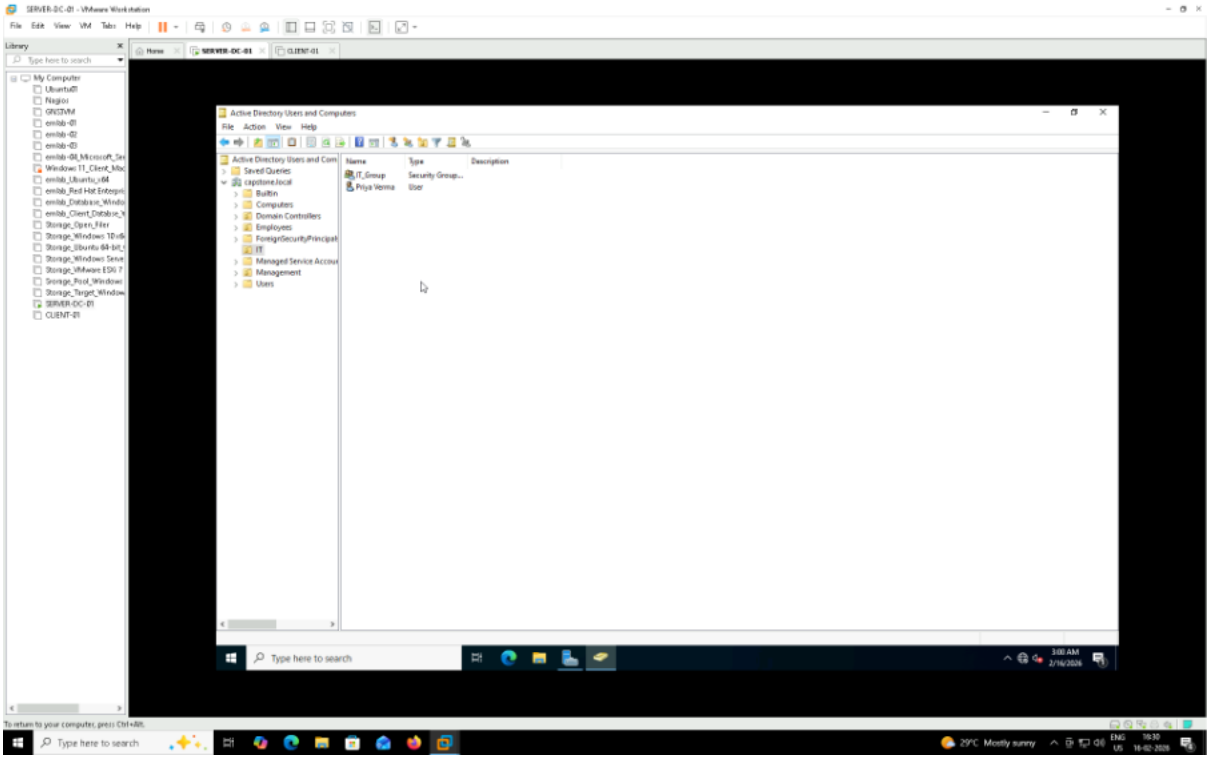


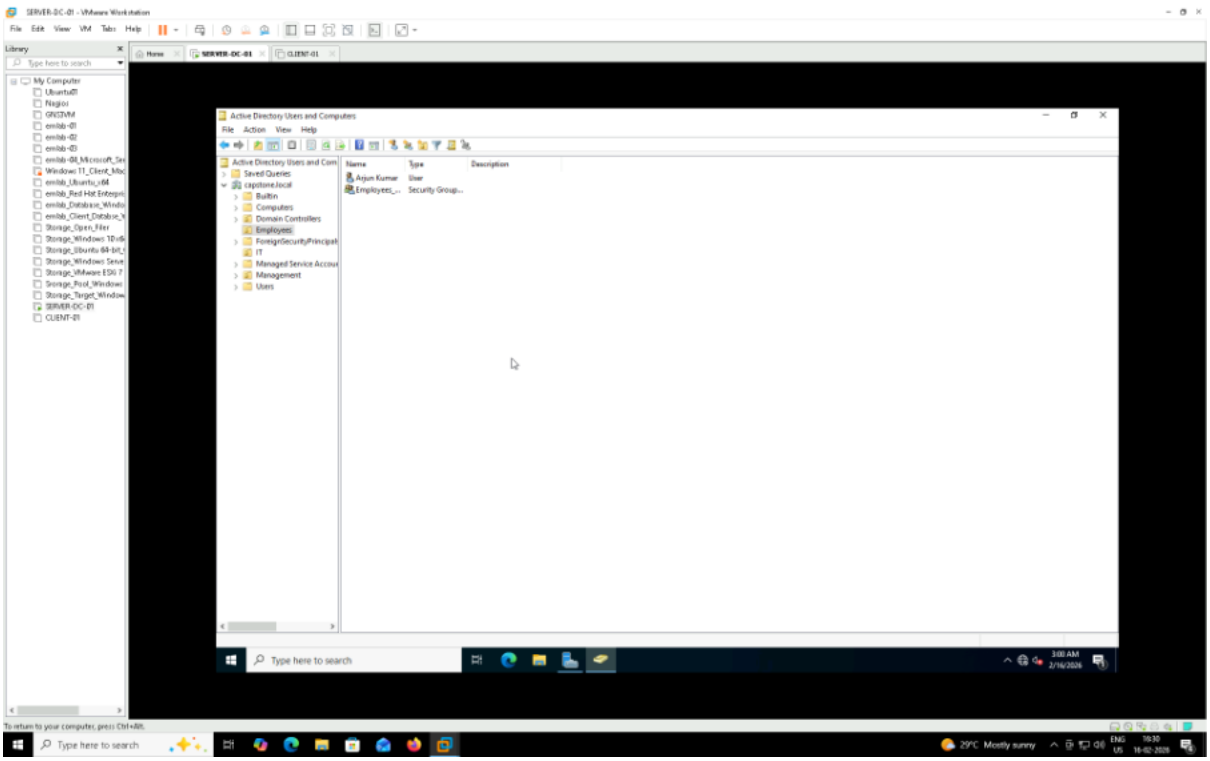
Security groups were created for each department and users were added to their respective groups. This allows centralized permission and policy assignment using role-based access control.



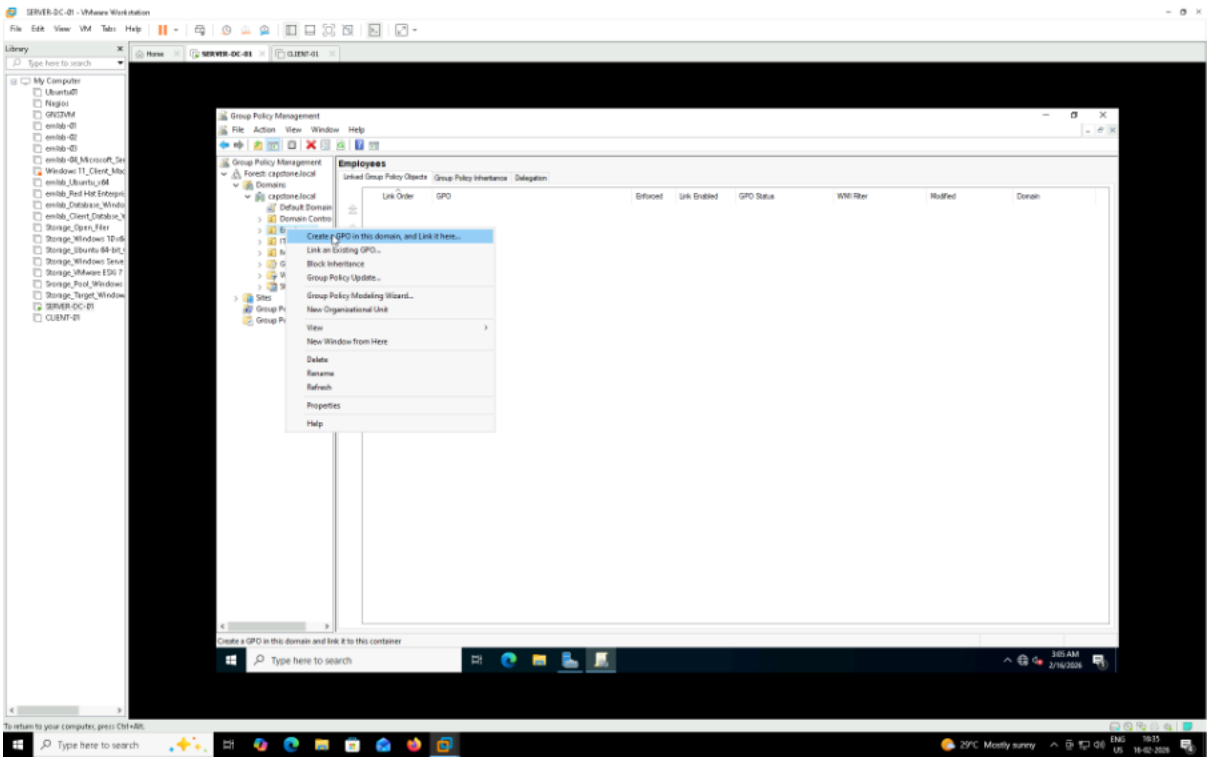


In the similar fashion, create for the rest of the OUs.

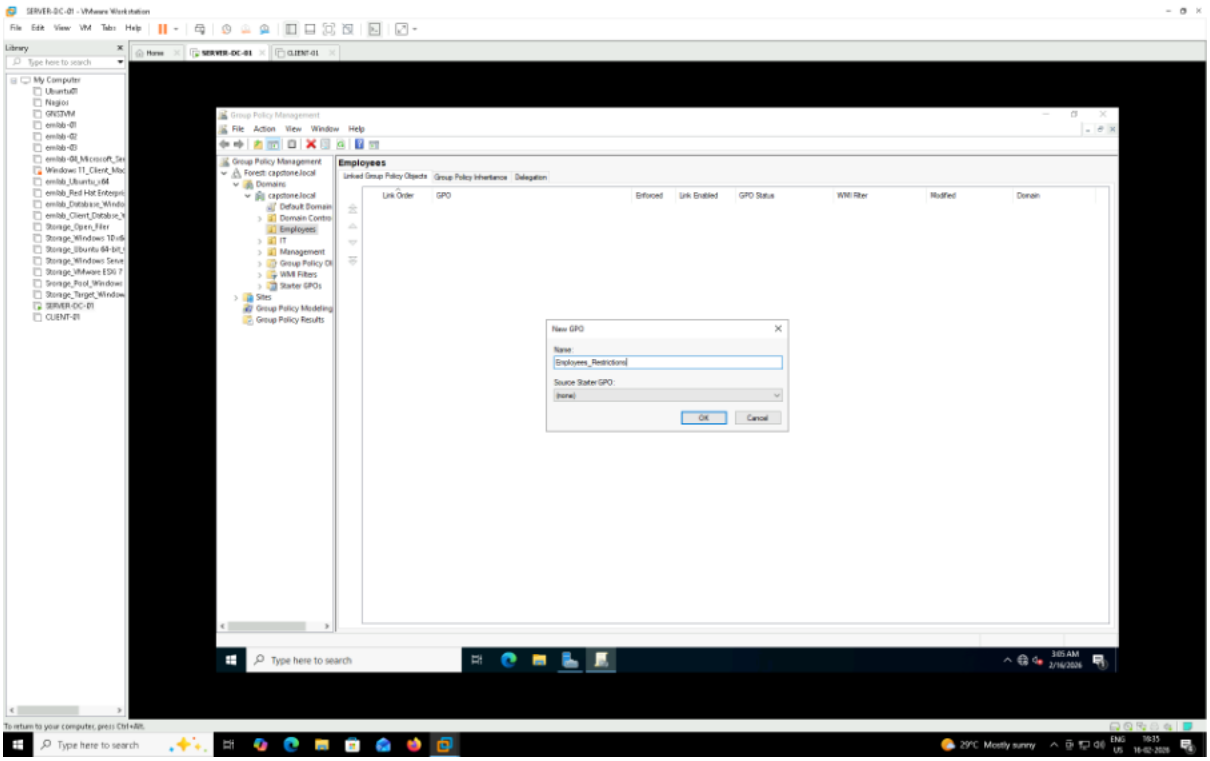


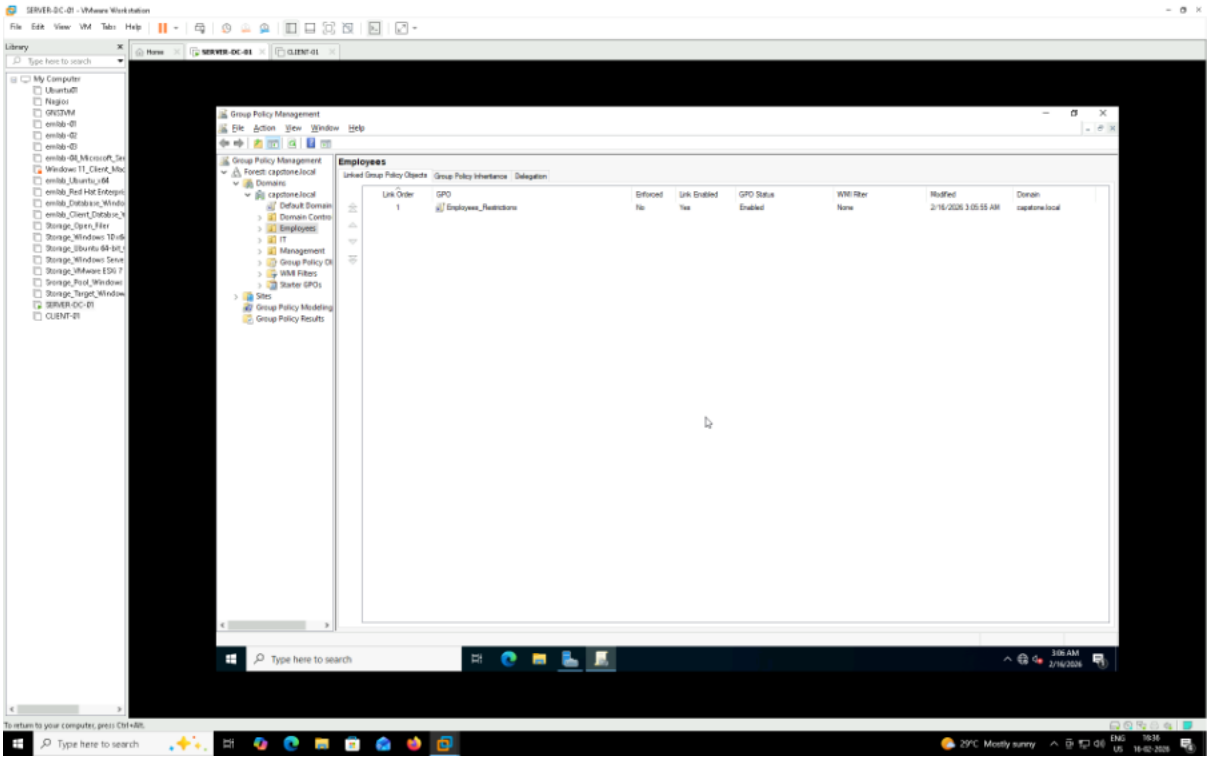


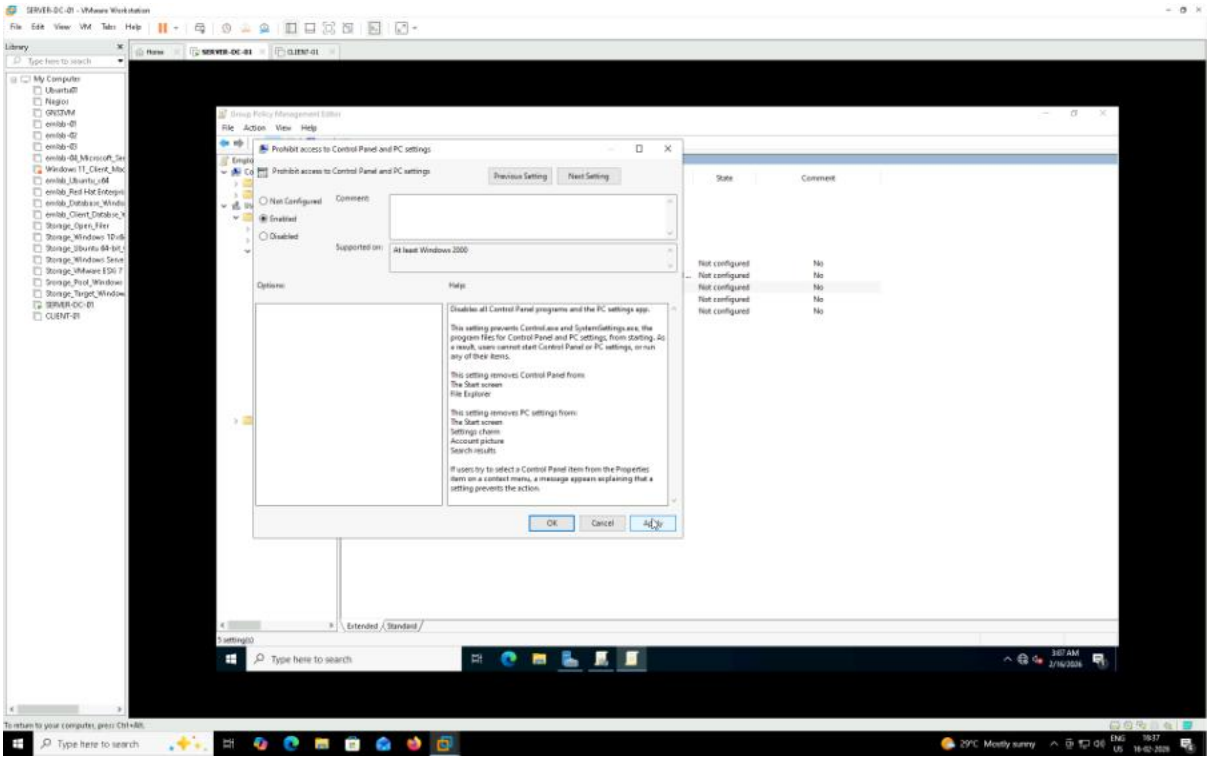
We will restrict the Employees so that they cannot open Control Panel / Settings.



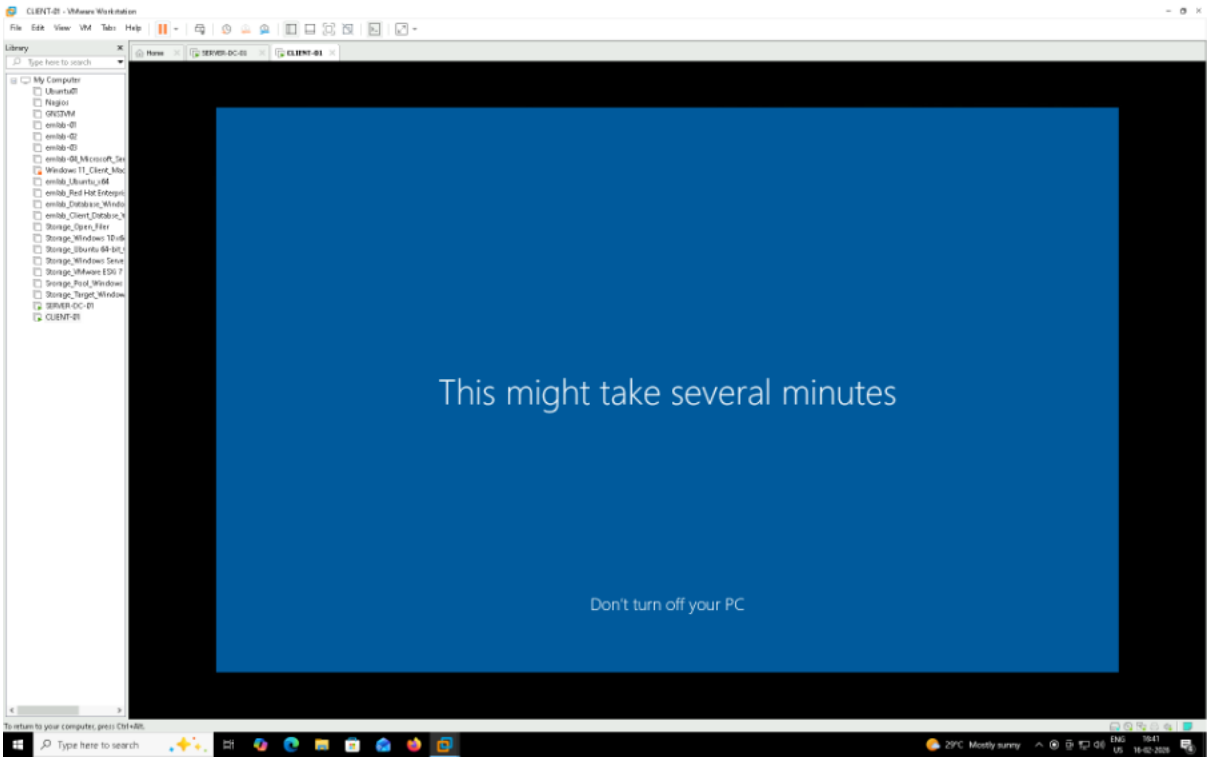
A Group Policy Object named Employees_Restrictions was created and linked to the Employees Organizational Unit. The policy was configured to prohibit access to Control Panel and PC Settings to restrict standard user configuration changes.



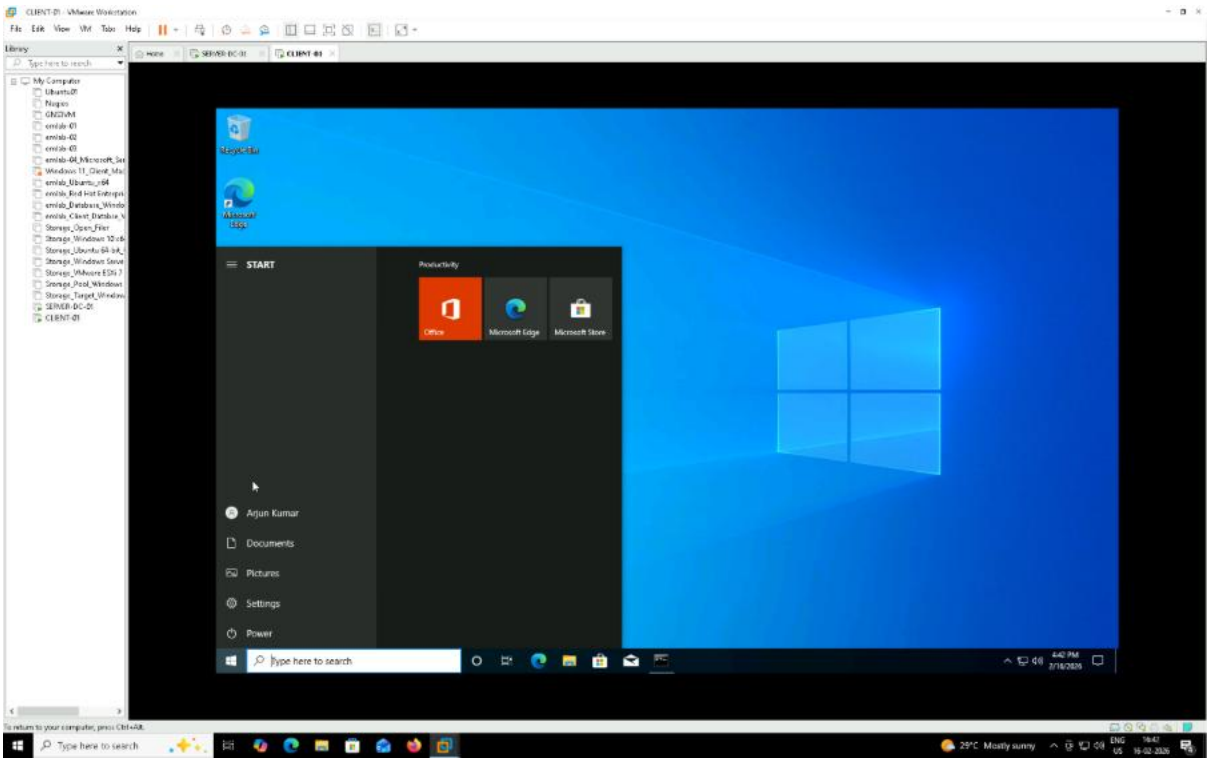

Group Policy Management
Employees
Group Policy Management Editor
Control Panel



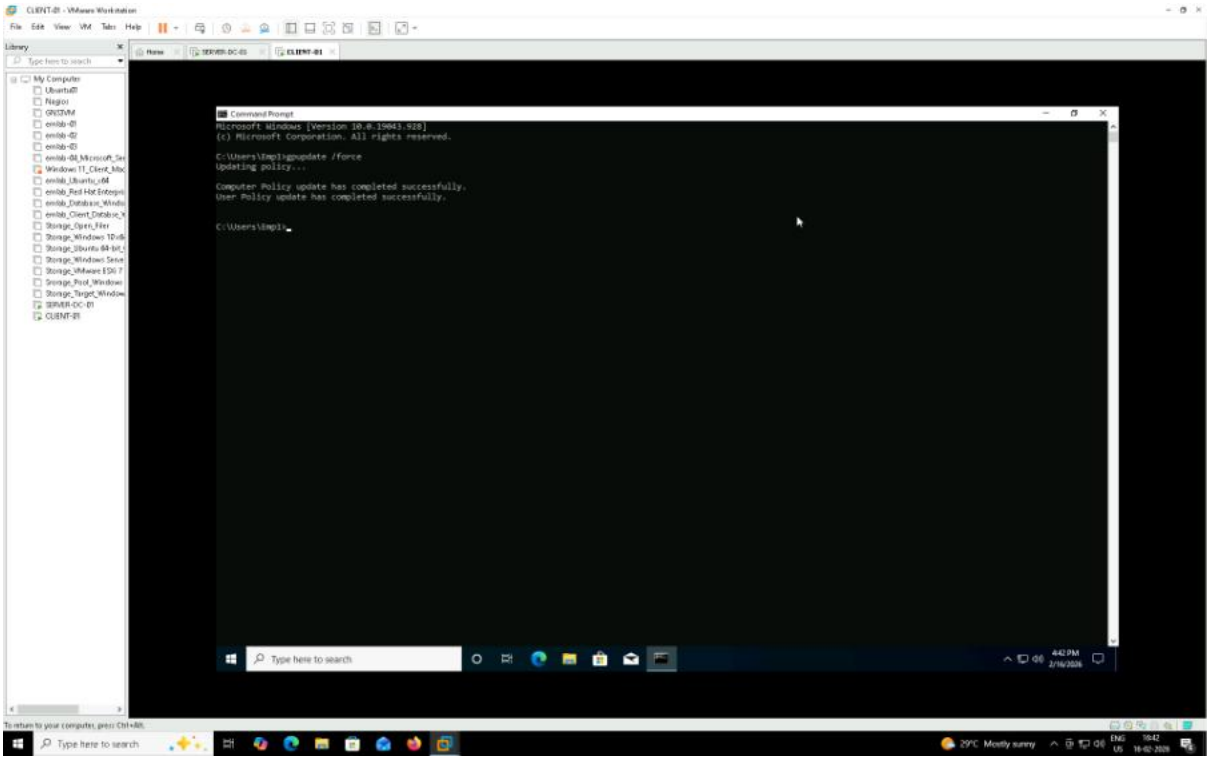
Logging in as Emp1 in CLIENT-01 machine. As we have logged in for the first time in the user account, it is getting setup.

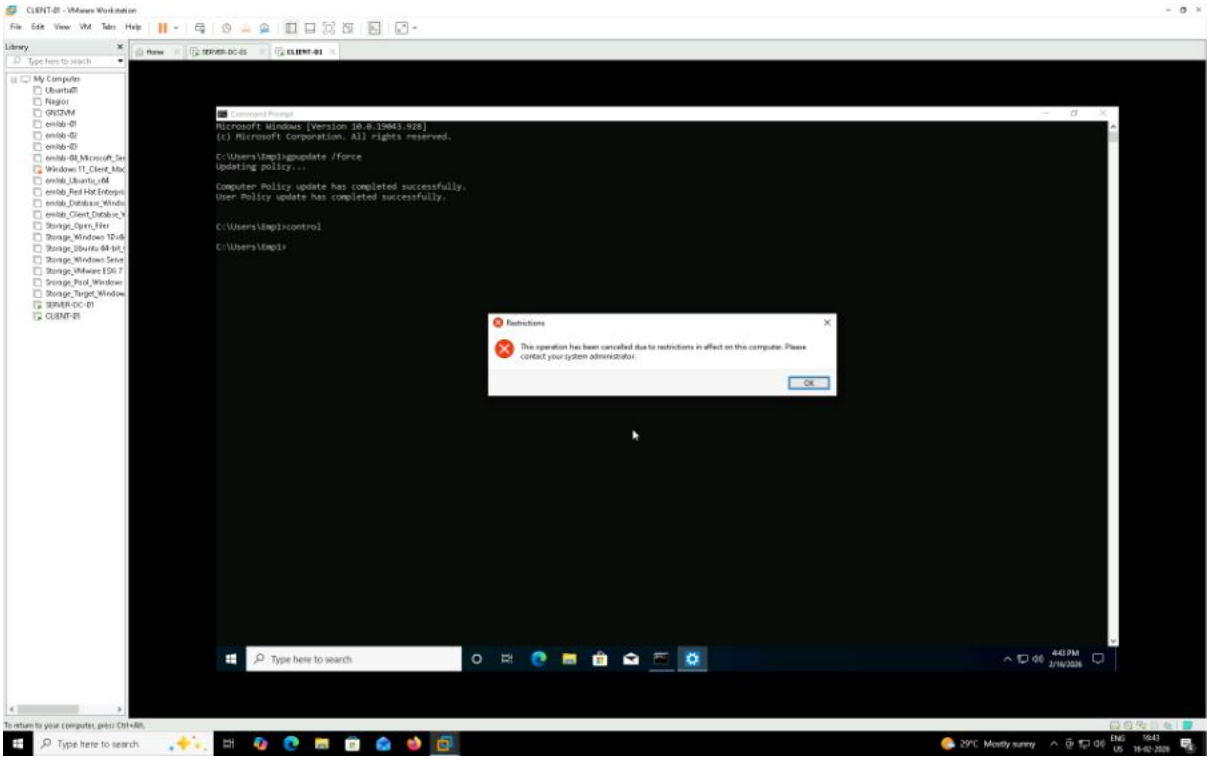


We have successfully logged into the created user.

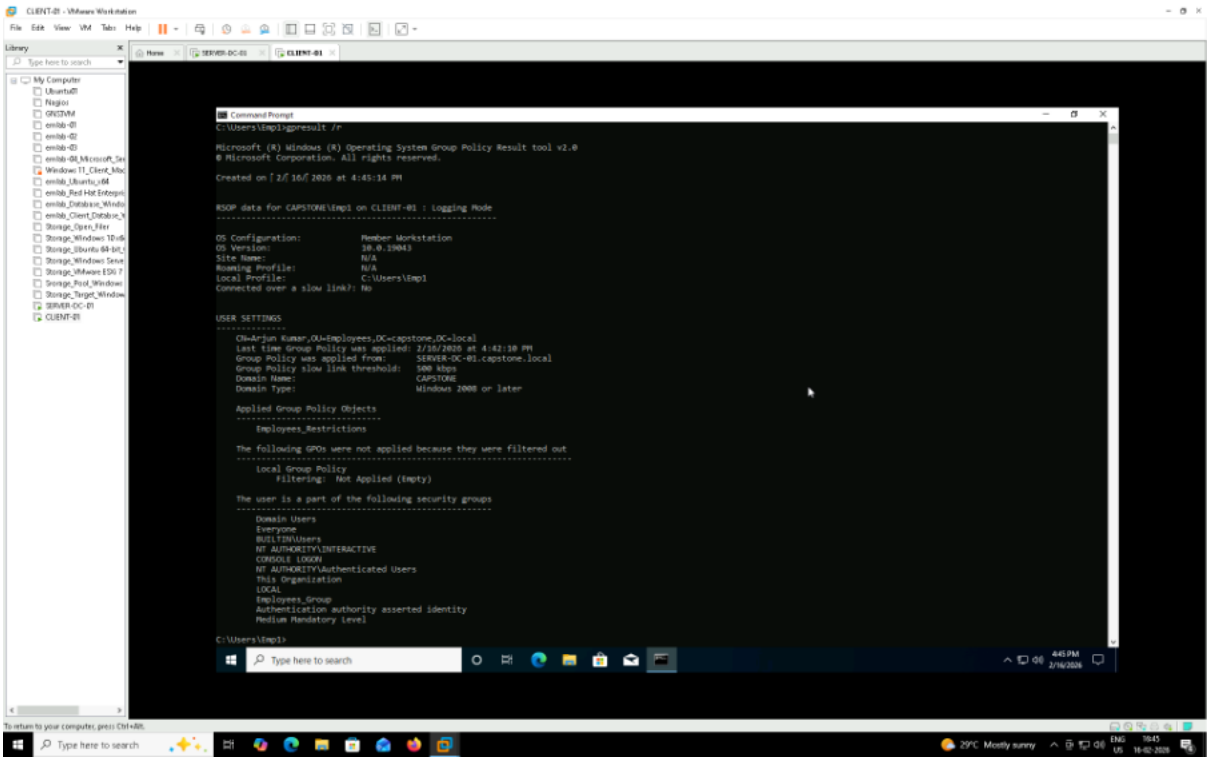


Updating the group policies that we have introduced.





After policy update, the employee user was unable to access Control Panel, confirming successful application of the Group Policy from the Domain Controller.



The gpresult command confirmed that the Employees_Restrictions Group Policy Object was applied to the client system, verifying the policy enforcement originated from the Domain Controller.

Conclusion: -

The Active Directory Domain Services (ADDS) implementation provided practical experience in configuring centralized identity and access management within a Windows Server Environment. The domain controller was successfully deployed, and core components such as Users, Groups and Organizational Units (OUs) were created and managed. Group Policy Objects (GPOs) were configured to enforce security and administrative controls across domain-joined systems.

The assignment demonstrated the importance of centralized authentication, authorization and policy management in enterprise environments. ADDS enhances security, simplifies administration, and improves scalability by allowing administrators to manage resources efficiently within a structured domain architecture.

Overall, the ADDS implementation strengthened understanding of enterprise directory services and domain-based network management.